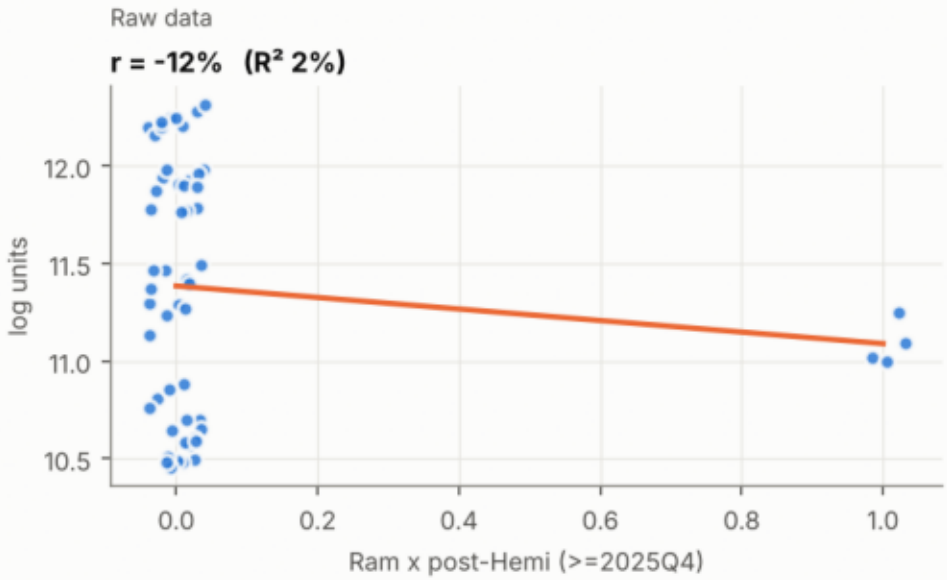
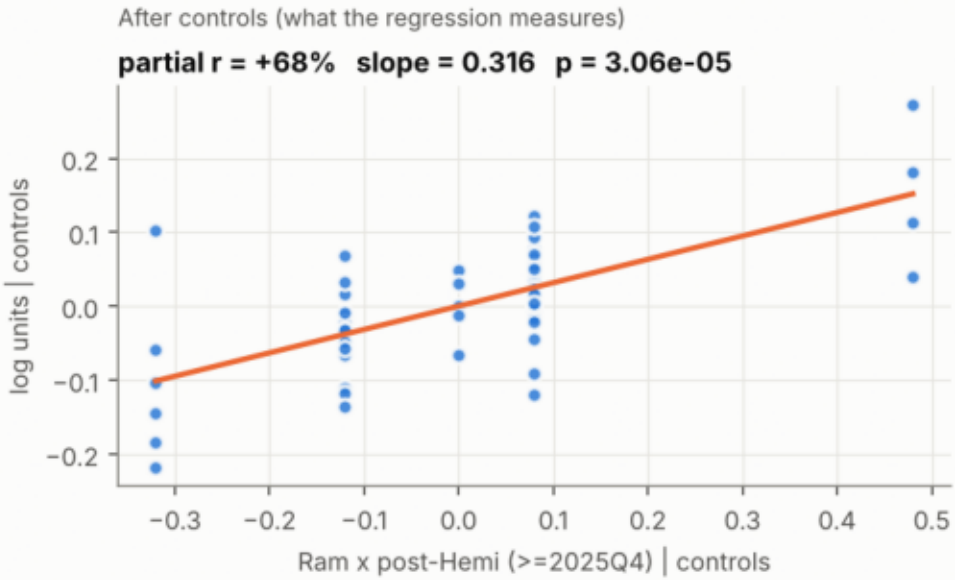
Stellantis powertrain demand - correlation charts for every regression

Each page: left = raw data with OLS line and Pearson r (as %); right = the same relationship after removing the regression's
controls / fixed effects / seasonality - its slope is the reported coefficient and its partial r is the regression's correlation.
r is shown as a percent (e.g. r = +62% means 0.62); R² is the share of variance explained.

86 charts. Tests with no data (no chart possible):
9. Conquest share ~ new powertrain launches - No conquest figures in any 2022-2026 Stellantis US sales release (searched); J.D. Power / S&P [...]
14. Model sales ~ (MSRP - segment ATP) - KBB segment ATP collected (data/processed/kbb_atp_monthly.csv), but model-level MSRP/transaction prices [...]
15. Hemi option price ~ Hemi share of Ram sales - Stellantis does not disclose Hemi take rates; only two option-price points exist, so no [...]
16. Listing discount % ~ powertrain - Requires the weekly dealer-listing scrape described in the project plan (not yet collected; CarGurus [...]
17. Discount % ~ model year, by powertrain - Requires listing scrape.
18. Stellantis incentive % ~ days' supply - Cox publishes brand incentives / days' supply only inside paid vAuto/Cox data or press charts; the [...]
19. Dealer discount minus OEM incentive ~ days listed - Requires listing scrape plus a history of OEM offers.
20. Forum price-paid reports ~ MSRP, by powertrain - Needs manual/consented collection from Ram Forum and Jeep Garage threads; not collected.
21. VIN exit-rate hazard model ~ powertrain + price band + metro - Needs the weekly listings scrape (VIN, powertrain via NHTSA vPIC decode, [...]
22. Days listed ~ powertrain | trim, price - Needs the weekly listings scrape (VIN, powertrain via NHTSA vPIC decode, price, dealer, first/last [...]
23. Days' supply by model ~ # powertrain variants - Cox model-level days' supply is not public (vAuto subscription).
24. Dealer inventory depth ~ powertrain mix - Needs the weekly listings scrape (VIN, powertrain via NHTSA vPIC decode, price, dealer, first/last [...]
25. Brand days' supply ~ production cuts - No public brand days' supply series; production cuts appear only as text in releases.
26. Prior-model-year share ~ powertrain - Needs the weekly listings scrape (VIN, powertrain via NHTSA vPIC decode, price, dealer, first/last [...]
27. Exit rate ~ price band, Stellantis vs Ford - Needs the weekly listings scrape (VIN, powertrain via NHTSA vPIC decode, price, dealer, [...]
29. Exit rate ~ distance from metro center - Needs the weekly listings scrape (VIN, powertrain via NHTSA vPIC decode, price, dealer, first/last [...]
30. Exit rate ~ color/package, same powertrain - Needs the weekly listings scrape (VIN, powertrain via NHTSA vPIC decode, price, dealer, [...]
34. Reddit price-complaint share ~ brand ATP - reddit.com returns 403 from this environment; needs Reddit API credentials (OAuth app).
35. Reddit Hemi sentiment ~ Ram retail growth - Same: Reddit API access required.
36. YouTube review views ~ launch-quarter sales - YouTube Data API v3 requires an API key; none configured.
37. CarGurus deal-rating mix ~ exit rate - Requires CarGurus listings scrape (site returns 406; check terms).
39. Dealer review sentiment ~ inventory turn - Google Places API key required, plus dealer inventory turns from the listings scrape.
42. Cherokee hybrid exit rate ~ gas prices - Needs listings scrape; national Cherokee hybrid sales exist for only 4 quarters.
46. Stellantis share ~ Texas/Southeast vs coastal states - State new-vehicle registration data by make is licensed (S&P Global [...]
53. Used values: Hemi vs Hurricane Ram 1500 - Needs used-listing scrape (or Manheim/J.D. Power valuation data by engine); Manheim's public index [...]
54. Used Jeep prices ~ new Jeep incentives - No public Jeep incentive series (see #18) and no brand-level used price index.

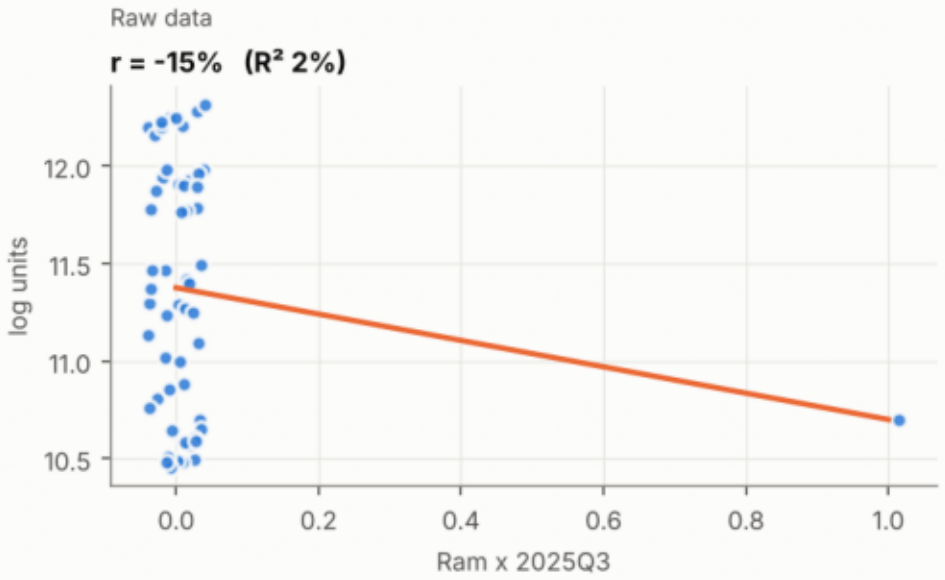
Test 1: Hemi return DiD: Ram LD (1500) vs F-150/Silverado/Sierra/Tundra

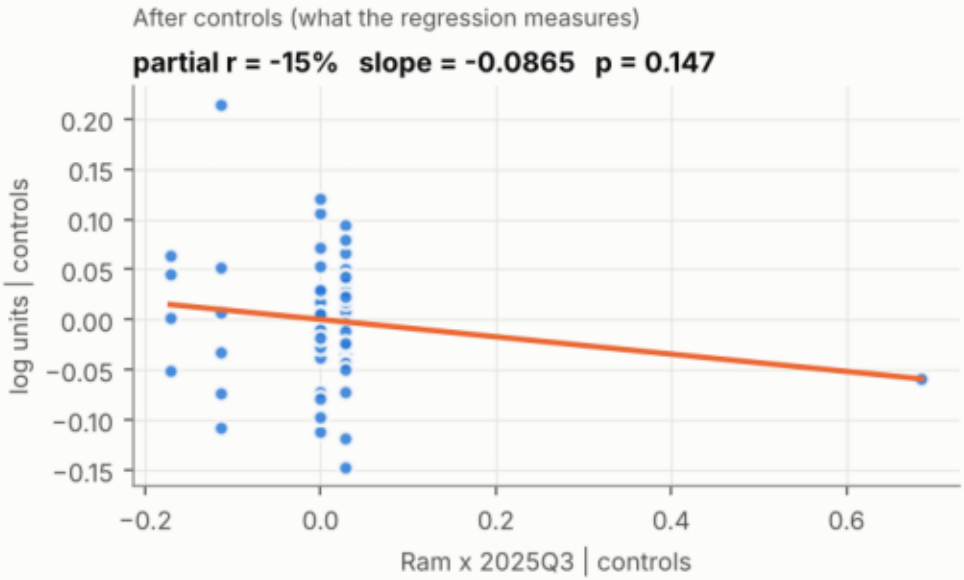




log(units) ~ Ram x Post(>=2025Q4) + Ram x 2025Q3 + unit FE + quarter FE; HC1 SE n = 55 model R² = 99% [RUN]

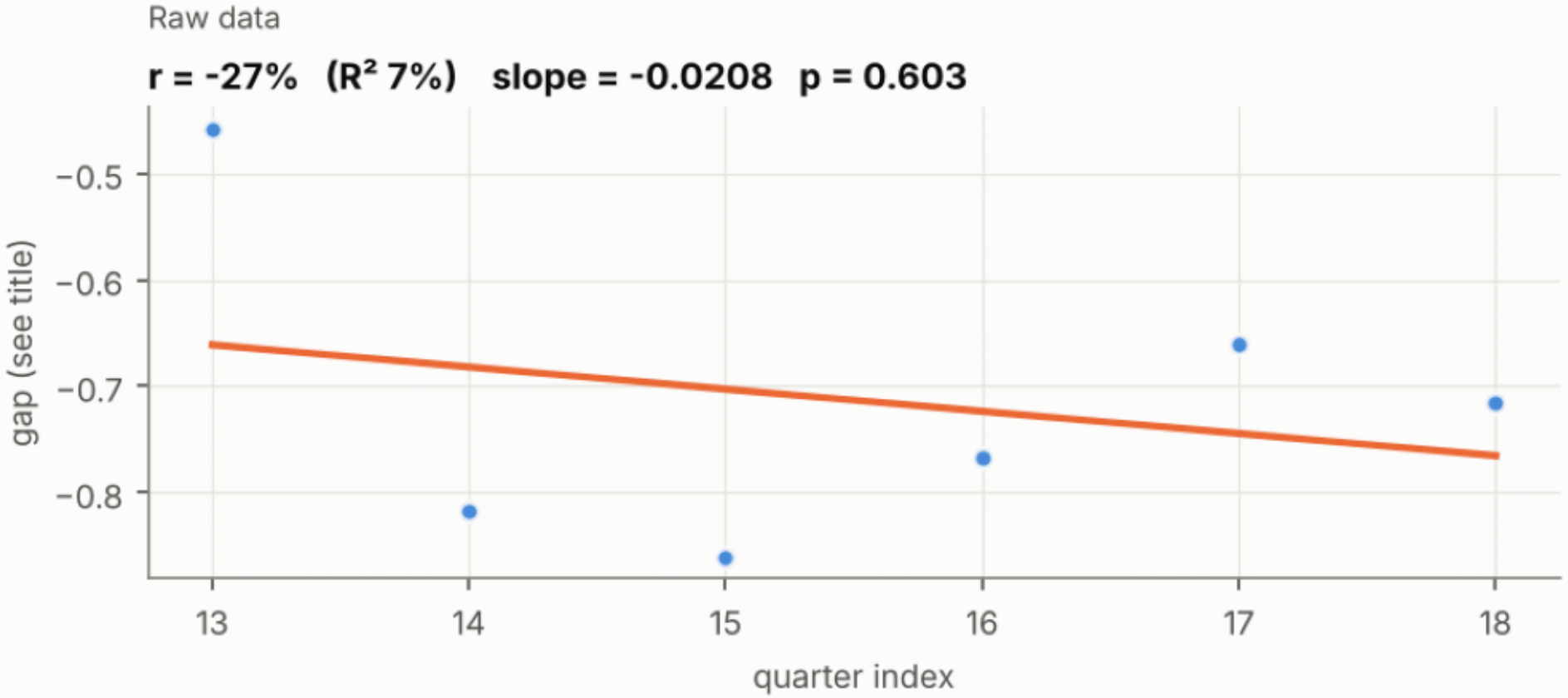
Test 1: Hemi return DiD: Ram LD (1500) vs F-150/Silverado/Sierra/Tundra





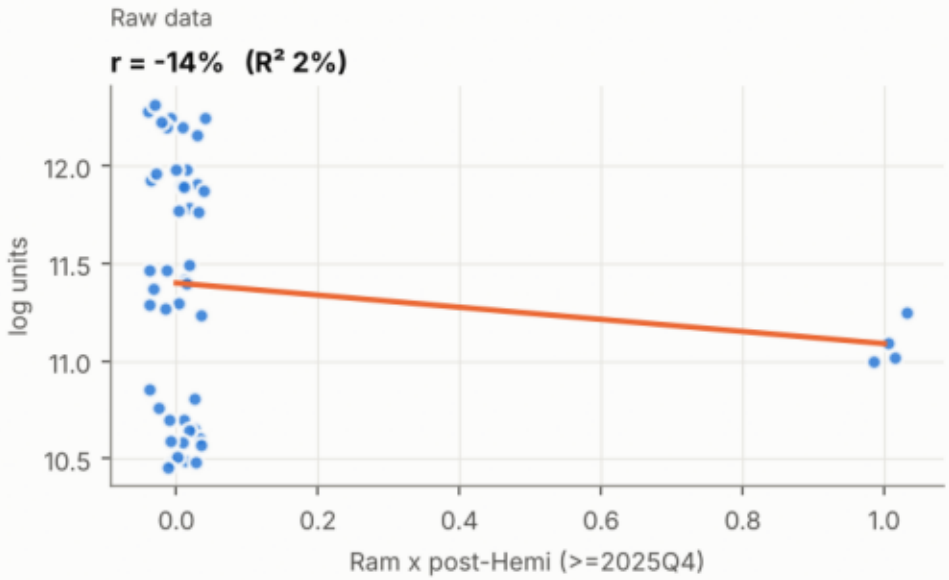
log(units) ~ Ram x Post(>=2025Q4) + Ram x 2025Q3 + unit FE + quarter FE; HC1 SE n = 55 model R² = 99% [RUN]

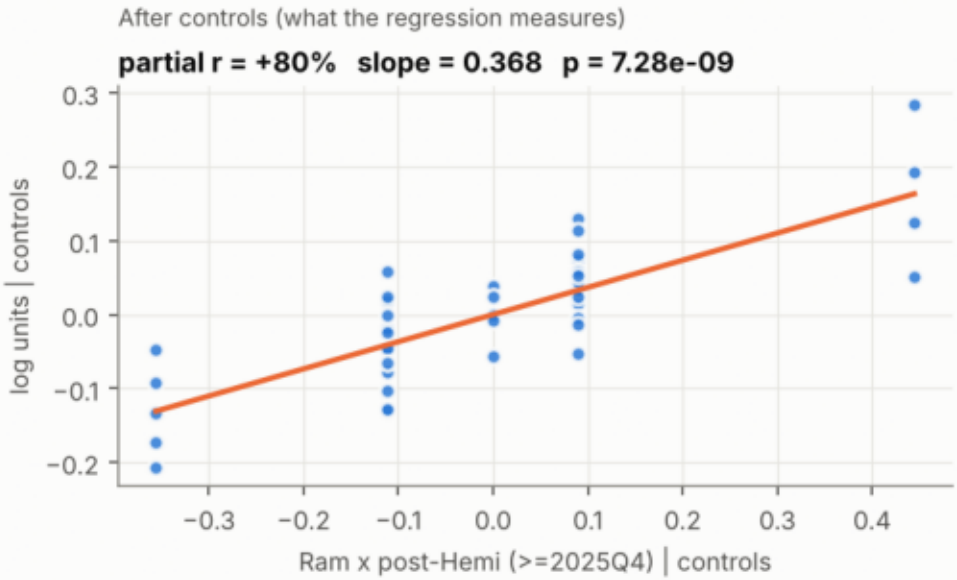
Test 1: Pre-trend: slope of Ram-vs-control log gap before 2025Q3 (Ram LD (1500) vs F-150/Silverado/Sierra/Tundra)



gap_t = log(Ram) - mean log(controls); gap ~ t over pre-period; HAC(1) n = 6 model R² = 7% [RUN]

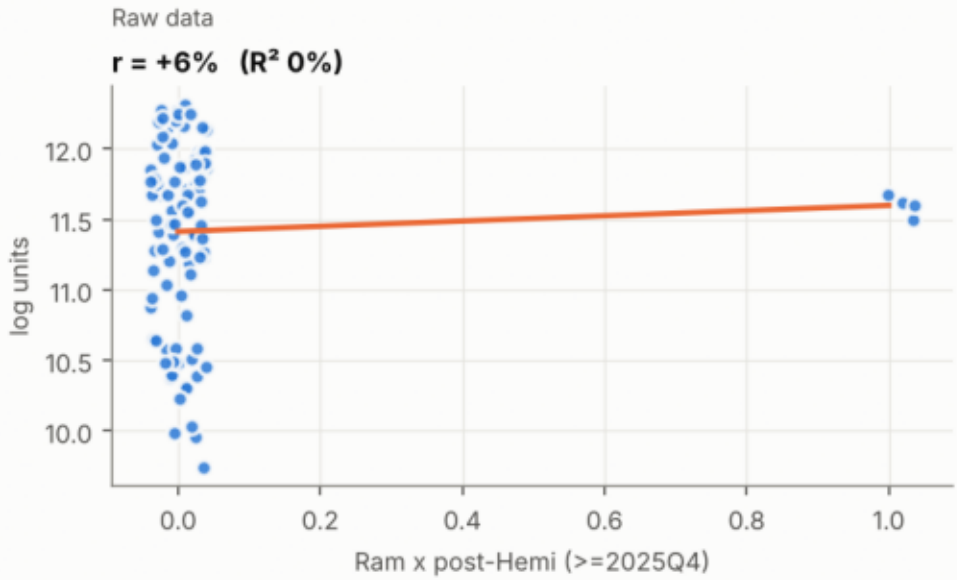
Test 1: Robustness: Ram LD DiD, pre-period 2024Q2-2025Q2 (Hurricane-transition quarters only)

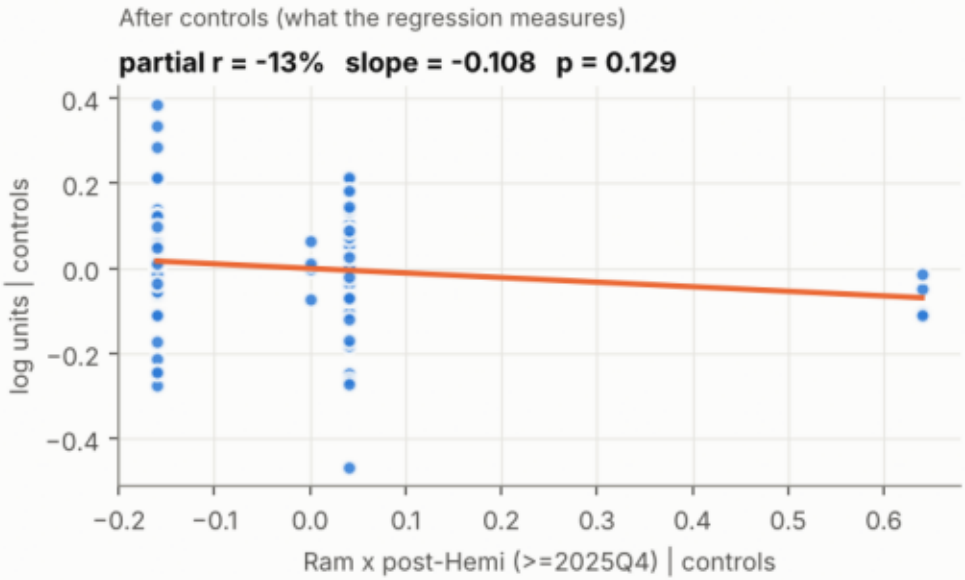




as above, dropping 2024Q1 n = 50 model R² = 99% [RUN]

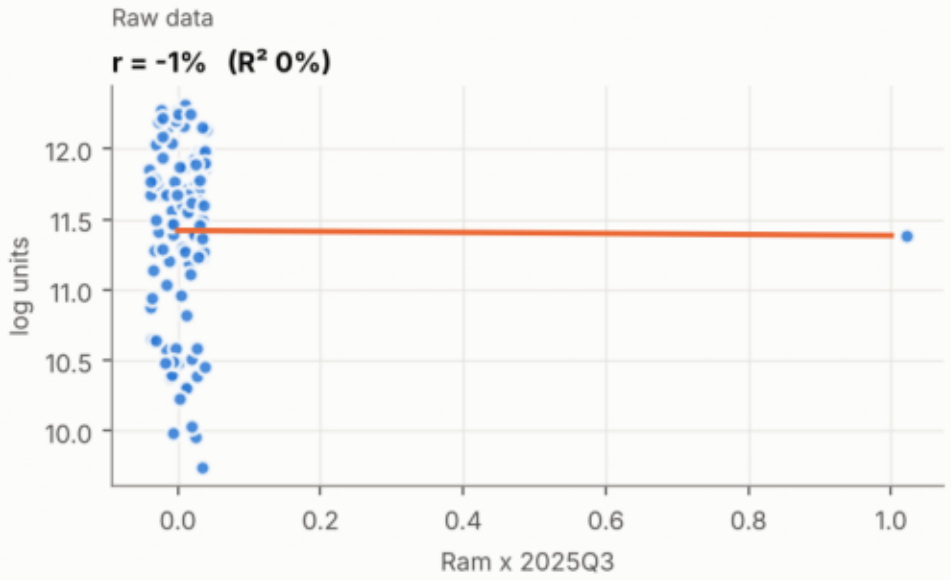
Test 1: Hemi return DiD: Ram pickups total vs same controls (longer pre-period)

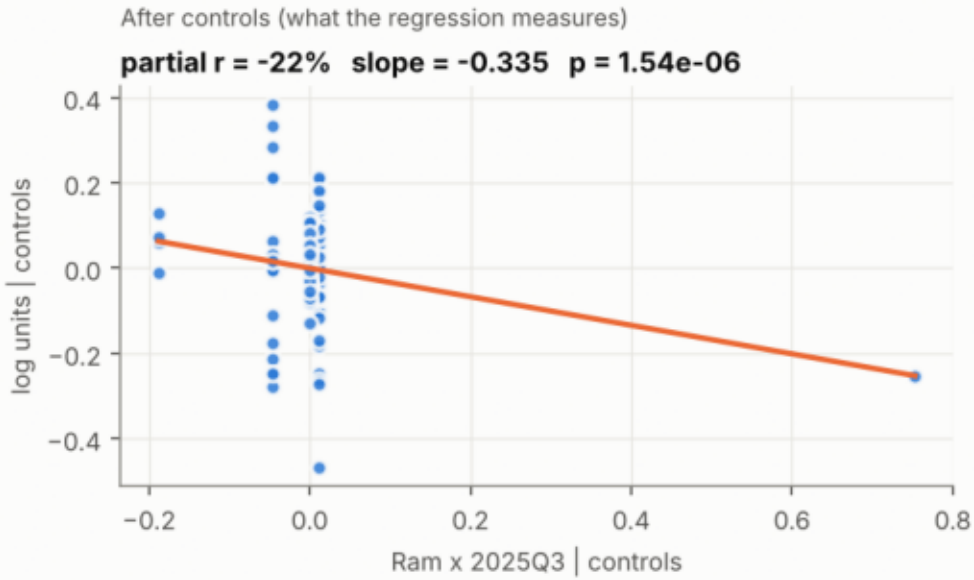




log(units) ~ Ram x Post(>=2025Q4) + Ram x 2025Q3 + unit FE + quarter FE; HC1 SE n = 105 model R² = 96% [RUN]

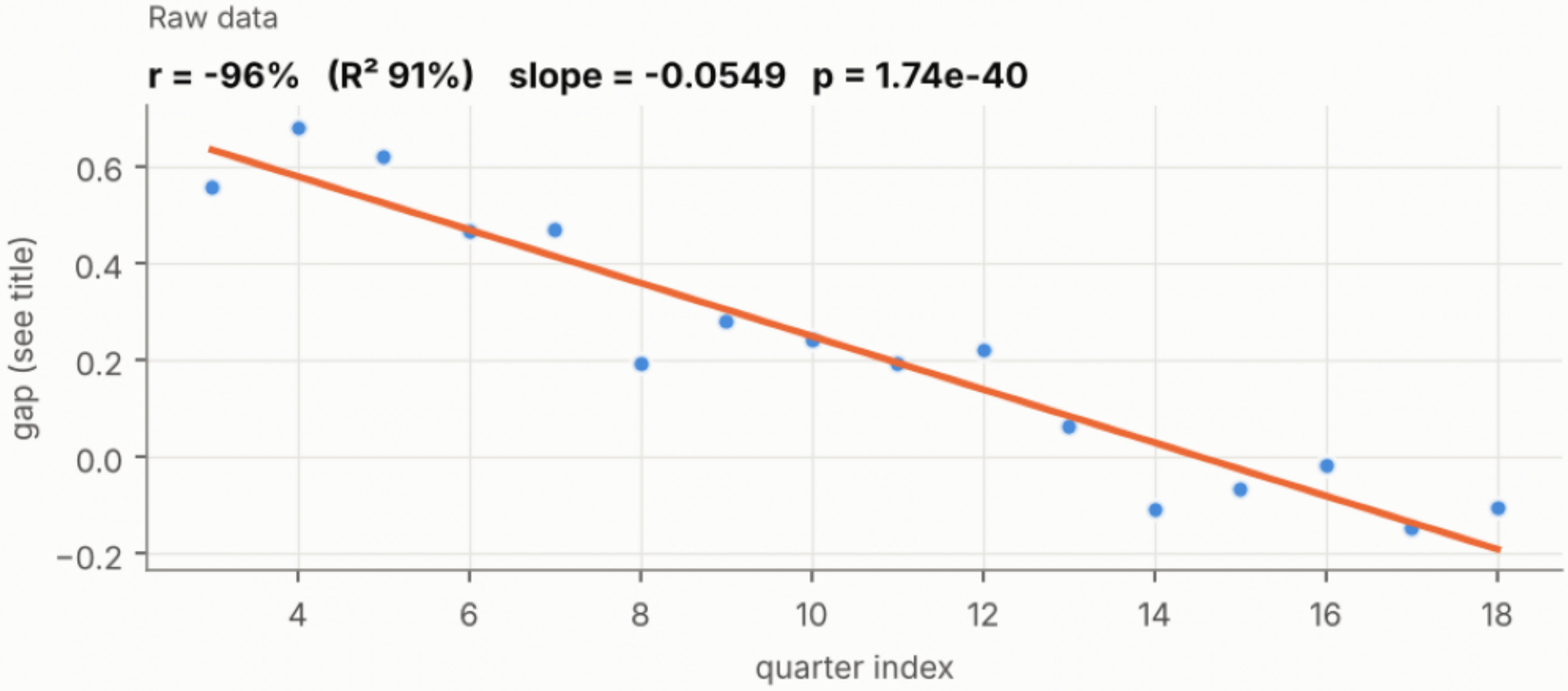
Test 1: Hemi return DiD: Ram pickups total vs same controls (longer pre-period)





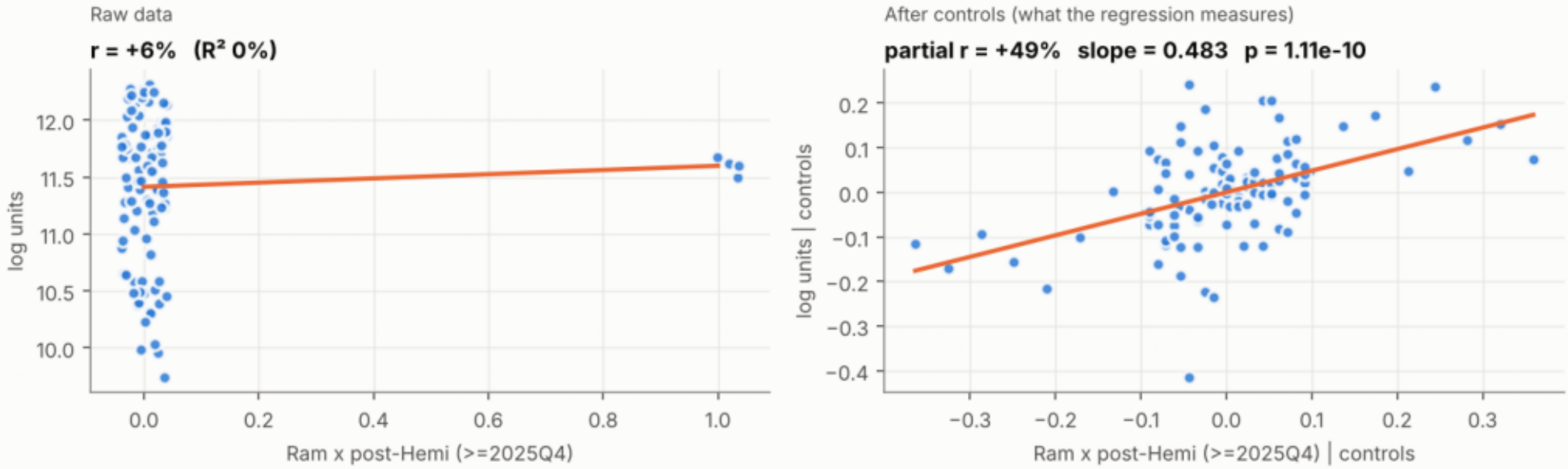
log(units) ~ Ram x Post(>=2025Q4) + Ram x 2025Q3 + unit FE + quarter FE; HC1 SE n = 105 model R² = 96% [RUN]

Test 1: Pre-trend: slope of Ram-vs-control log gap before 2025Q3 (Ram pickups total vs same controls (longer pre-period))



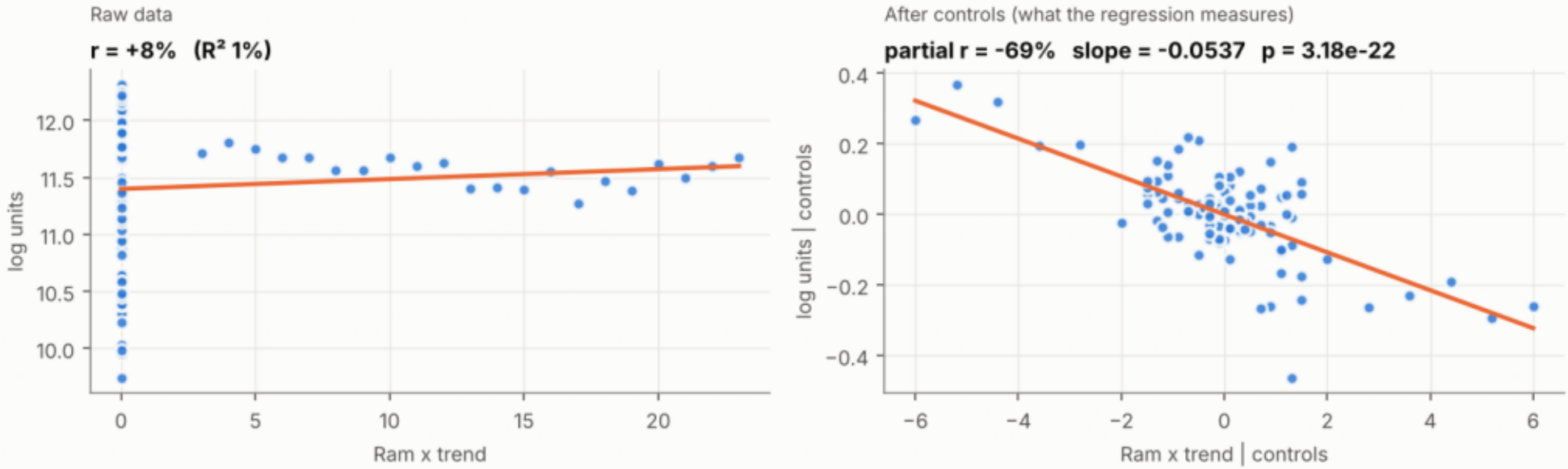
gap_t = log(Ram) - mean log(controls); gap ~ t over pre-period; HAC(1) n = 16 model R² = 91% [RUN]

Test 1: Robustness: Ram pickups total DiD with Ram-specific linear trend



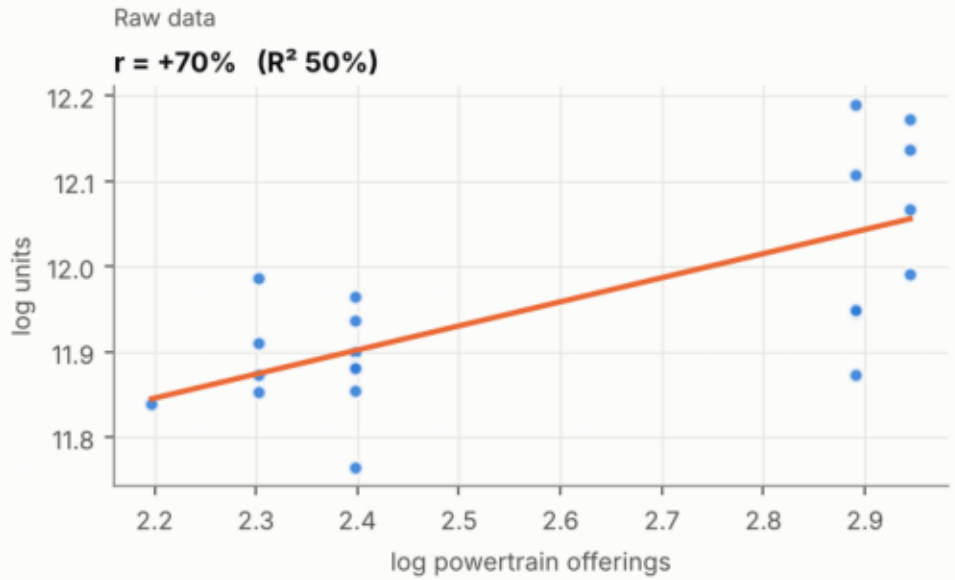
+ Ram x linear trend (pre-period share loss of -5.5%/qtr) n = 105 model R² = 98% [RUN]

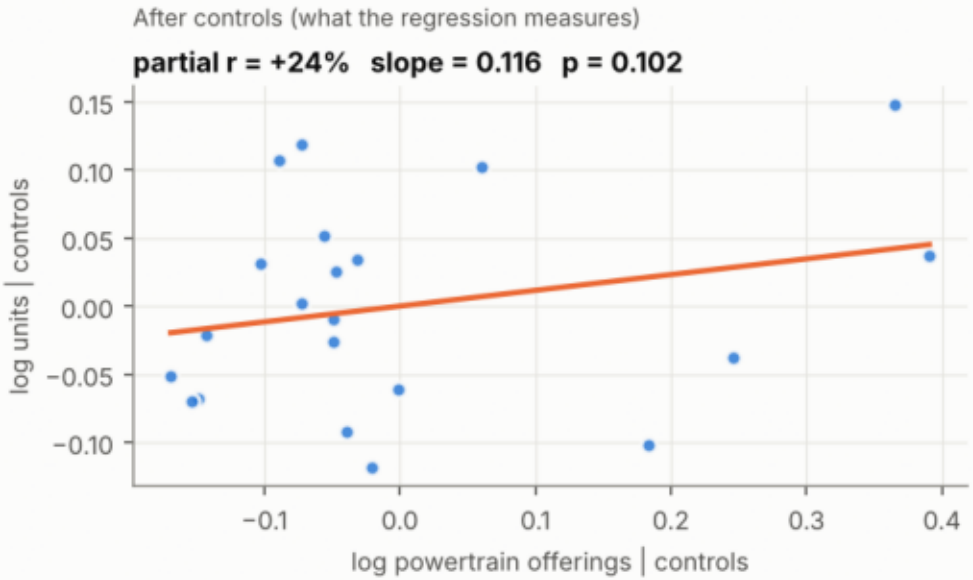
Test 1: Robustness: Ram pickups total DiD with Ram-specific linear trend



+ Ram x linear trend (pre-period share loss of -5.5%/qtr) n = 105 model R² = 98% [RUN]

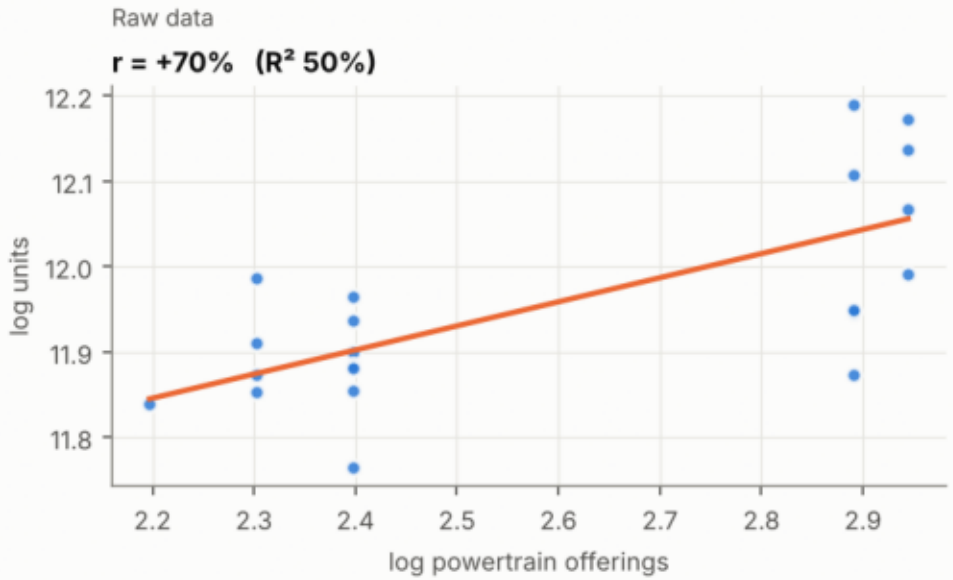
Test 2: Jeep brand sales ~ # powertrain offerings (model x powertrain)

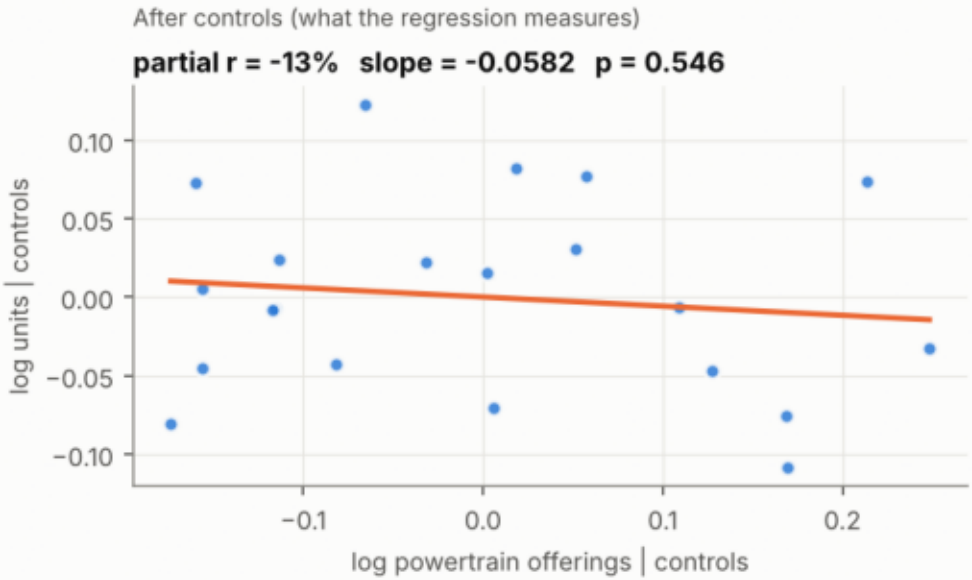




log(Jeep units) ~ log(powertrain offerings) + log(US light-vehicle sales) + quarter-of-year; HAC(2) n = 21 model R² = 60% [RUN]

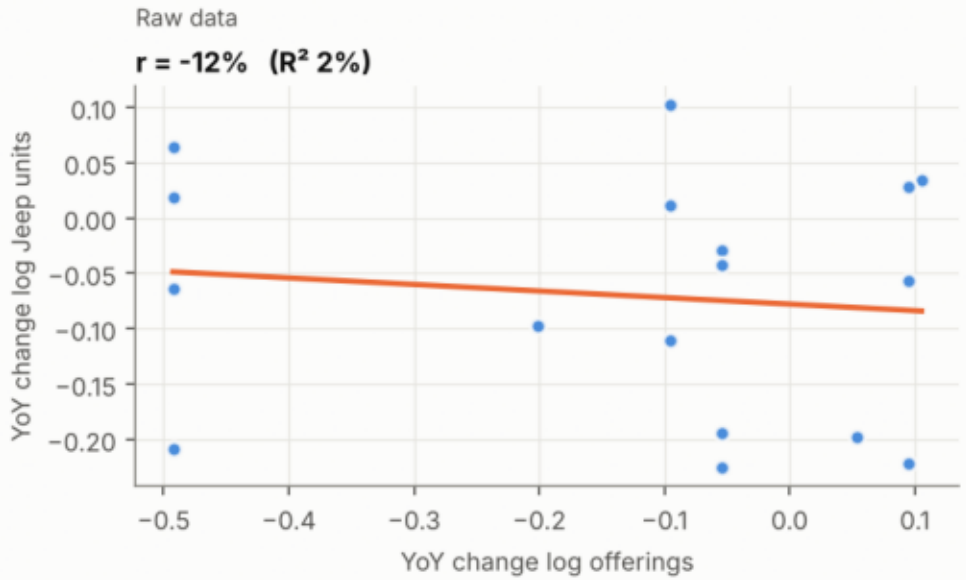
Test 2: Same, with linear trend

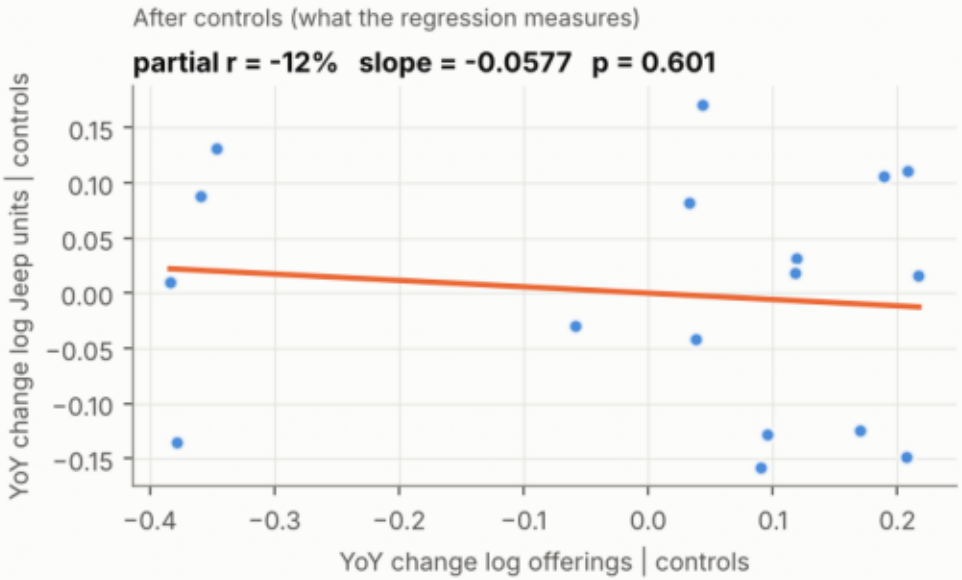




+ linear trend; HAC(2) n = 21 model R² = 72% [RUN]

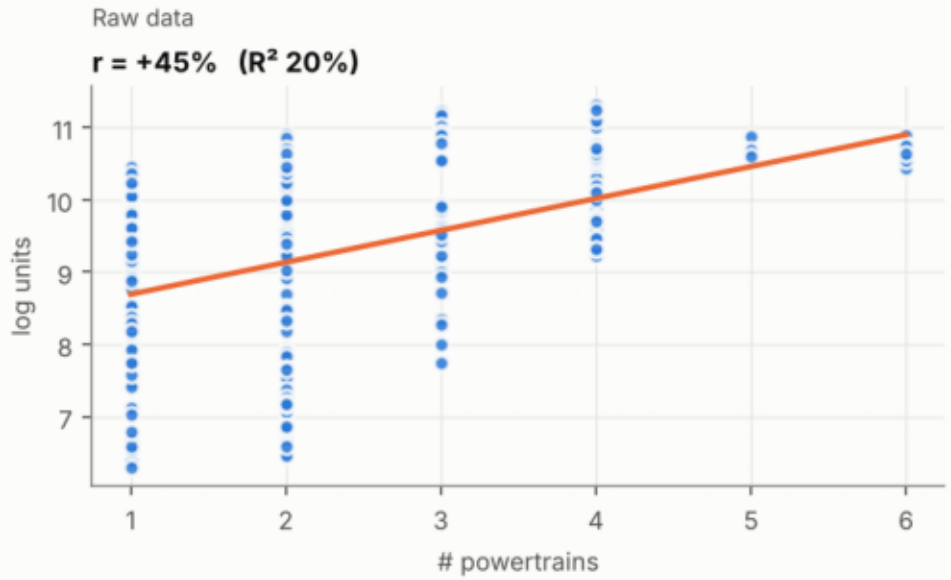
Test 2: YoY growth version

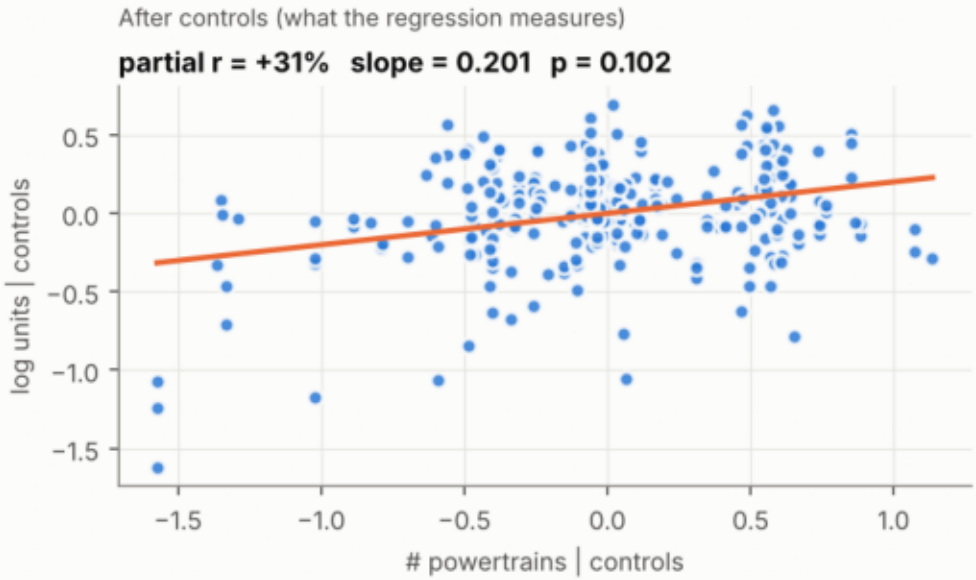




d4 log(Jeep) ~ d4 log(offerings) + d4 log(industry); HAC(2) n = 17 model R^2 = 2% [RUN]

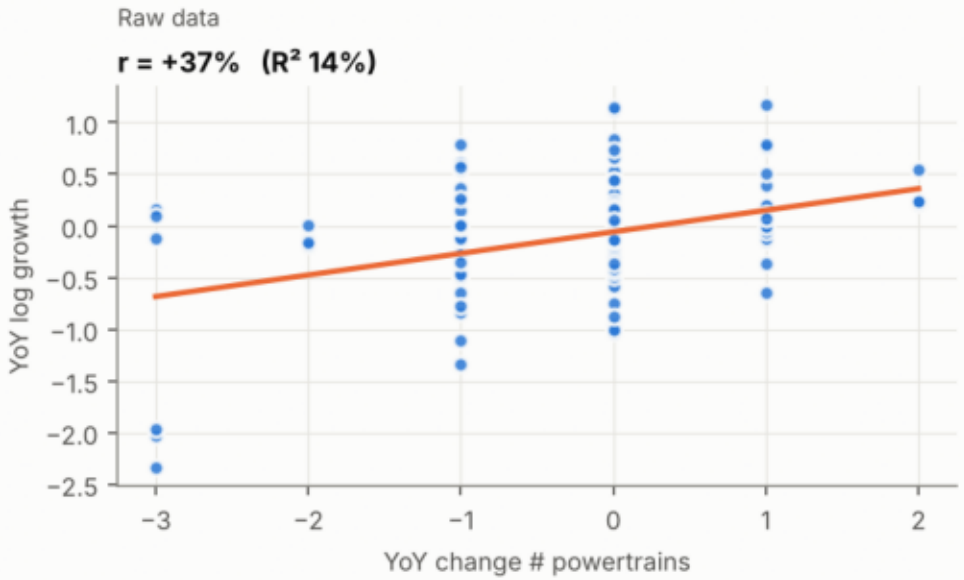
Test 3: Model sales ~ powertrain count (Stellantis nameplate panel)

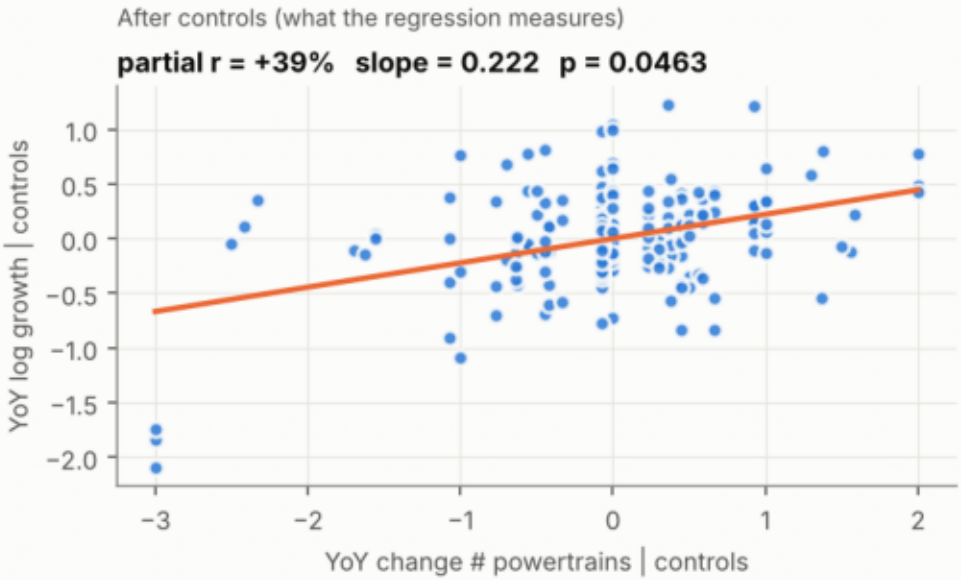




log(units) ~ n_powertrains + model FE + quarter FE; SE clustered by model n = 265 model R² = 94% [RUN]

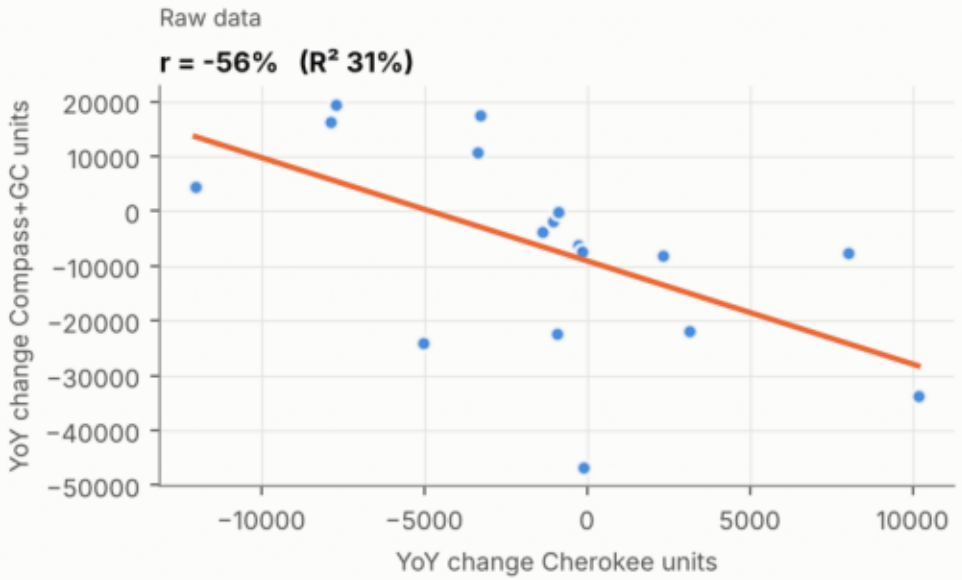
Test 3: YoY growth ~ change in powertrain count

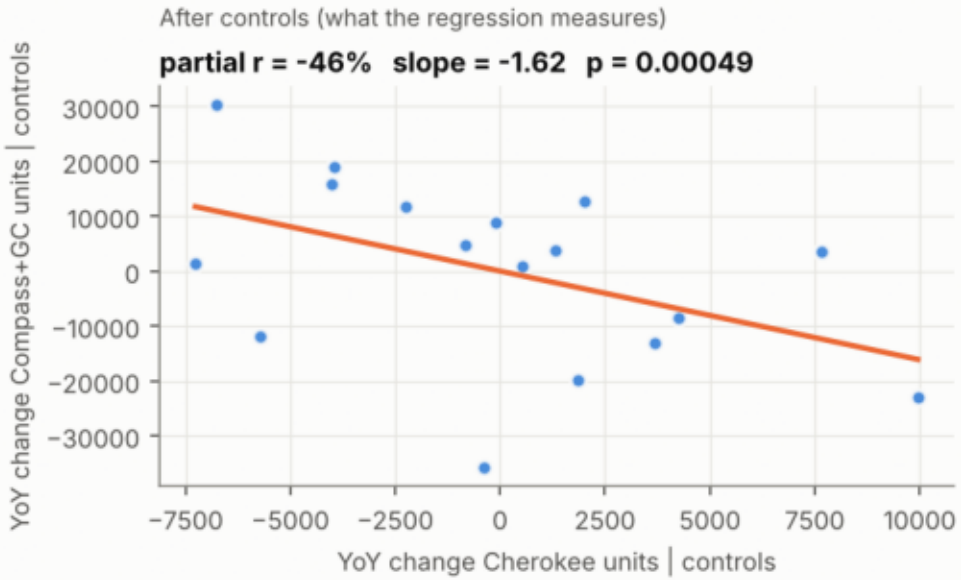




d4 log(units) ~ d4 n_powertrains + quarter FE; cluster model n = 187 model R² = 22% [RUN]

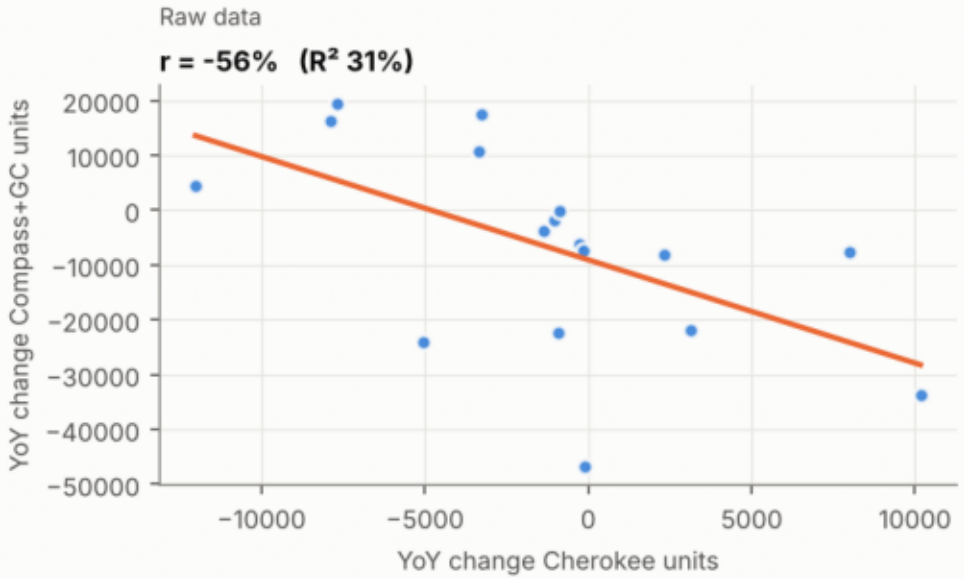
Test 4: Cannibalization: YoY change Compass+Grand Cherokee units ~ YoY change Cherokee units

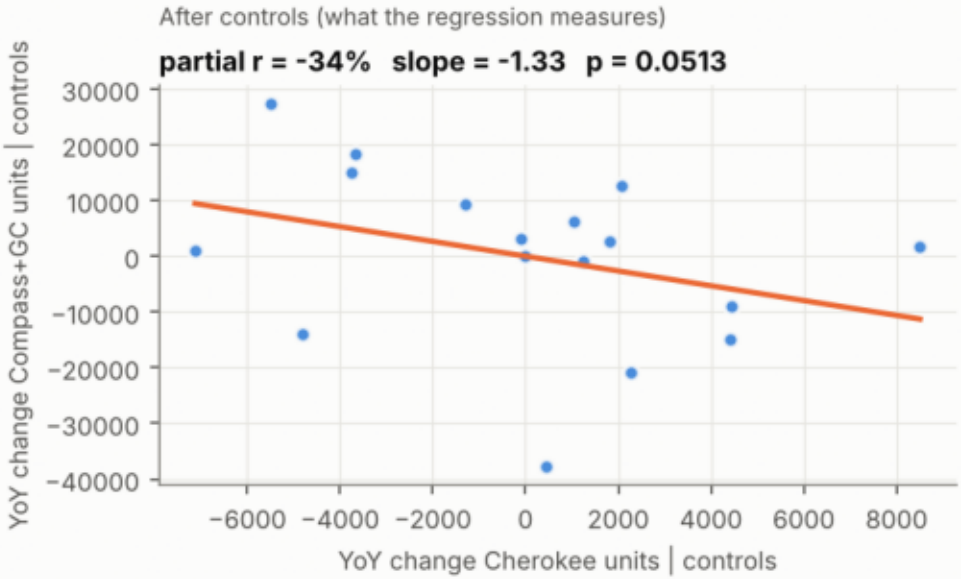




d4 (Compass+GC) ~ d4 Cherokee + d4 rival compact/midsize SUV units; HAC(2). Coef -1 = full cannibalization, 0 = new demand n = 17 model R² = 33% [RUN]

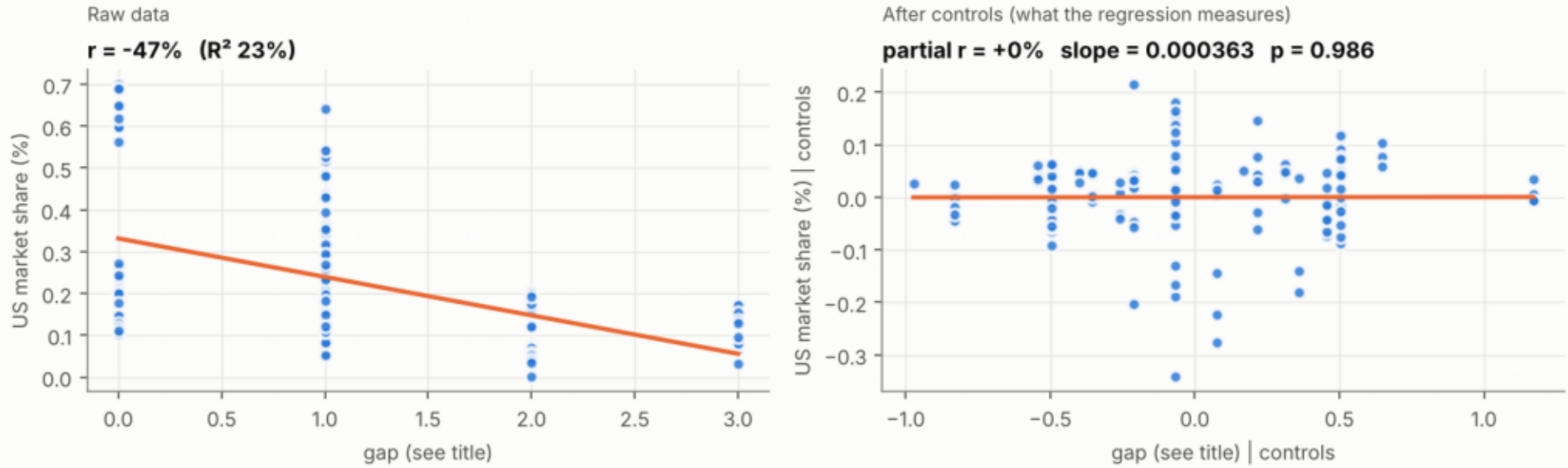
Test 4: Same, controlling for Q3-2026 production shutdown





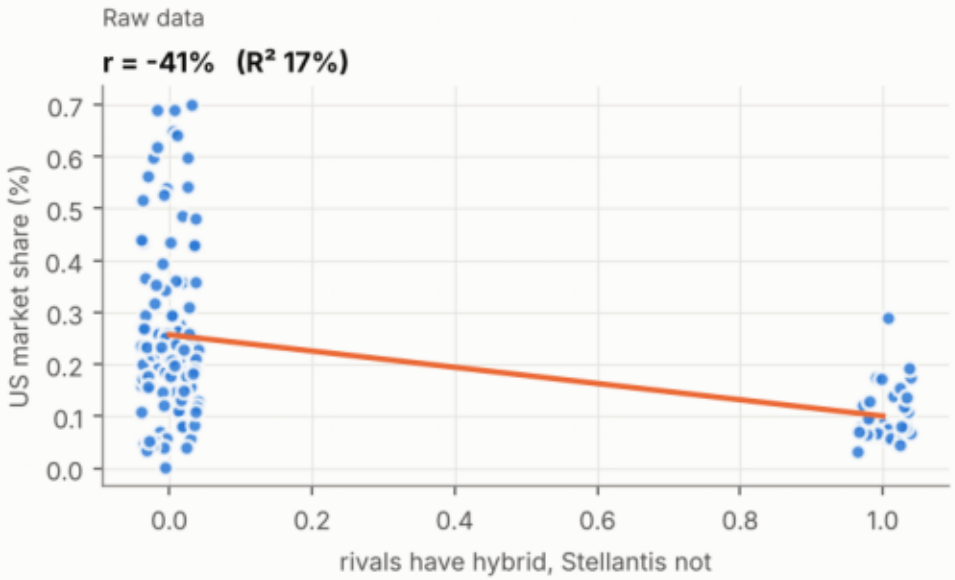
+ dummy 2026Q3 n = 17 model R² = 35% [RUN]

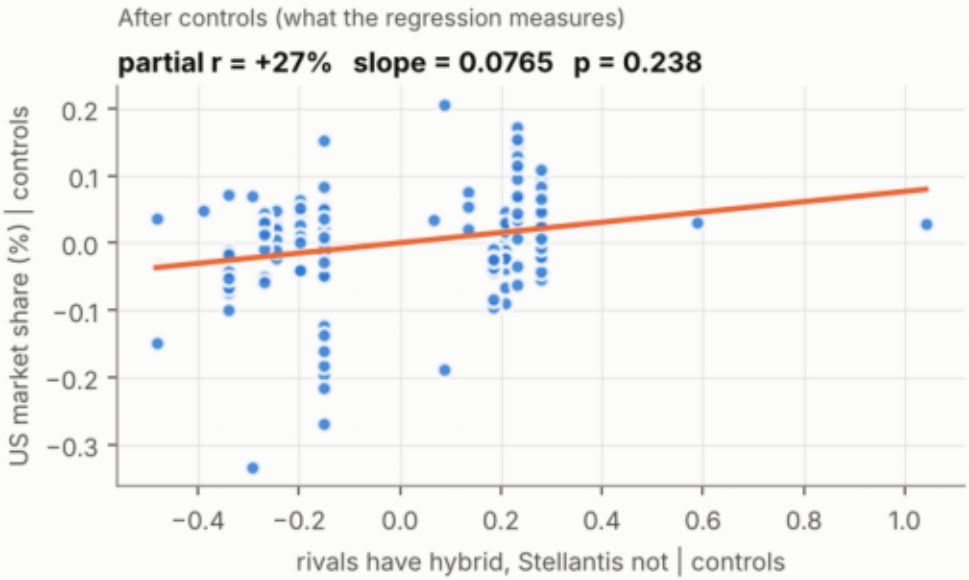
Test 5: Stellantis segment share ~ powertrain coverage gap vs rivals



share ~ #types (HEV/PHEV/EV/V8/diesel) offered by a rival but not Stellantis + segment FE + quarter FE; cluster segment n = 147 model R² = 78% [RUN]

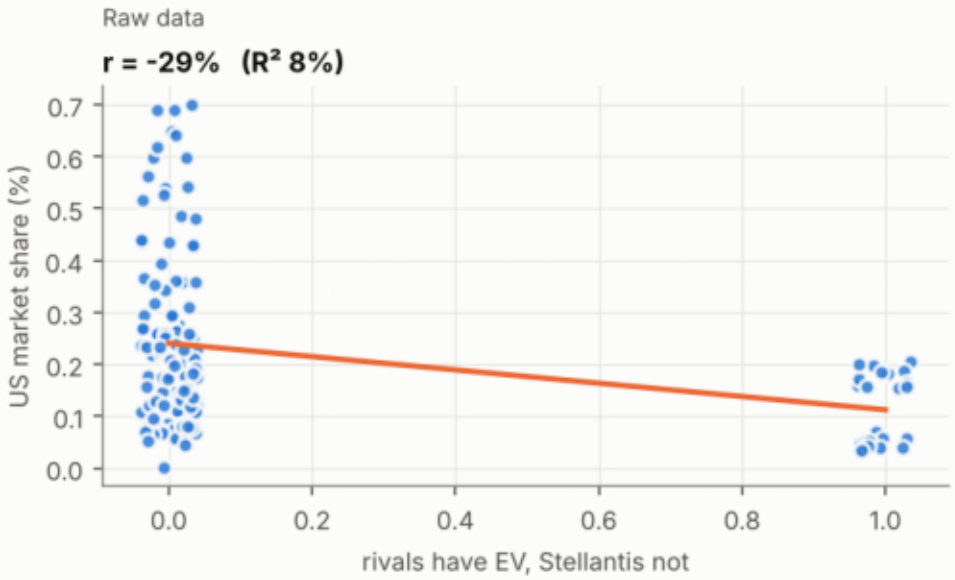
Test 5: Same, by missing type

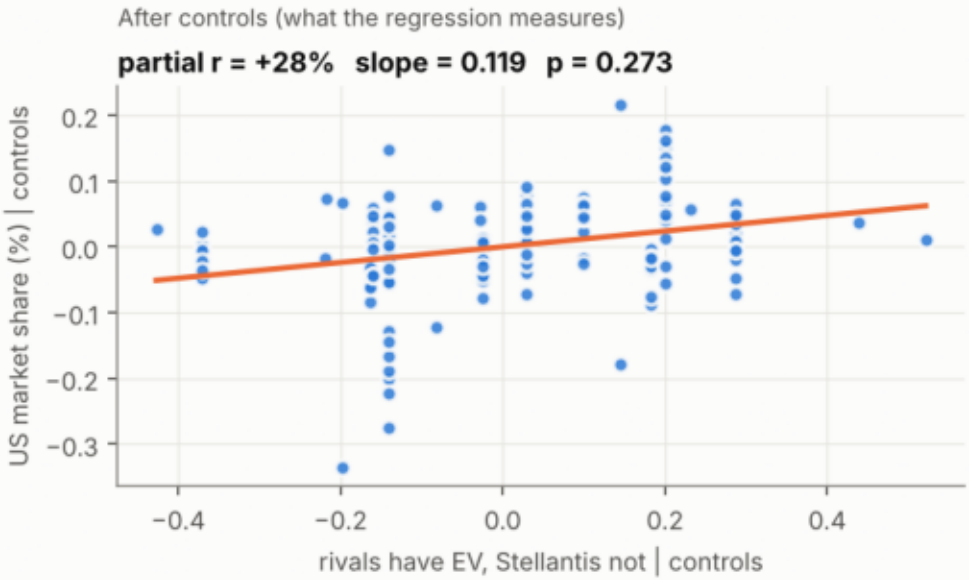




share ~ missing hybrid + missing EV + FE n = 147 model R² = 80% [RUN]

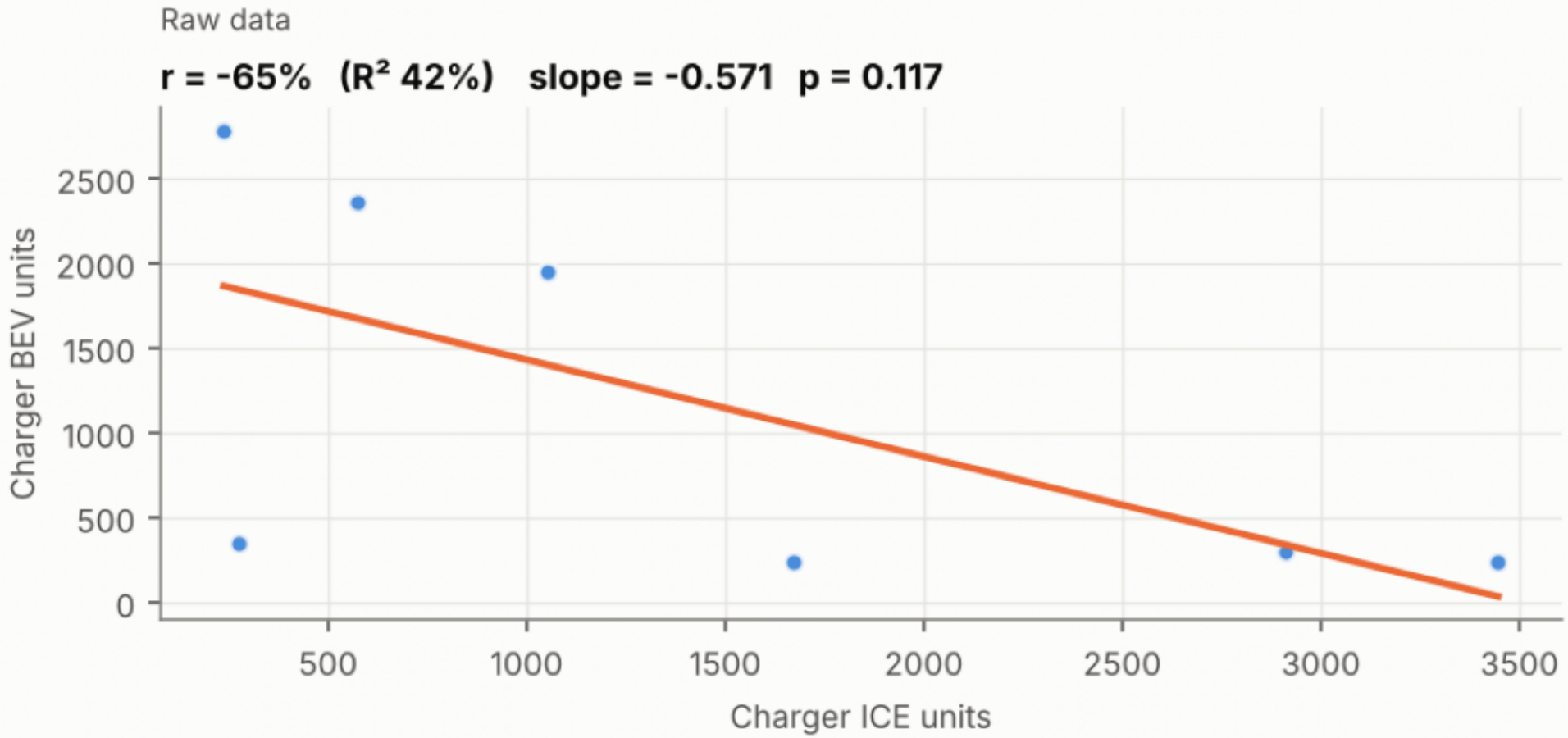
Test 5: Same, by missing type





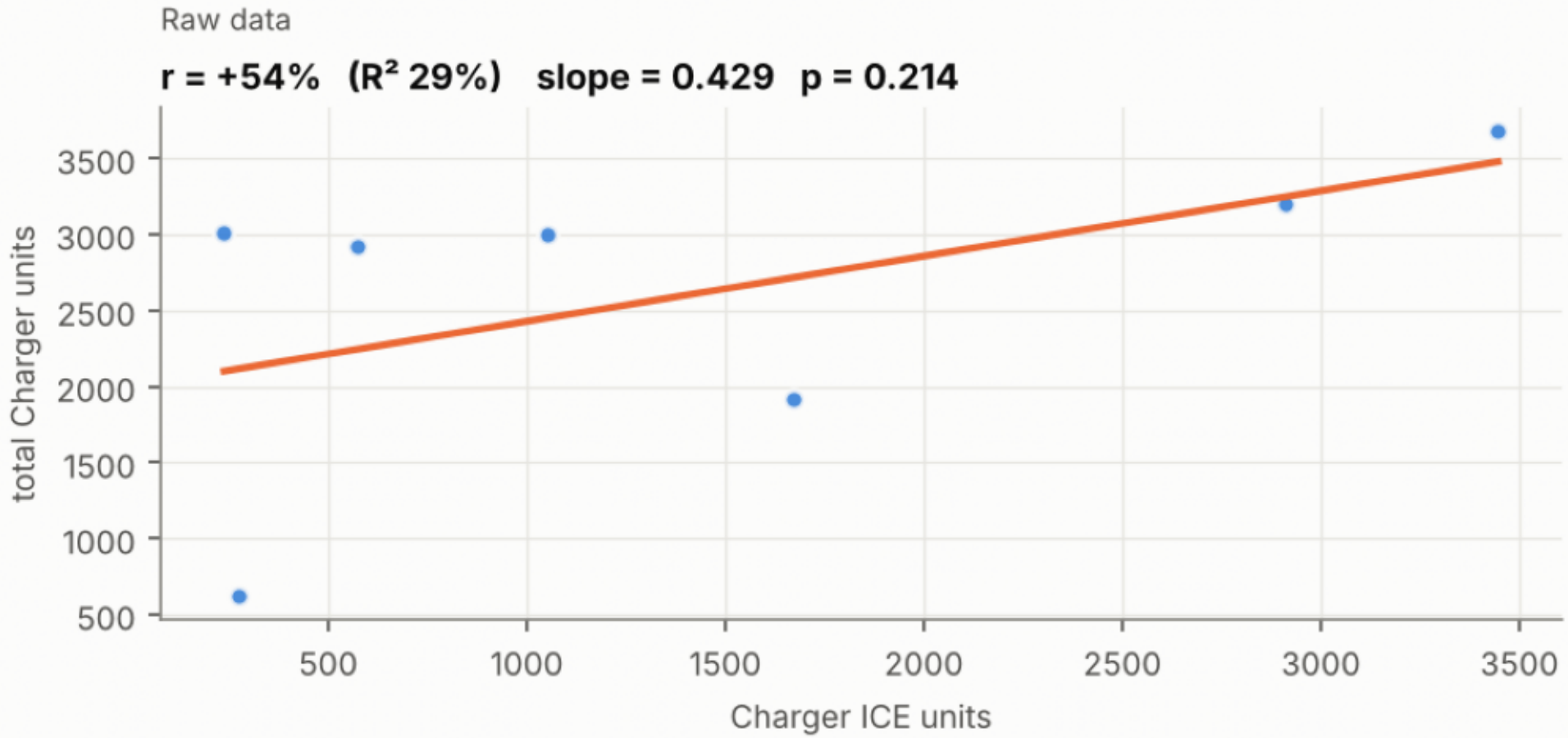
share ~ missing hybrid + missing EV + FE n = 147 model R² = 80% [RUN]

Test 6: Charger BEV units ~ Charger ICE (Sixpack) units, 2025Q1-2026Q3



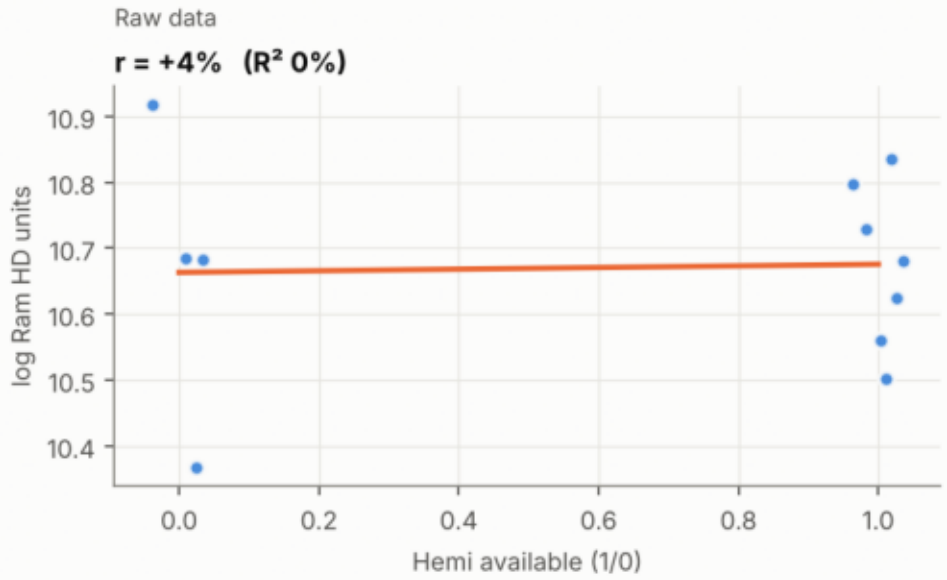
BEV_t ~ ICE_t; HAC(1). Negative = ICE replaced EV demand n = 7 model R² = 42% [RUN]

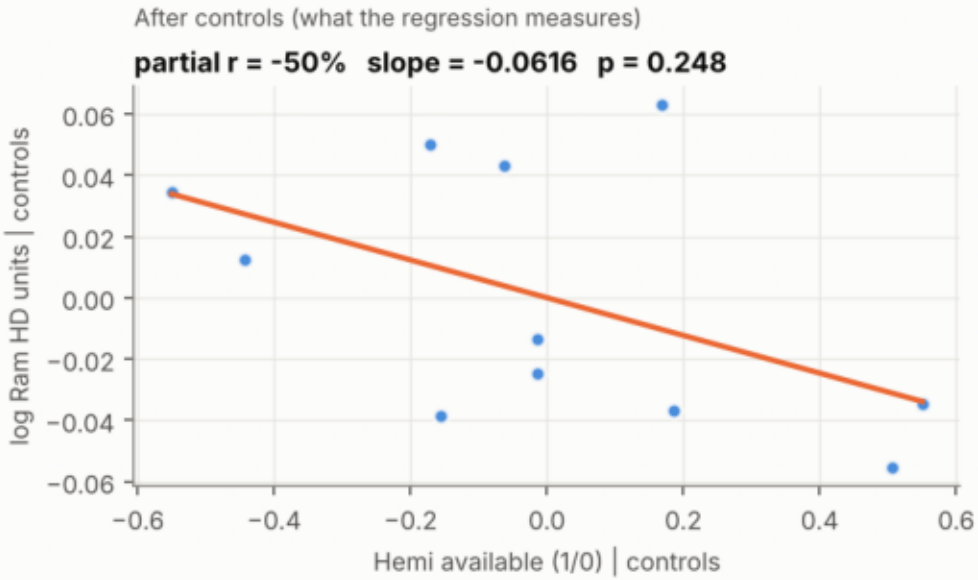
Test 6: Total Charger units ~ ICE units (coef 1 = purely additive)



total_t ~ ICE_t; HAC(1) n = 7 model R² = 29% [RUN]

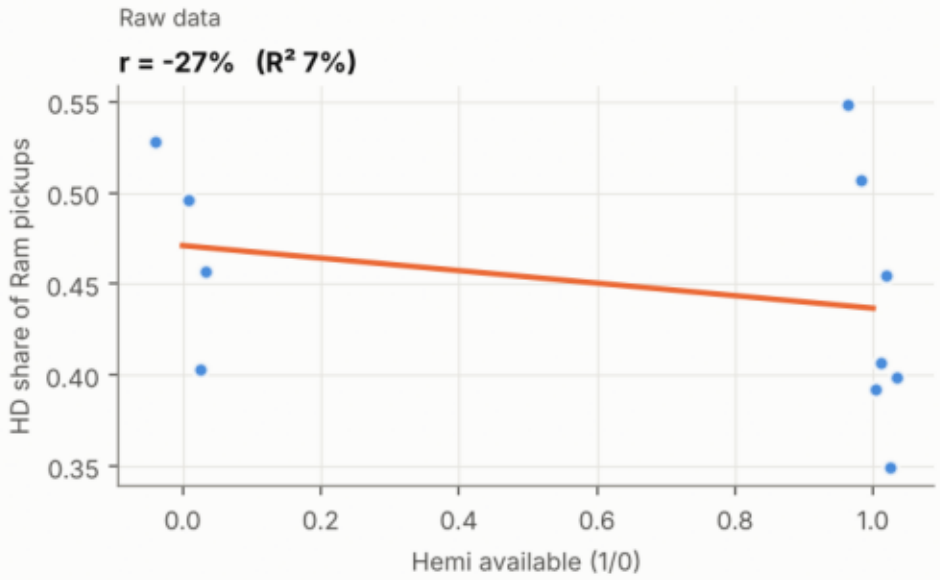
Test 7: Ram HD units ~ Hemi available in Ram 1500

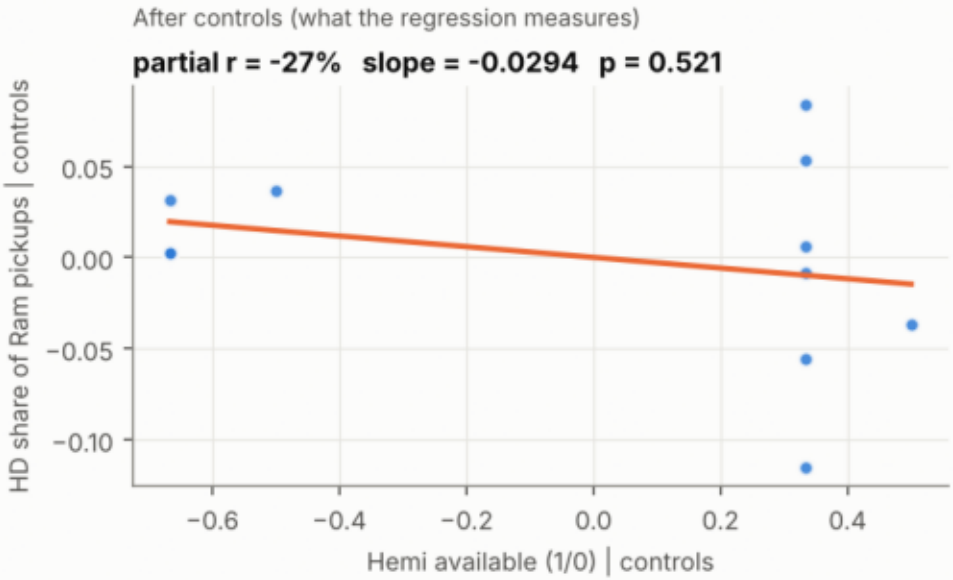




log(HD) ~ Hemi-available dummy (MY lineup) + log(F/Silverado/Sierra units) + quarter-of-year; HAC(1) n = 11 model R² = 95% [RUN]

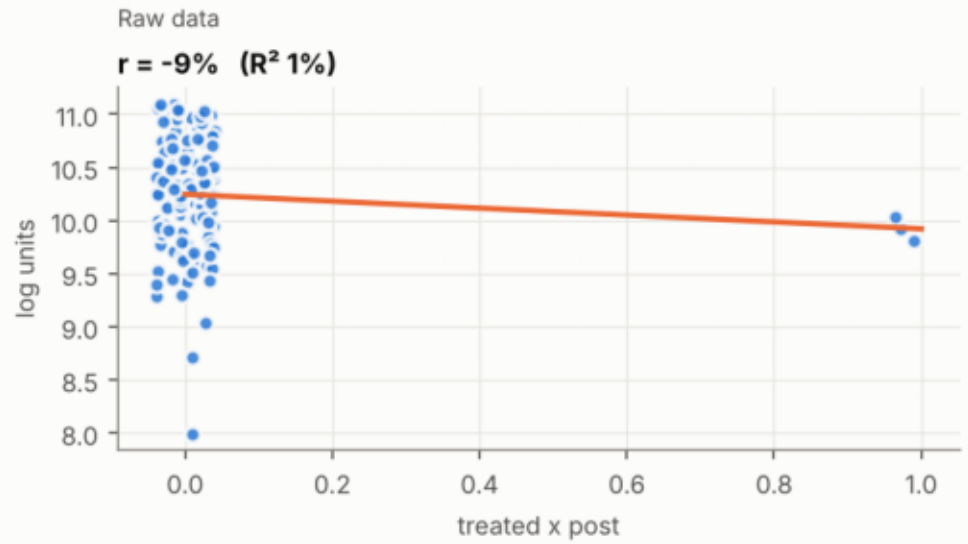
Test 7: HD share of Ram pickups ~ Hemi availability

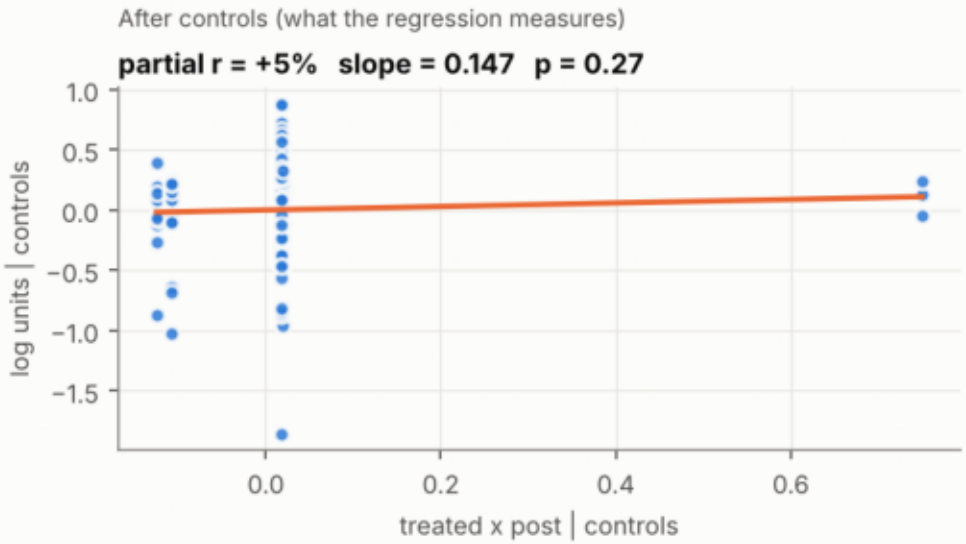




HD/(HD+LD) ~ Hemi + quarter-of-year; HAC(1) n = 11 model R² = 33% [RUN]

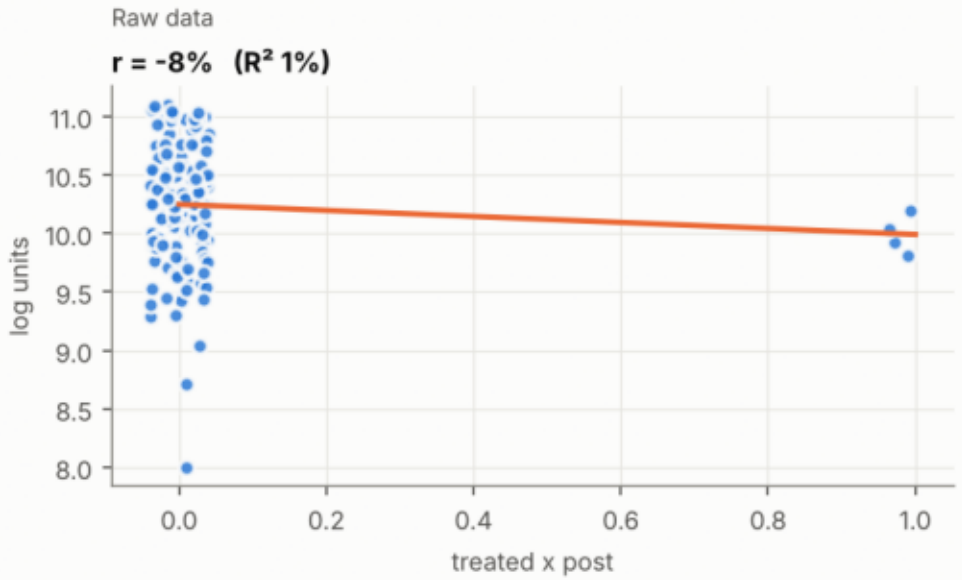
Test 8: Durango vs 3-row SUV rivals after Charger ICE launch, post from 2026Q1 (first quarter Charger ICE >1,000 units)





log(units) ~ Durango x Post + unit FE + quarter FE; HC1 n = 167 model R² = 46% [RUN]

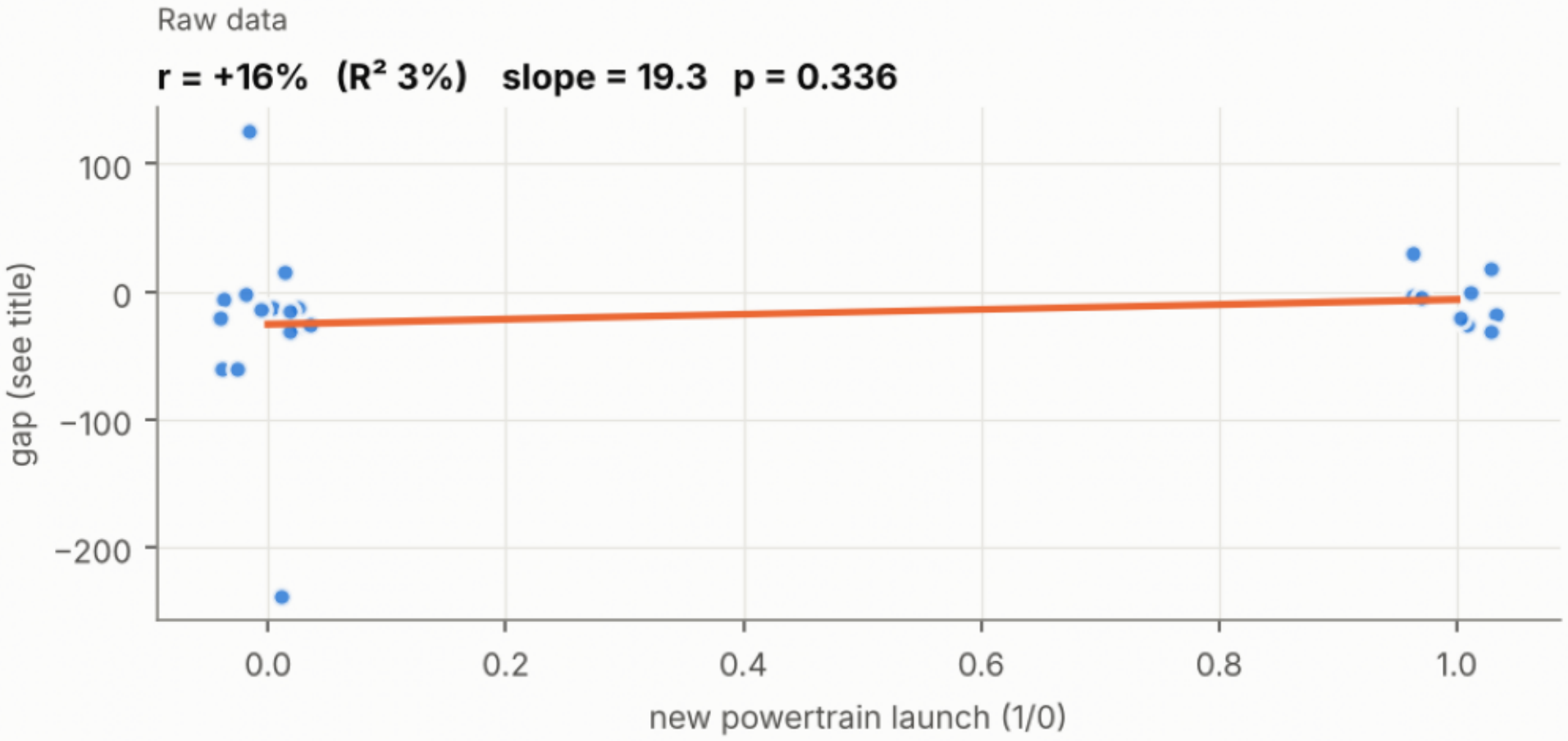
Test 8: Durango vs 3-row SUV rivals after Charger ICE launch, post from 2025Q4





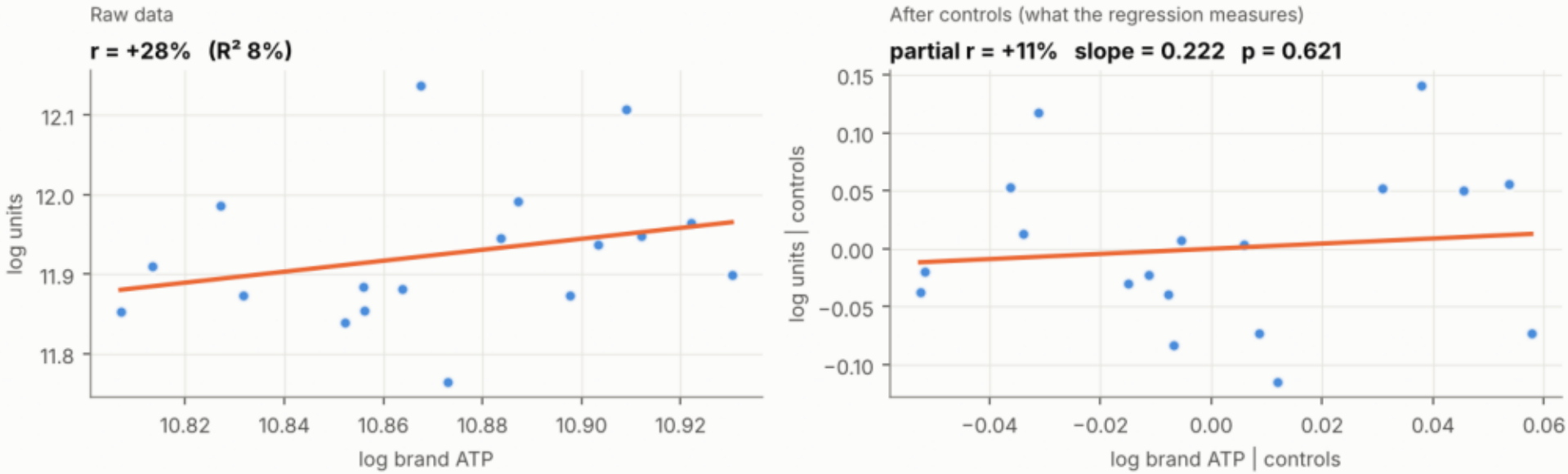
log(units) ~ Durango x Post + unit FE + quarter FE; HC1 n = 167 model R² = 46% [RUN]

Test 10: Total-minus-retail YoY growth gap ~ new powertrain launch



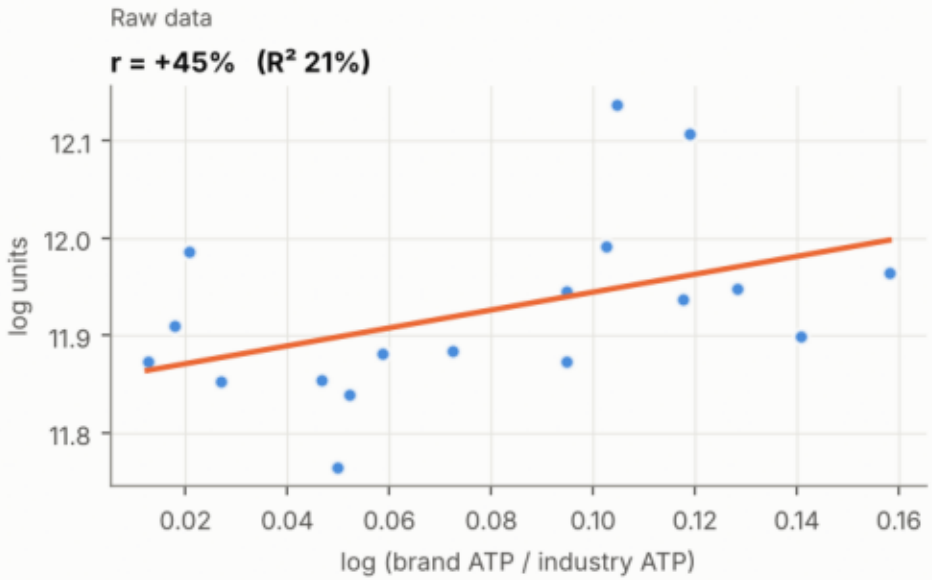
(total YoY % - retail YoY %) ~ launch dummy; HC1. Positive gap = non-retail (fleet) growing faster n = 24 model R² = 3% [RUN (proxy)]

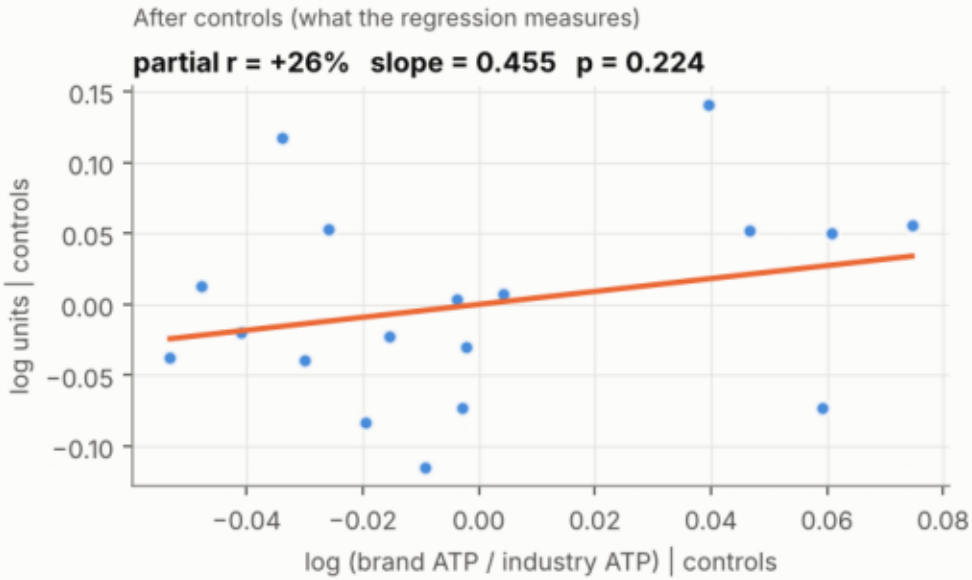
Test 11: Jeep quarterly sales ~ Jeep KBB ATP (log-log)



log(Jeep units) ~ log(Jeep ATP) + log(US light-vehicle sales, BEA) + quarter-of-year; HAC(2) n = 18 model R² = 44% [RUN]

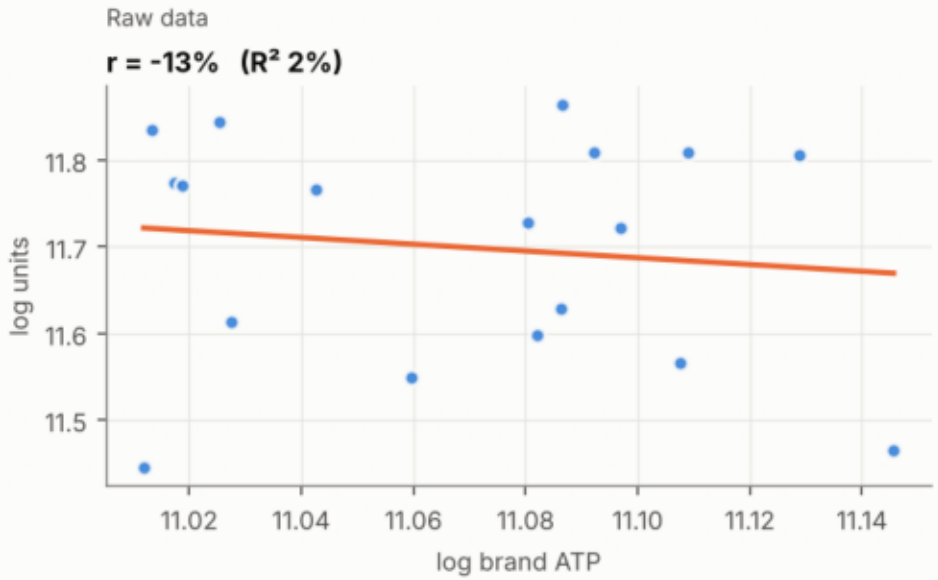
Test 11: Jeep sales ~ Jeep ATP relative to industry ATP

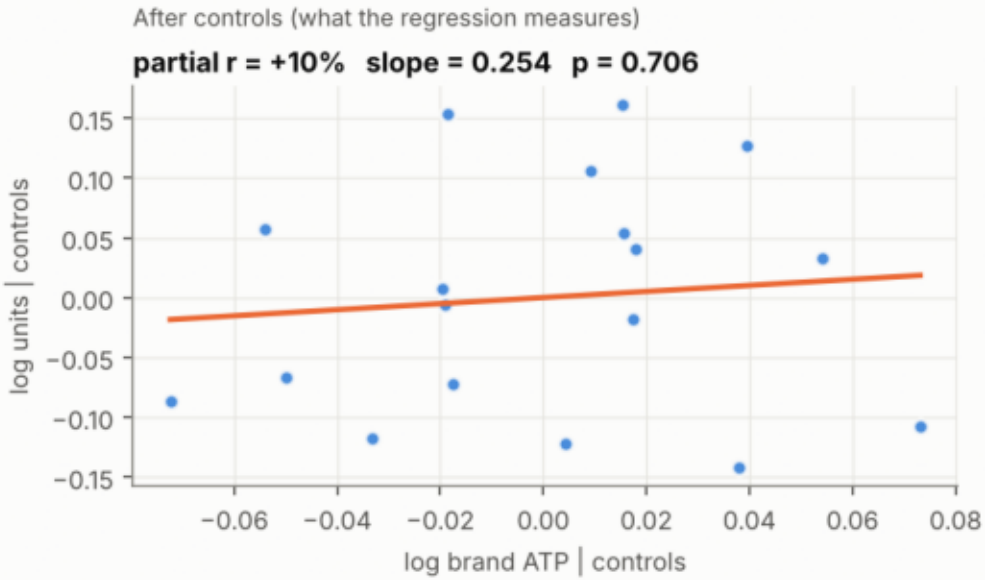




log(Jeep) ~ log(Jeep ATP / industry ATP) + log(industry units) + quarter-of-year; HAC(2) n = 18 model R² = 47% [RUN]

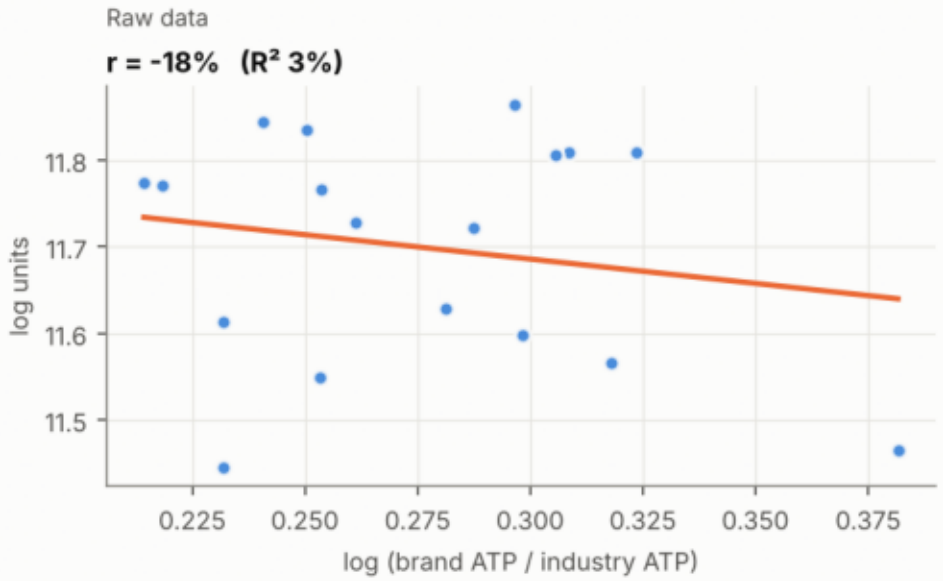
Test 12: Ram quarterly sales ~ Ram KBB ATP (log-log)

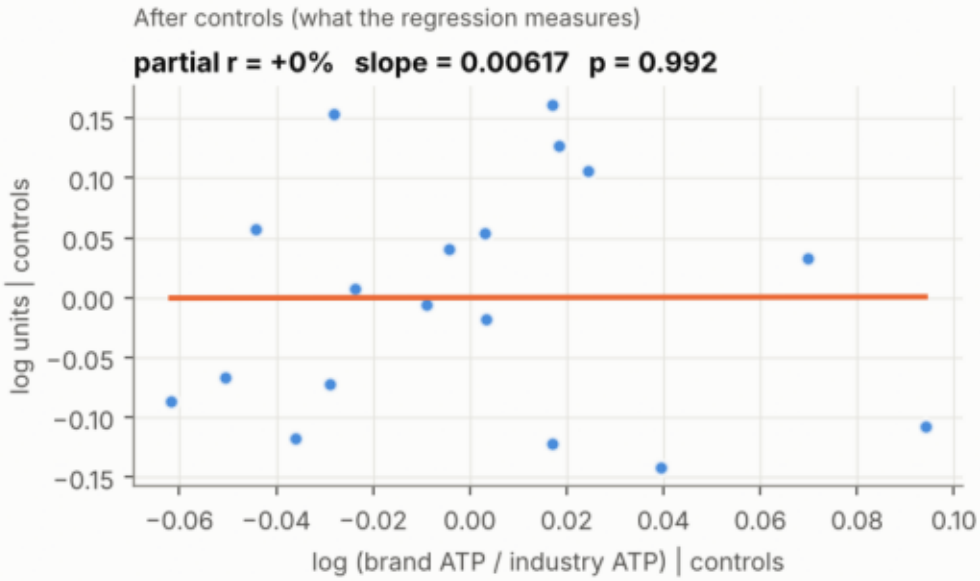




log(Ram units) ~ log(Ram ATP) + log(US light-vehicle sales, BEA) + quarter-of-year; HAC(2) n = 18 model R² = 46% [RUN]

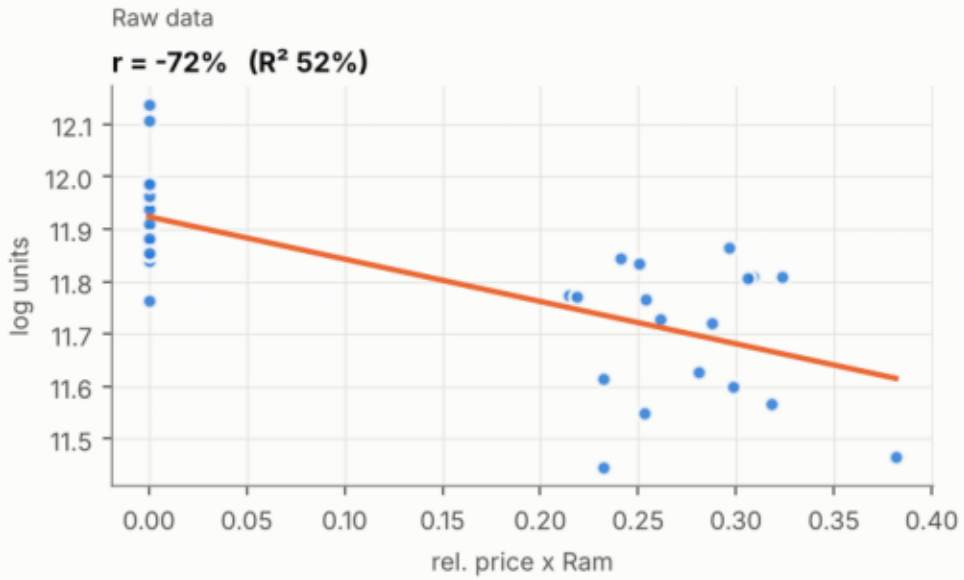
Test 12: Ram sales ~ Ram ATP relative to industry ATP

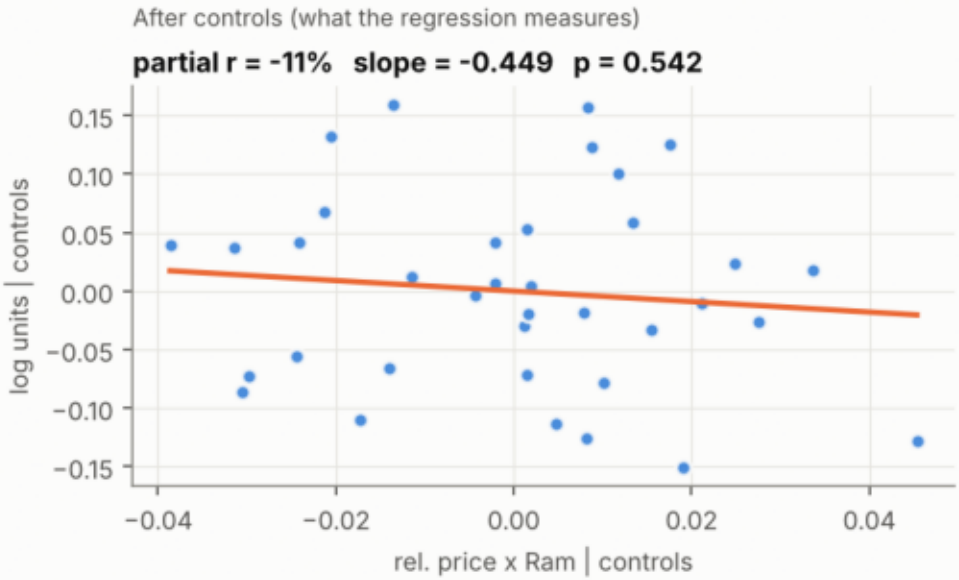




log(Ram) ~ log(Ram ATP / industry ATP) + log(industry units) + quarter-of-year; HAC(2) n = 18 model R² = 45% [RUN]

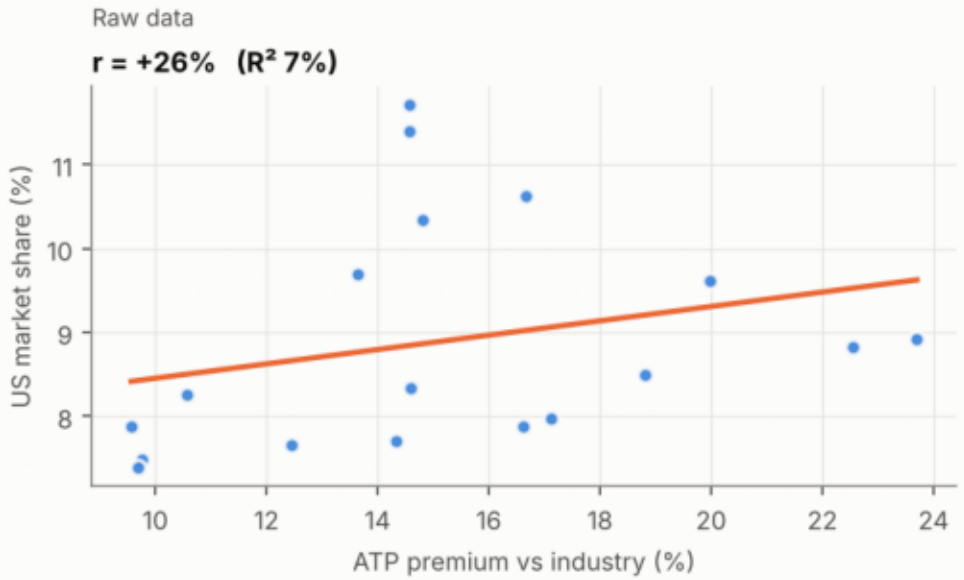
Test 12: Difference in relative-price coefficient, Ram minus Jeep

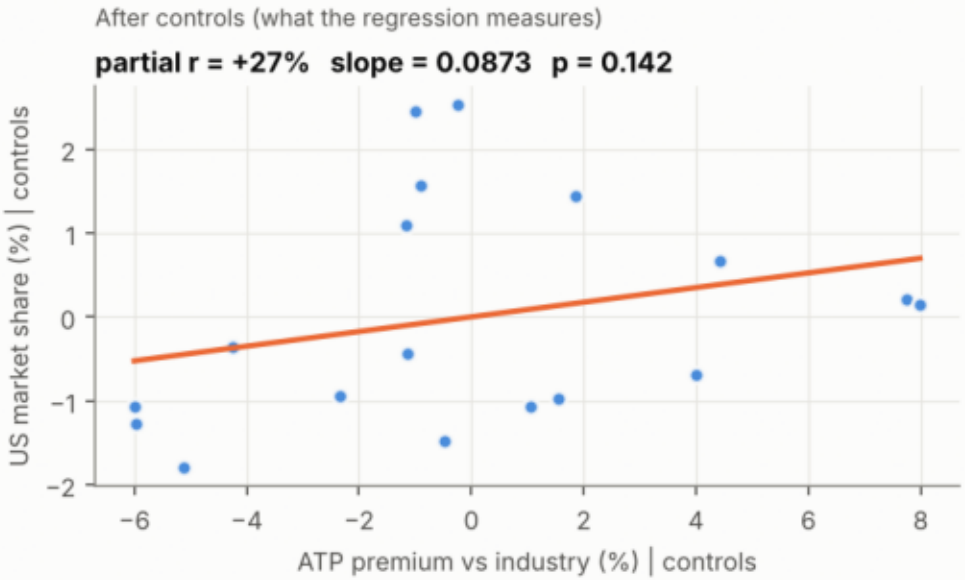




stacked Jeep+Ram; log(units) ~ rel. price + rel. price x Ram + brand-specific controls; HAC(2). Positive = Ram less price-sensitive n = 36 model R² = 74% [RUN]

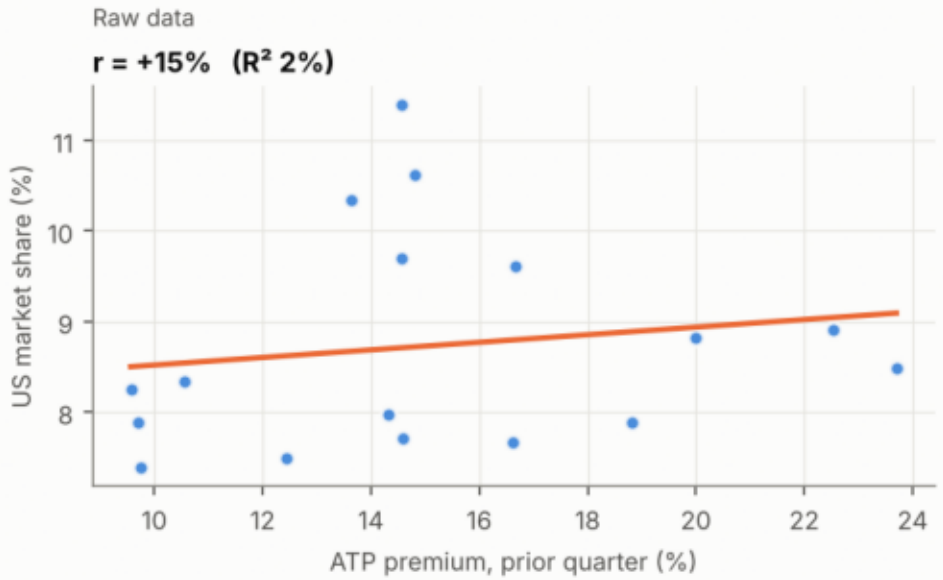
Test 13: Stellantis US market share (%) ~ Stellantis ATP premium over industry (%)

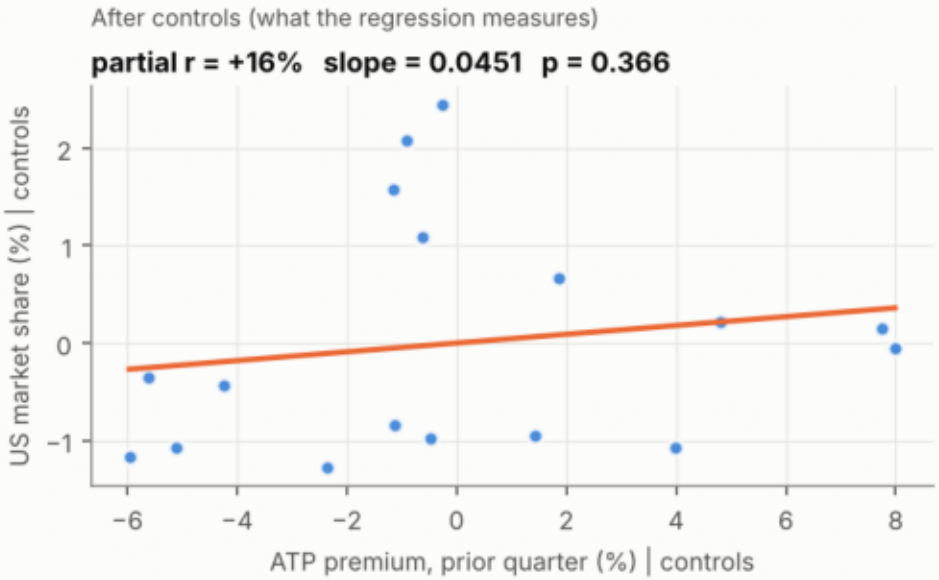




share_t ~ premium_t + quarter-of-year; HAC(2) n = 18 model R² = 10% [RUN]

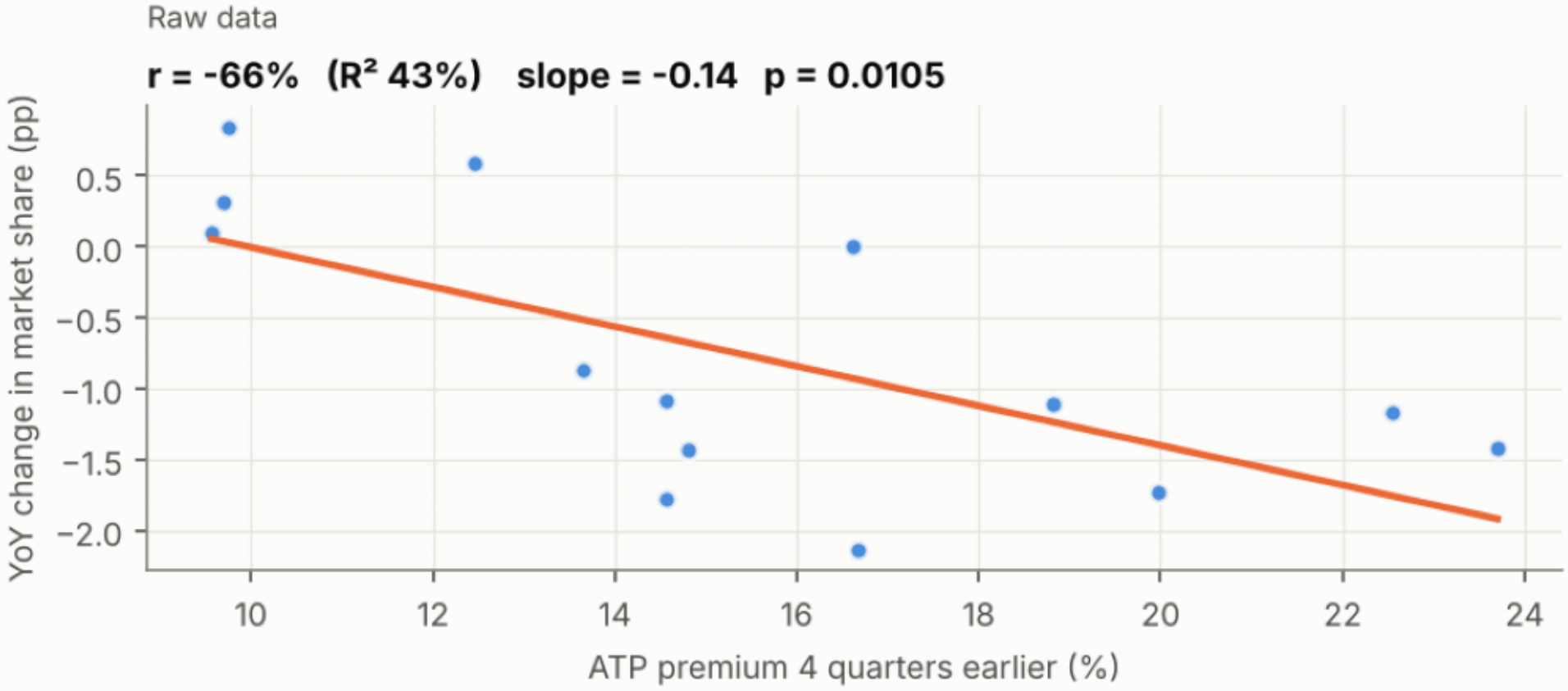
Test 13: Same, premium lagged one quarter



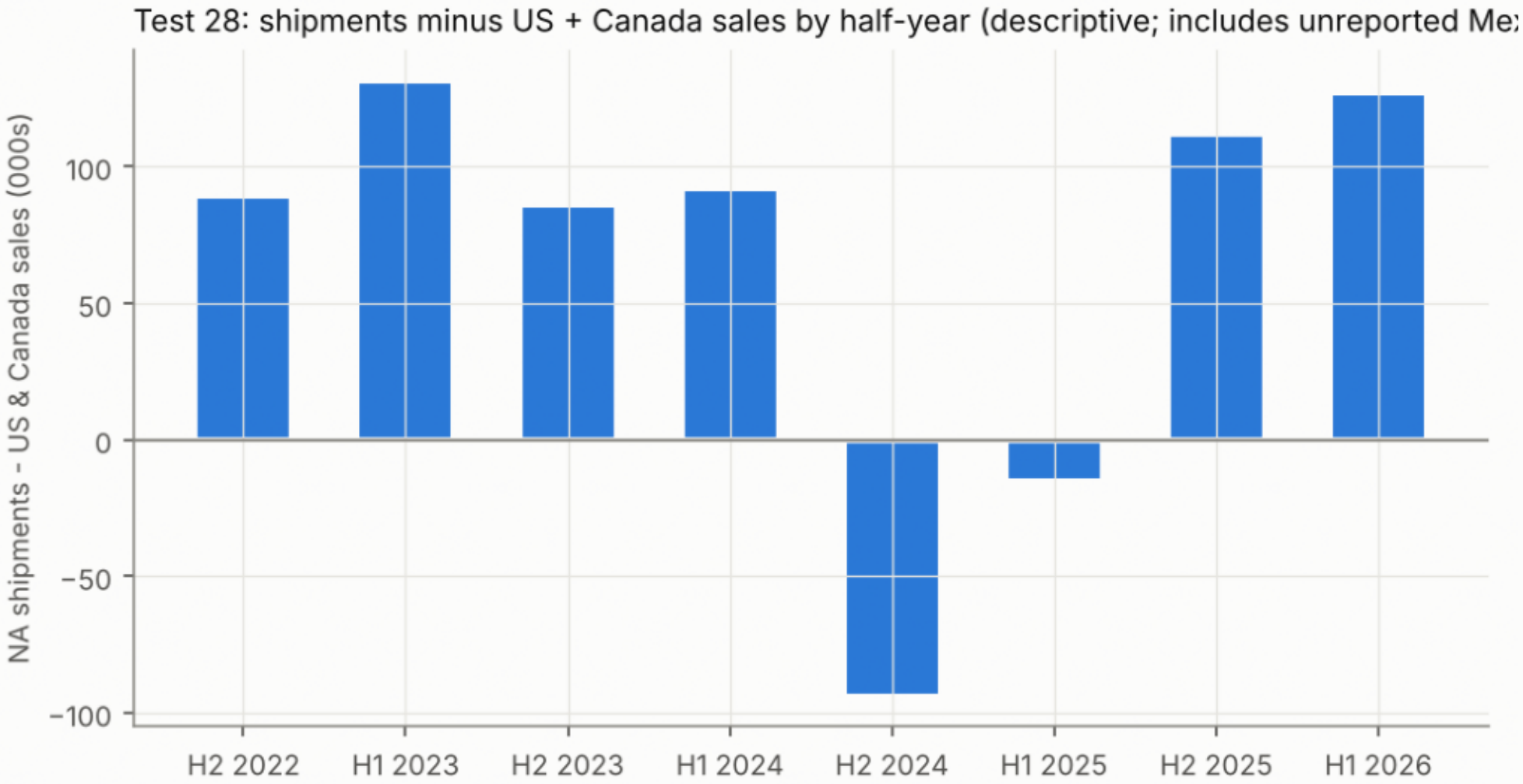


share_t ~ premium_{t-1} n = 17 model R² = 4% [RUN]

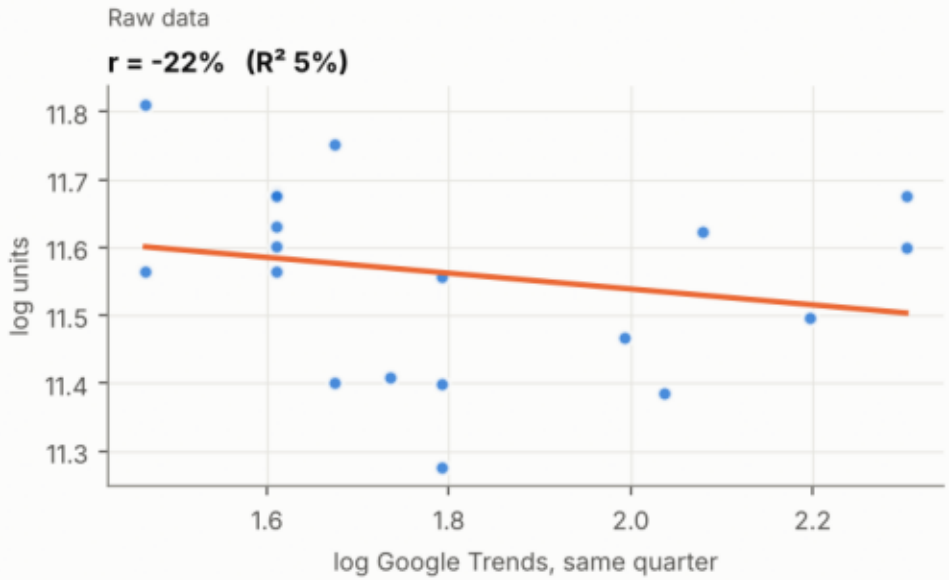
Test 13: Dynamic: YoY change in share (pp) ~ premium four quarters earlier

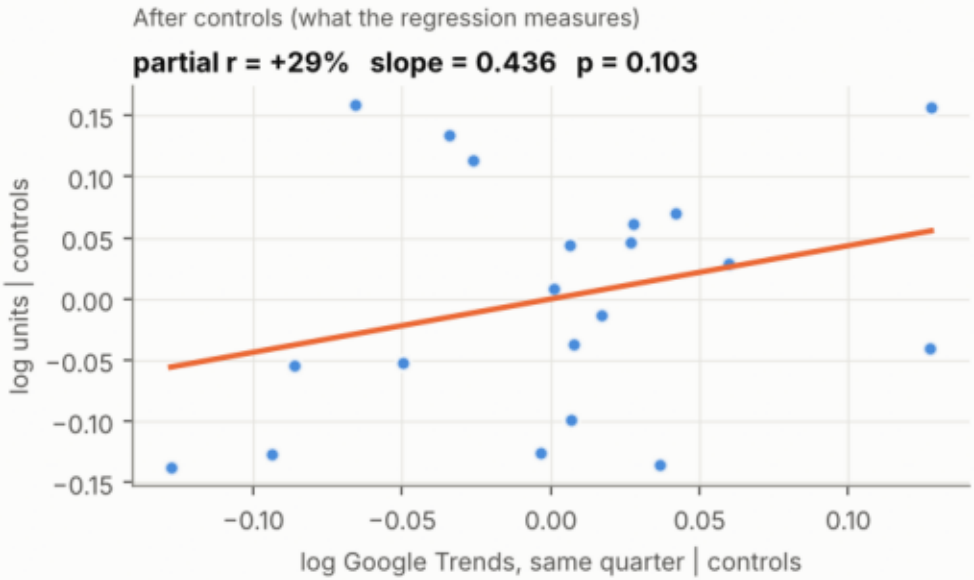


share_t - share_{t-4} ~ premium_{t-4}; HAC(2) (n<15 → classical SE). Negative = a high premium is followed by share loss n = 14 model R² = 43% [RUN]


Test 28: shipments minus US + Canada sales by half-year (descriptive; includes unreported Mex
NA shipments - US & Canada sales (000s)
100
50
0
−50
−100
H2 2022
H1 2023
H2 2023
H1 2024
H2 2024
H1 2025
H2 2025
H1 2026

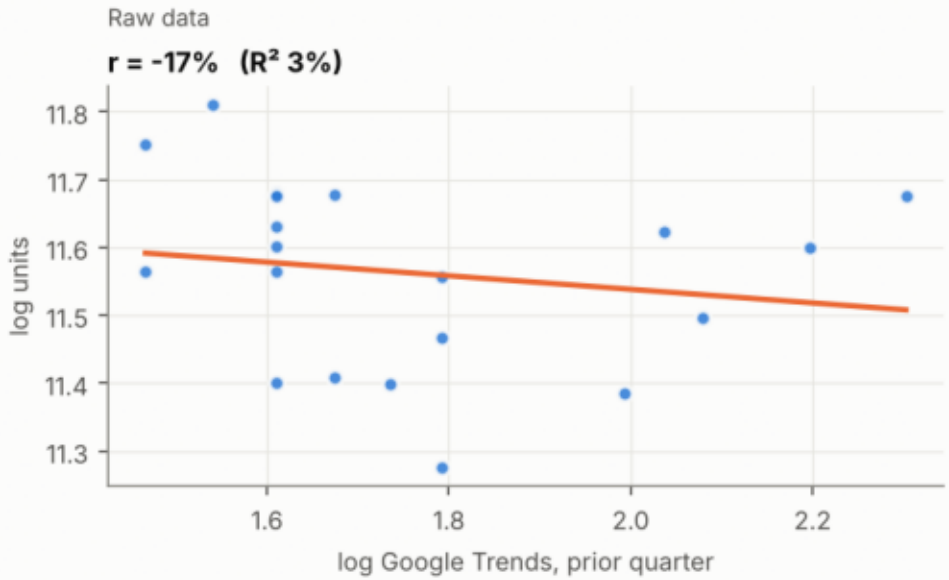
Test 31: Trends "Ram Hemi" ~ Ram pickups sales (current + 1-quarter lag)

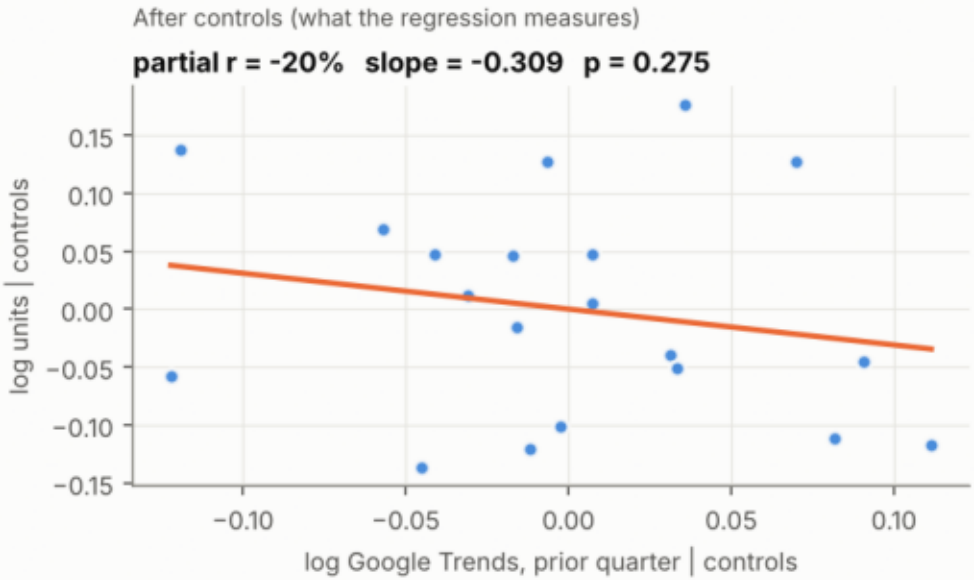




log(units) ~ log(Trends_t) + log(Trends_{t-1}) + log(industry) + quarter-of-year; HAC(2) n = 20 model R² = 54% [RUN]

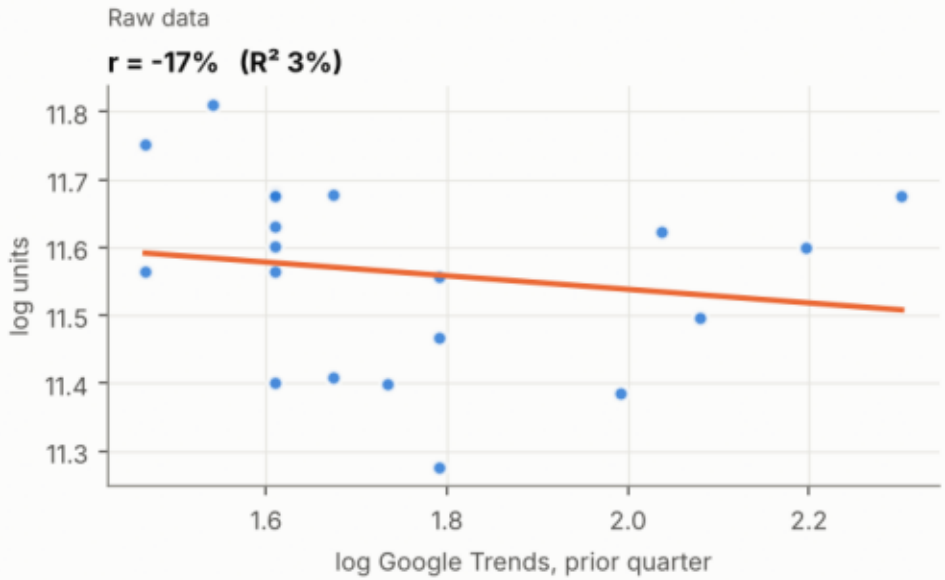
Test 31: Trends "Ram Hemi" ~ Ram pickups sales (current + 1-quarter lag)

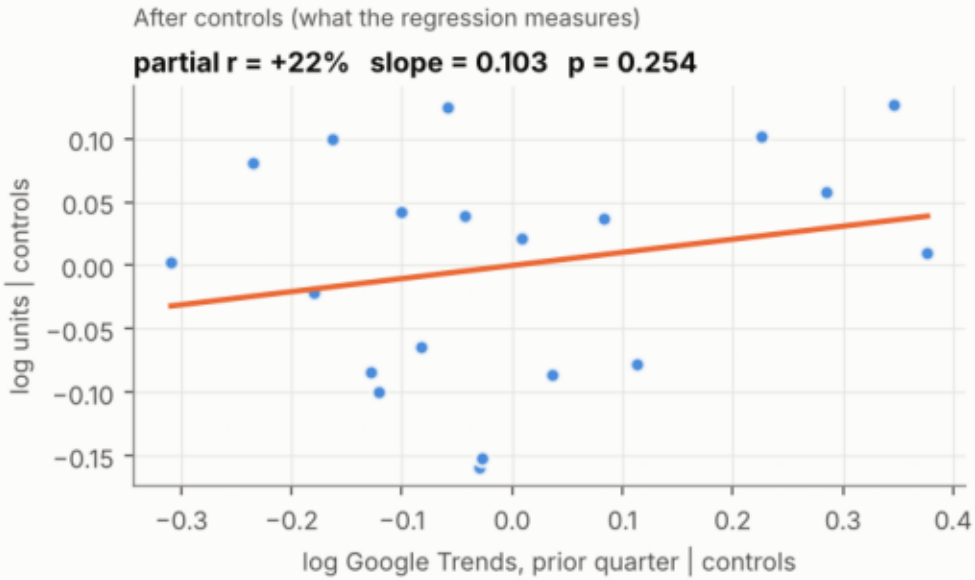




log(units) ~ log(Trends_t) + log(Trends_{t-1}) + log(industry) + quarter-of-year; HAC(2) n = 20 model R² = 54% [RUN]

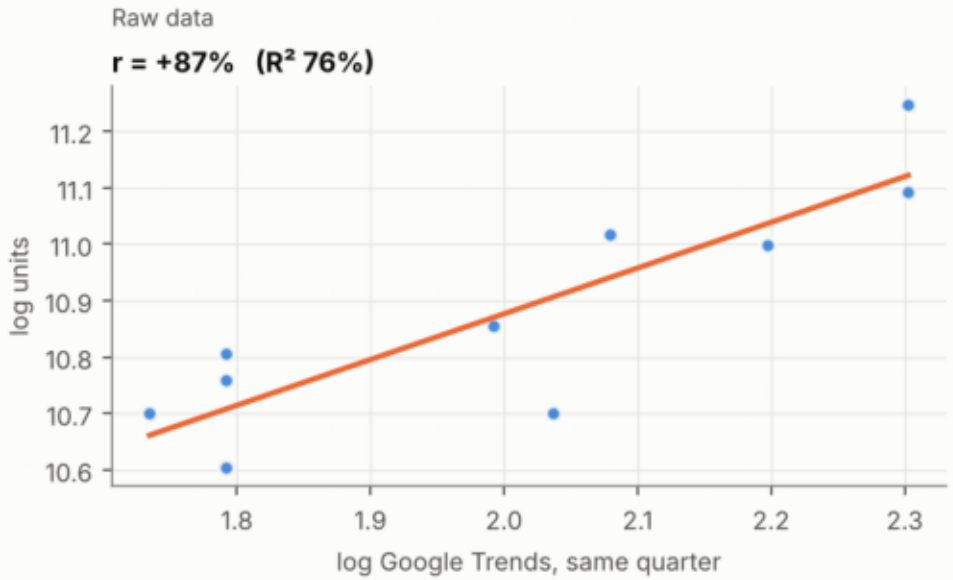
Test 31: Granger-style: Ram pickups sales ~ own lag + Trends "Ram Hemi" lag

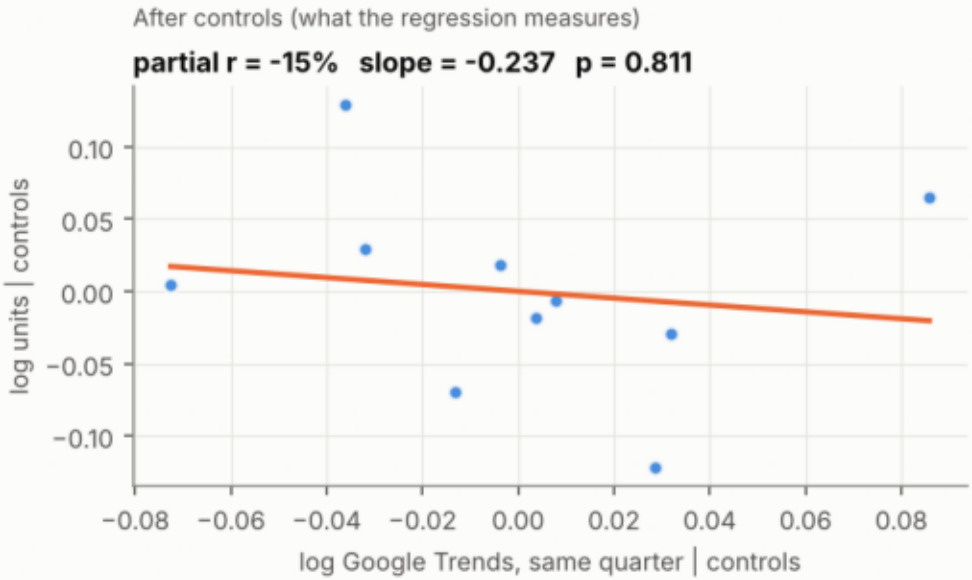




log(units_t) ~ log(units_{t-1}) + log(Trends_{t-1}) + log(industry) + quarter-of-year n = 20 model R² = 61% [RUN]

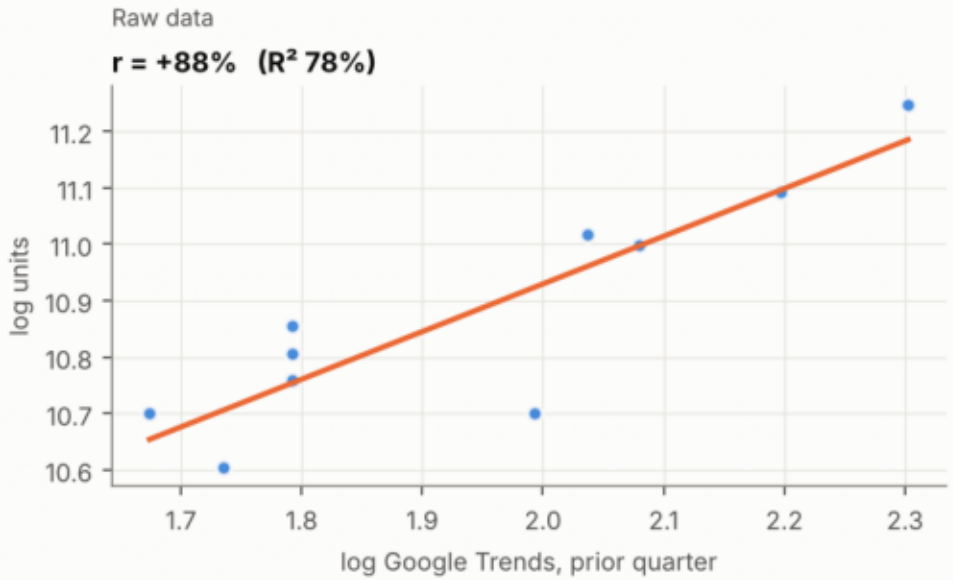
Test 31: Trends "Ram Hemi" ~ Ram 1500 sales (current + 1-quarter lag)

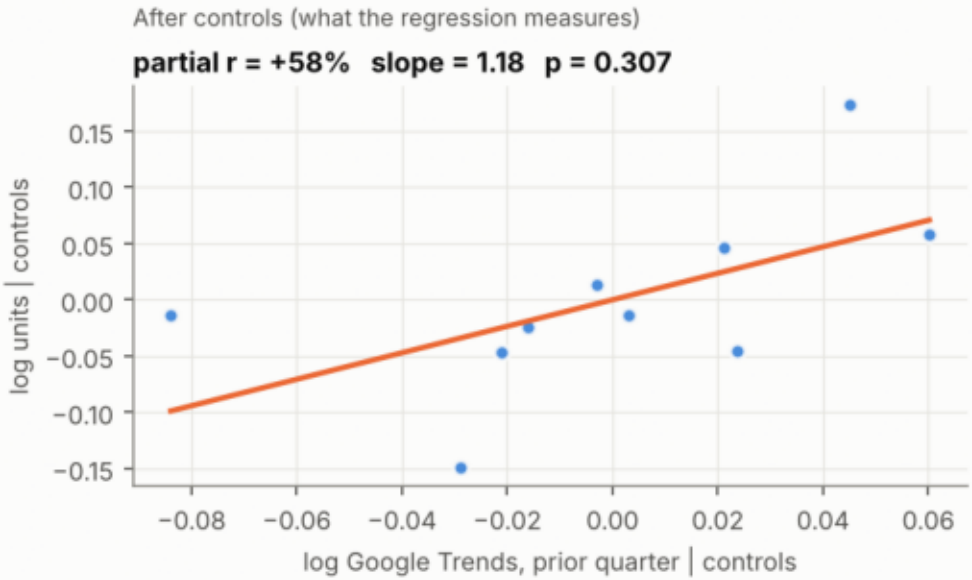




log(units) ~ log(Trends_t) + log(Trends_{t-1}) + log(industry) + quarter-of-year; HAC(2) n = 10 model R² = 89% [RUN]

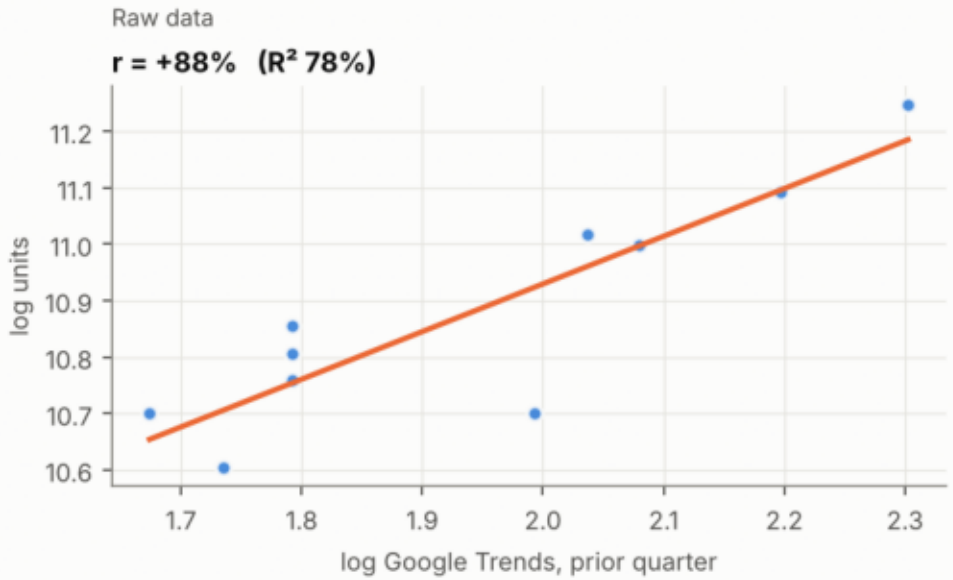
Test 31: Trends "Ram Hemi" ~ Ram 1500 sales (current + 1-quarter lag)

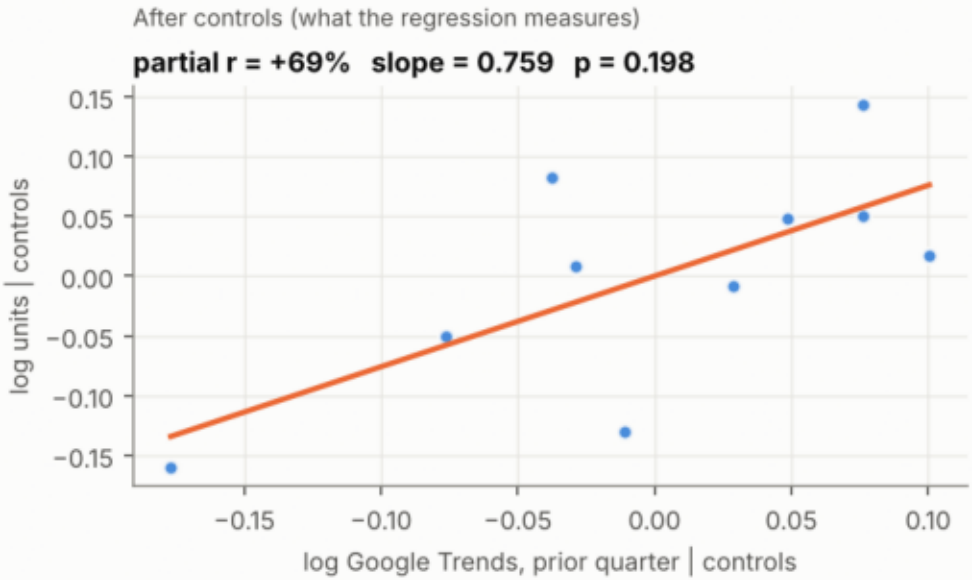




log(units) ~ log(Trends_t) + log(Trends_{t-1}) + log(industry) + quarter-of-year; HAC(2) n = 10 model R² = 89% [RUN]

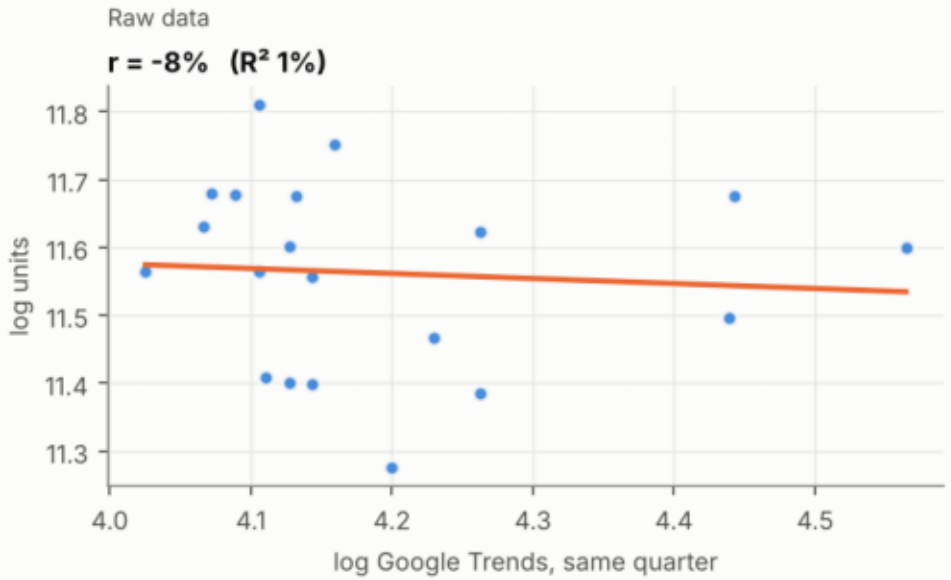
Test 31: Granger-style: Ram 1500 sales ~ own lag + Trends "Ram Hemi" lag

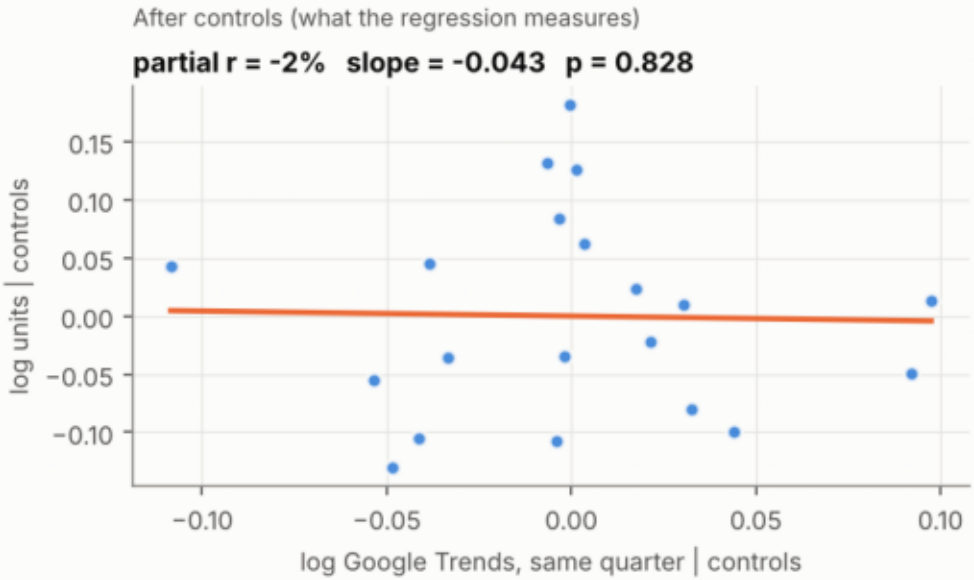




log(units_t) ~ log(units_{t-1}) + log(Trends_{t-1}) + log(industry) + quarter-of-year n = 10 model R² = 89% [RUN]

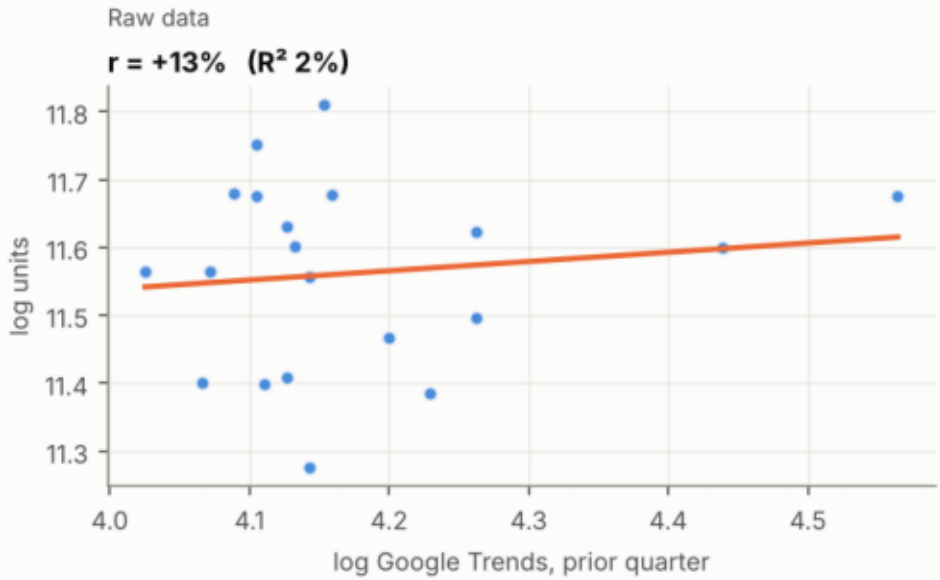
Test 31: Trends "Ram 1500" ~ Ram pickups sales (current + 1-quarter lag)

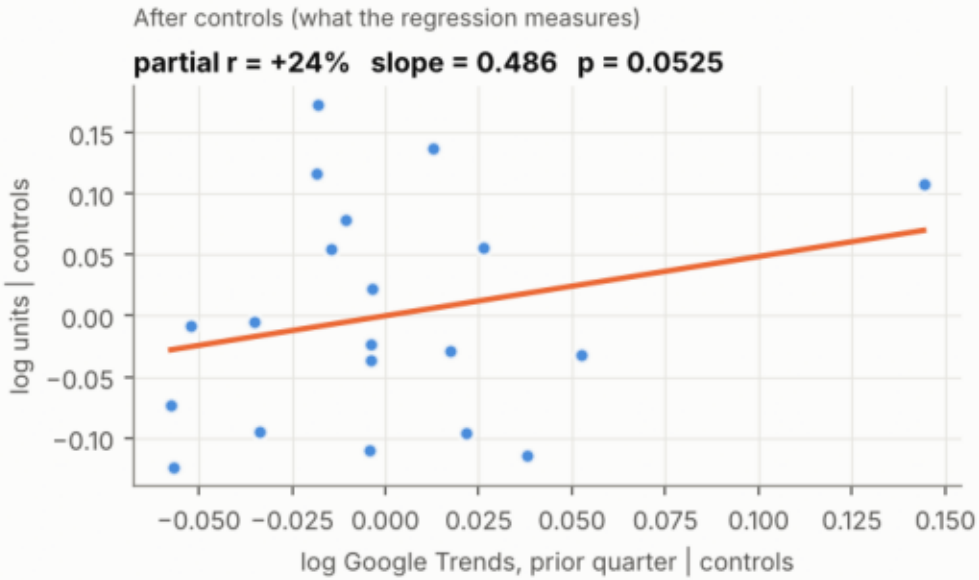




log(units) ~ log(Trends_t) + log(Trends_{t-1}) + log(industry) + quarter-of-year; HAC(2) n = 20 model R² = 60% [RUN]

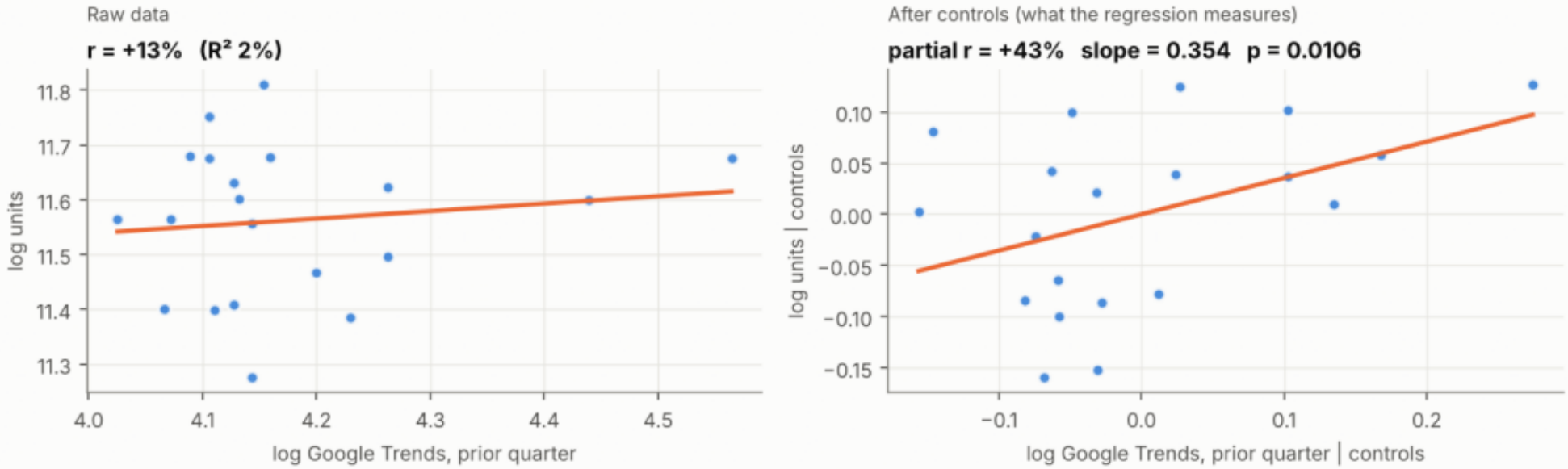
Test 31: Trends "Ram 1500" ~ Ram pickups sales (current + 1-quarter lag)





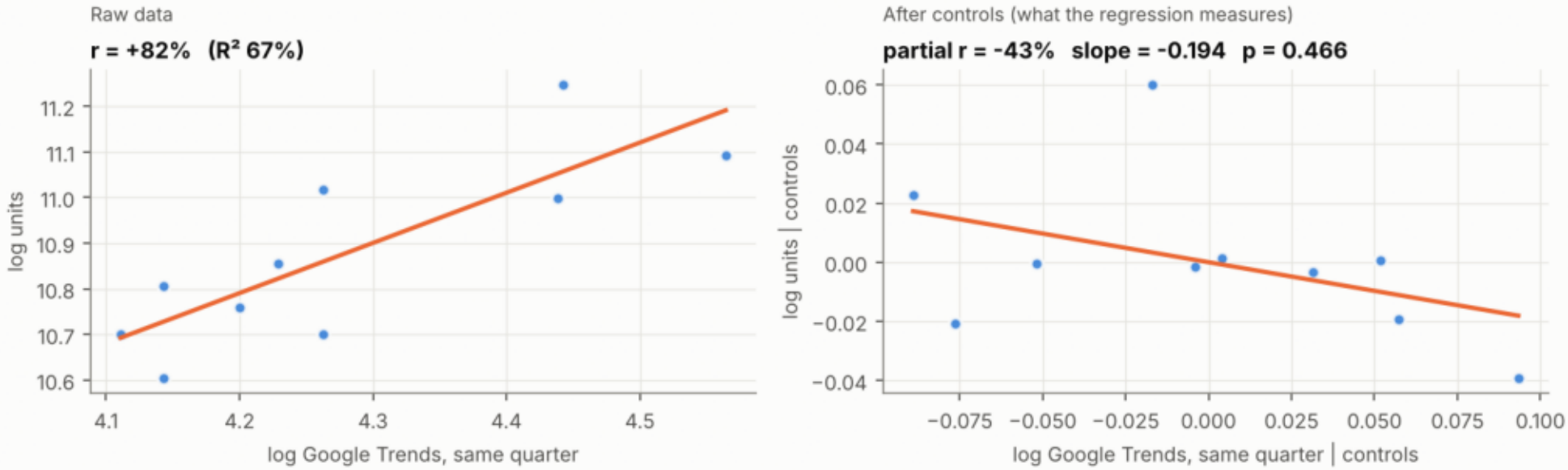
log(units) ~ log(Trends_t) + log(Trends_{t-1}) + log(industry) + quarter-of-year; HAC(2) n = 20 model R² = 60% [RUN]

Test 31: Granger-style: Ram pickups sales ~ own lag + Trends "Ram 1500" lag



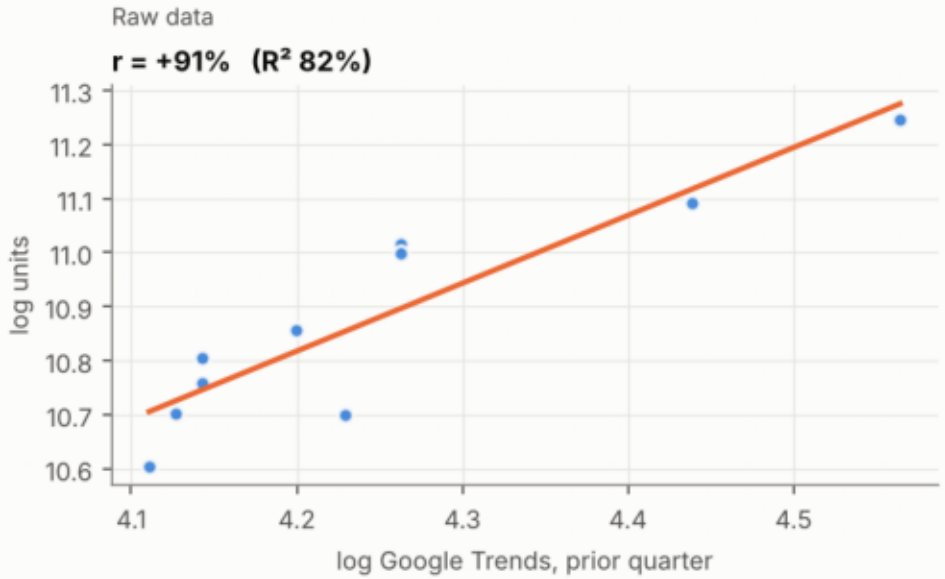
log(units_t) ~ log(units_{t-1}) + log(Trends_{t-1}) + log(industry) + quarter-of-year n = 20 model R² = 67% [RUN]

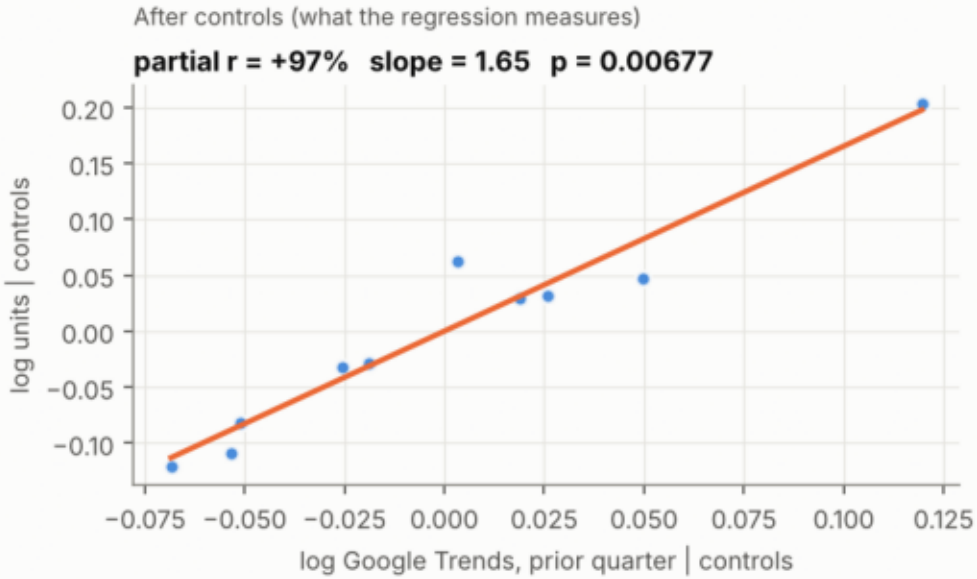
Test 31: Trends "Ram 1500" ~ Ram 1500 sales (current + 1-quarter lag)



log(units) ~ log(Trends_t) + log(Trends_{t-1}) + log(industry) + quarter-of-year; HAC(2) n = 10 model R² = 99% [RUN]

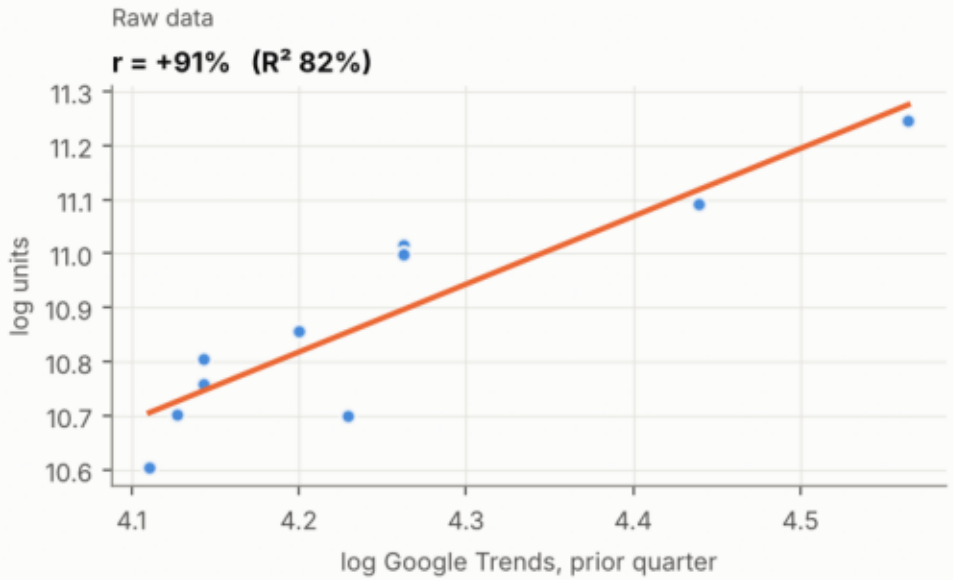
Test 31: Trends "Ram 1500" ~ Ram 1500 sales (current + 1-quarter lag)

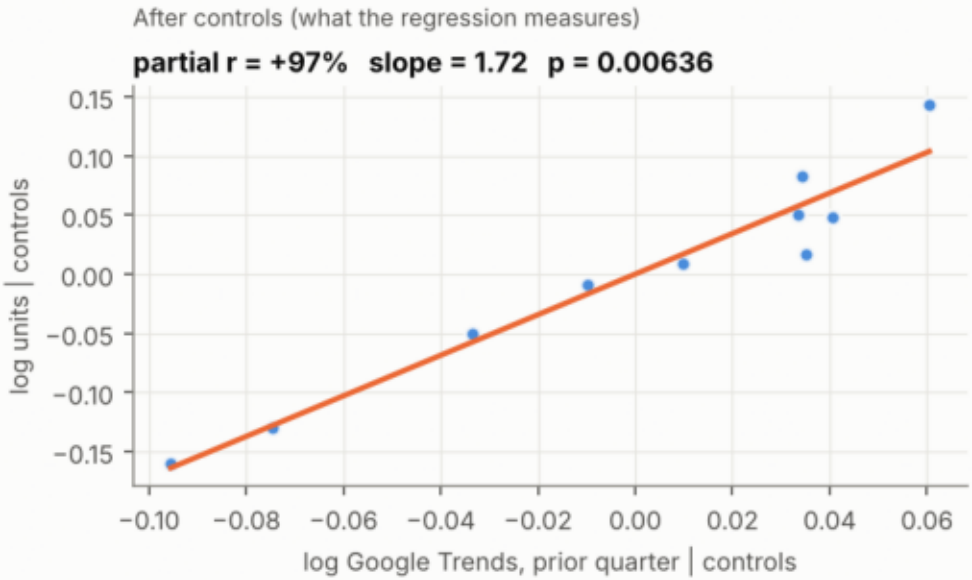




log(units) ~ log(Trends_t) + log(Trends_{t-1}) + log(industry) + quarter-of-year; HAC(2) n = 10 model R² = 99% [RUN]

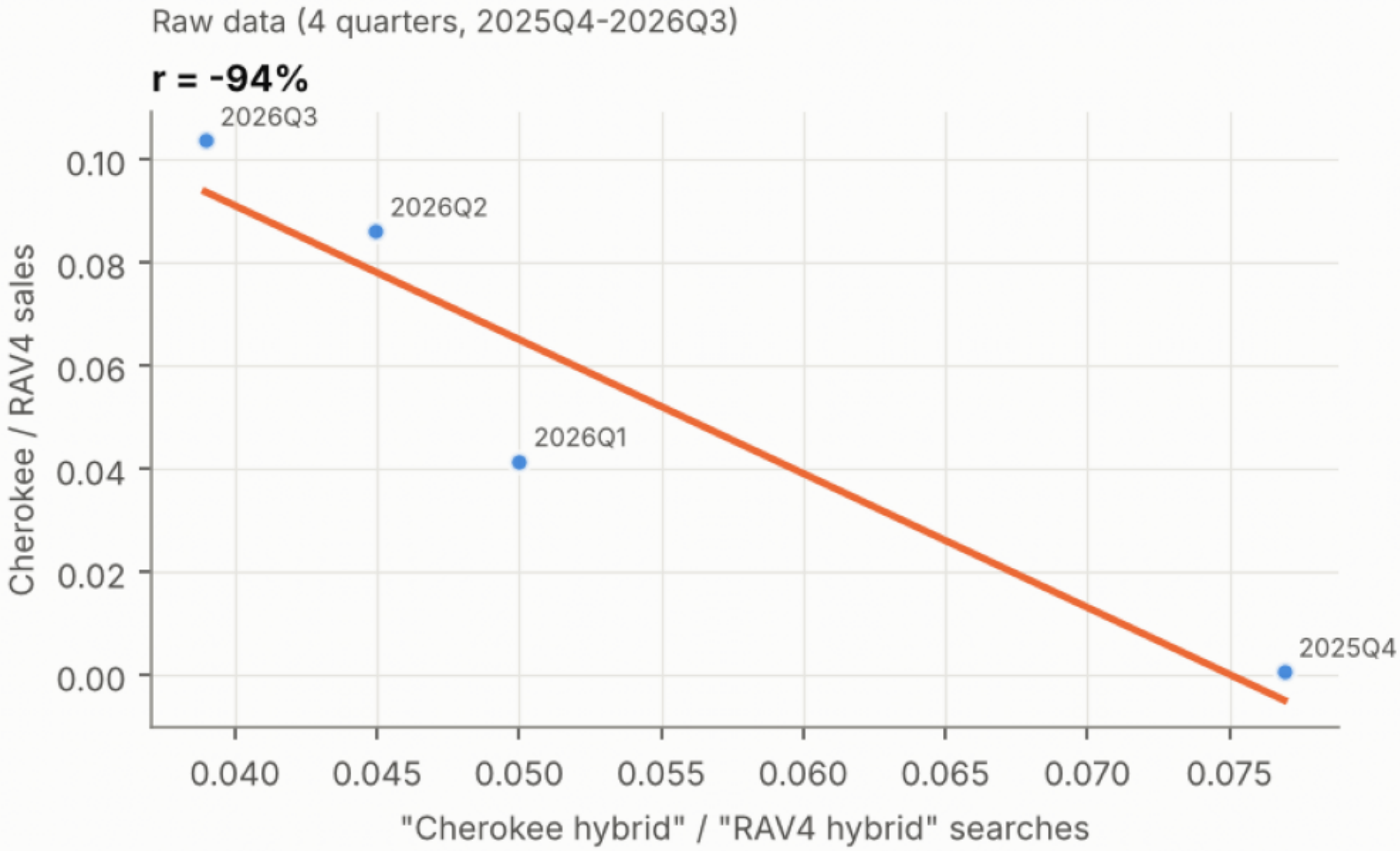
Test 31: Granger-style: Ram 1500 sales ~ own lag + Trends "Ram 1500" lag



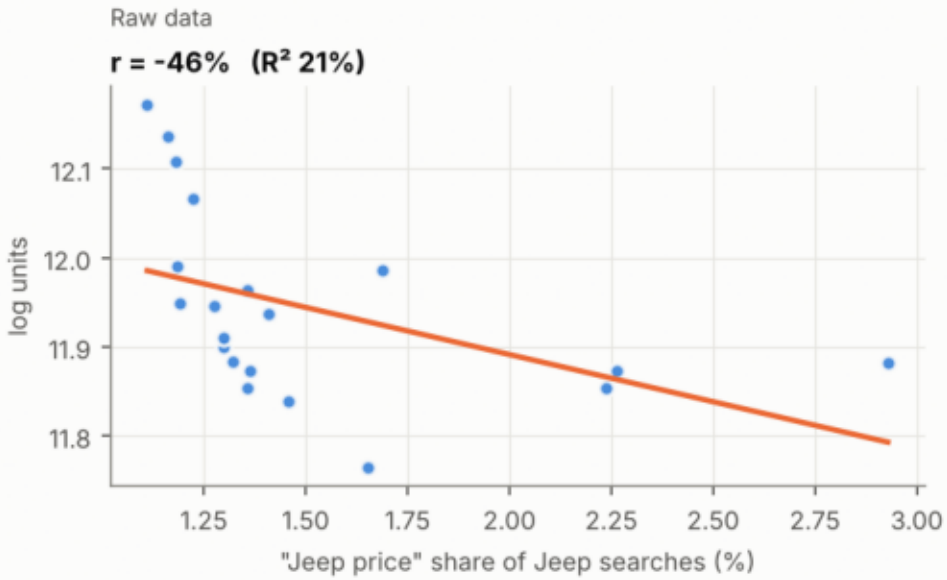


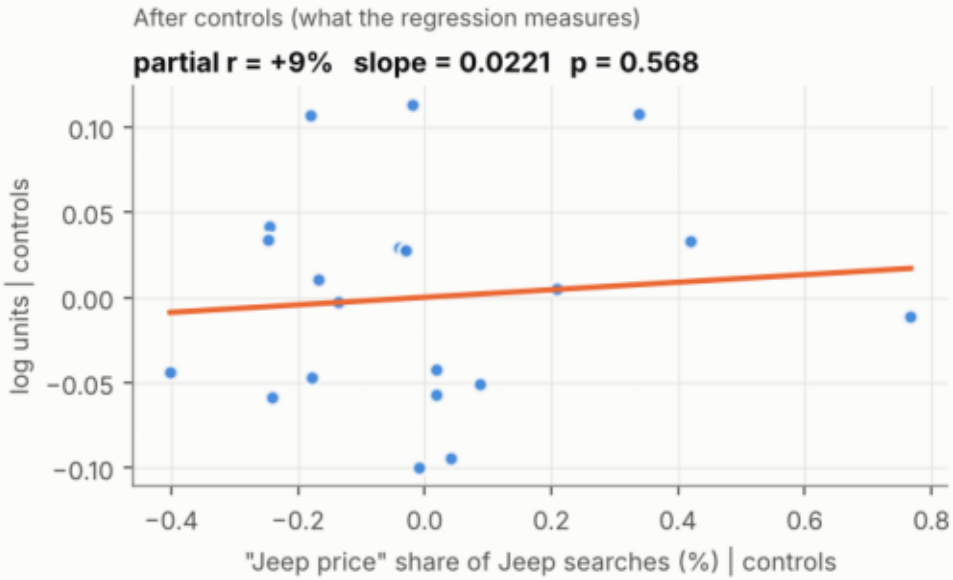
log(units_t) ~ log(units_{t-1}) + log(Trends_{t-1}) + log(industry) + quarter-of-year n = 10 model R² = 99% [RUN]

Test 32: Cherokee vs RAV4 hybrid interest gap (descriptive, n = 4)



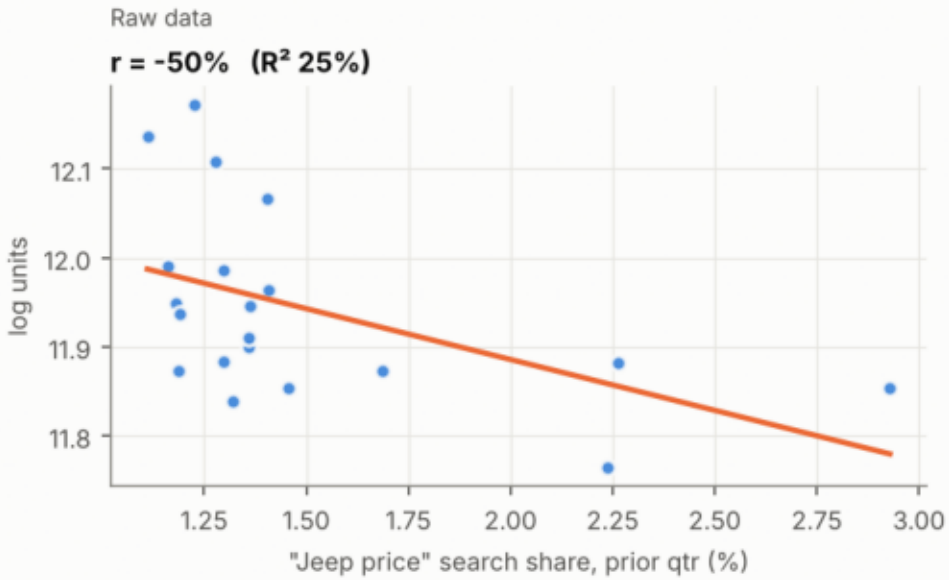
Test 33: Jeep sales ~ "Jeep price" search share (% of "Jeep" searches)

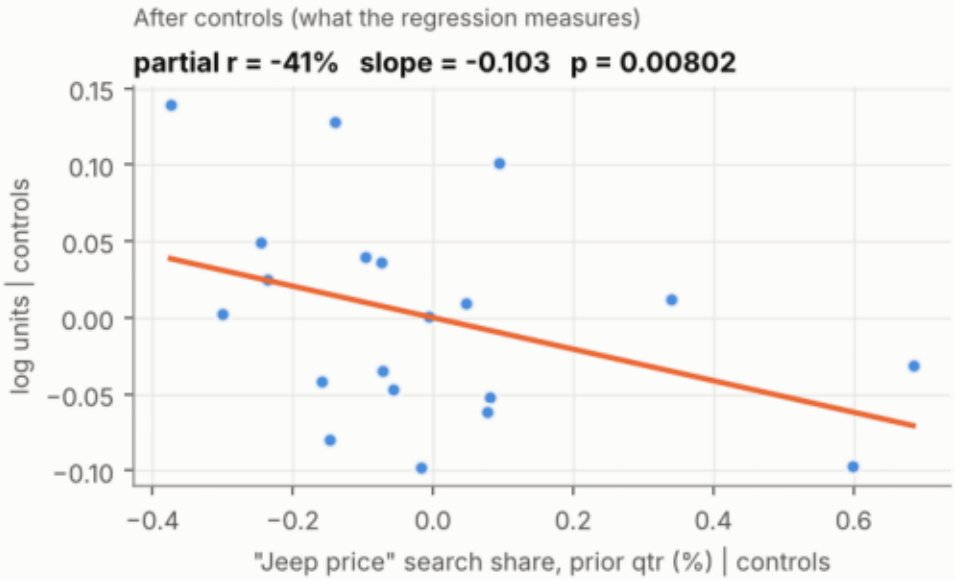




log(Jeep) ~ price-search share_t + share_{t-1} + log(industry) + quarter-of-year; HAC(2) n = 20 model R² = 65% [RUN]

Test 33: Jeep sales ~ "Jeep price" search share (% of "Jeep" searches)

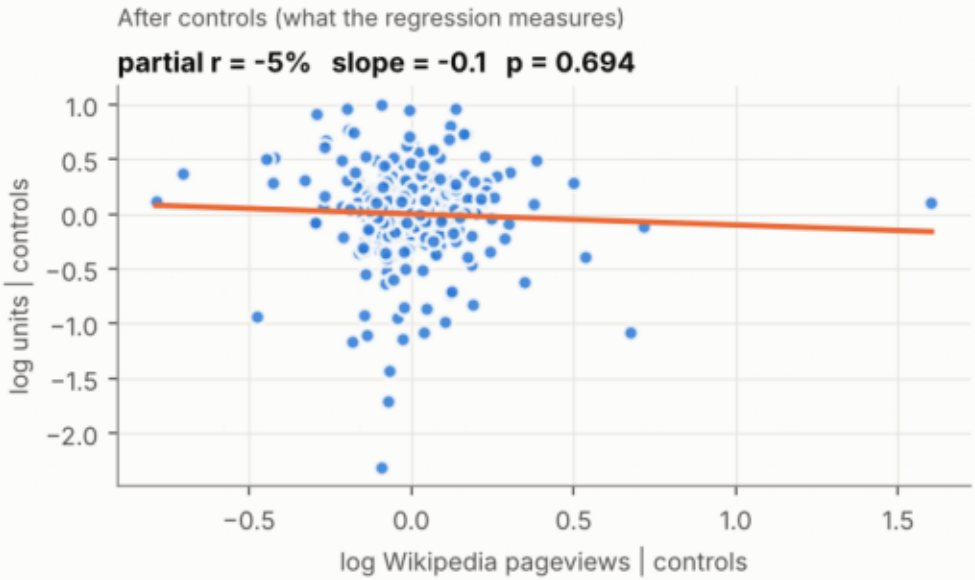




log(Jeep) ~ price-search share_t + share_{t-1} + log(industry) + quarter-of-year; HAC(2) n = 20 model R² = 65% [RUN]

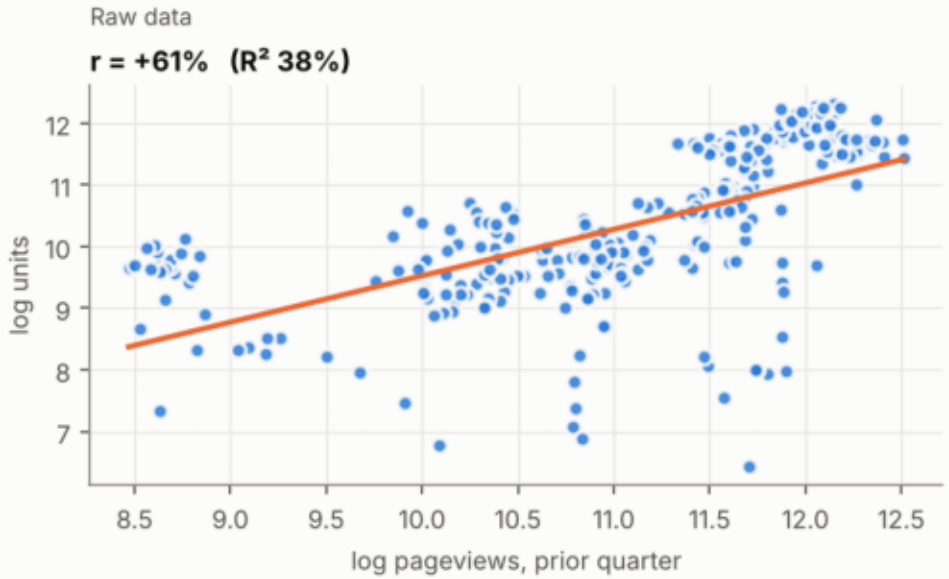
Test 38: Model sales ~ Wikipedia pageviews (model panel)

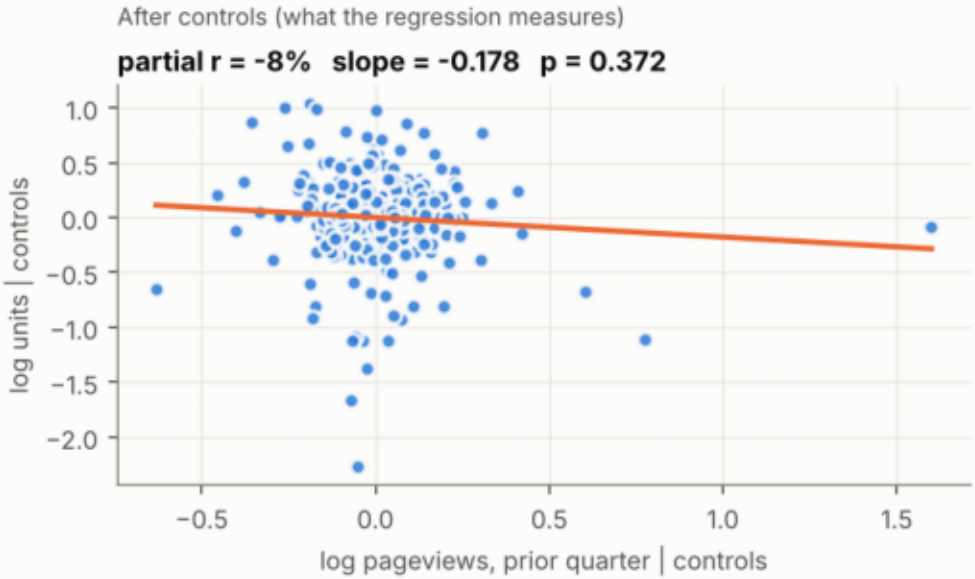




log(units) ~ log(pageviews) + model FE + quarter FE; cluster model n = 301 model R² = 89% [RUN]

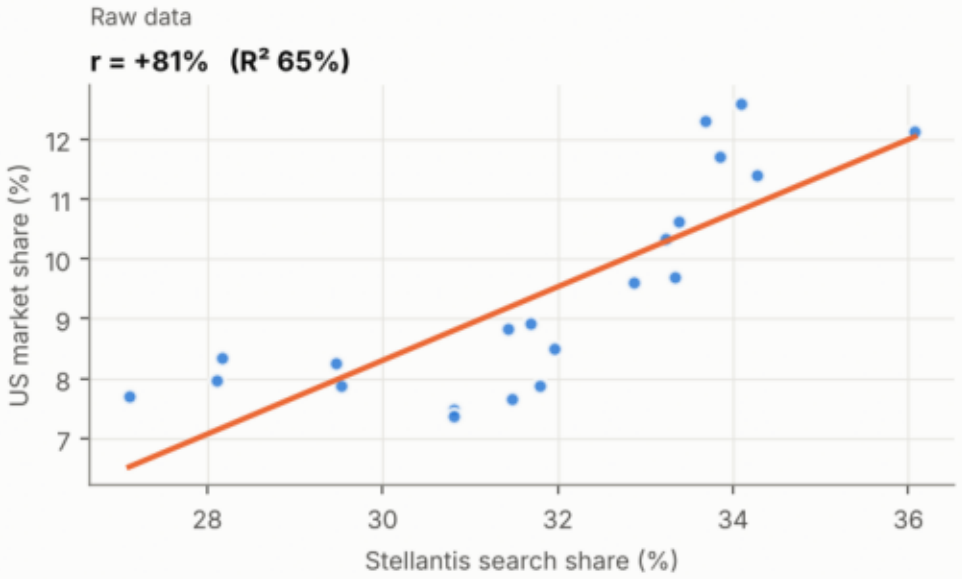
Test 38: Same, pageviews lagged one quarter

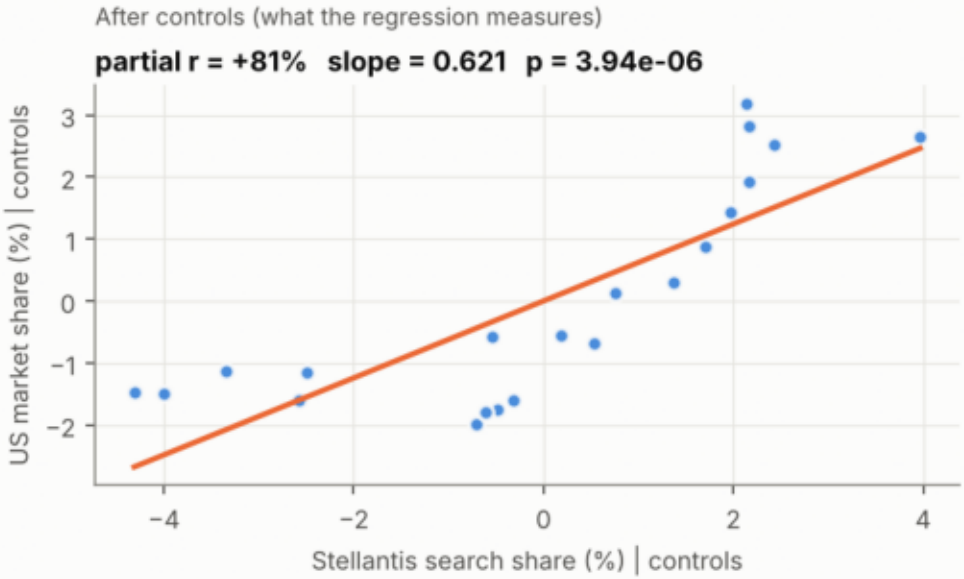




log(units_t) ~ log(views_{t-1}) + FE n = 288 model R² = 89% [RUN]

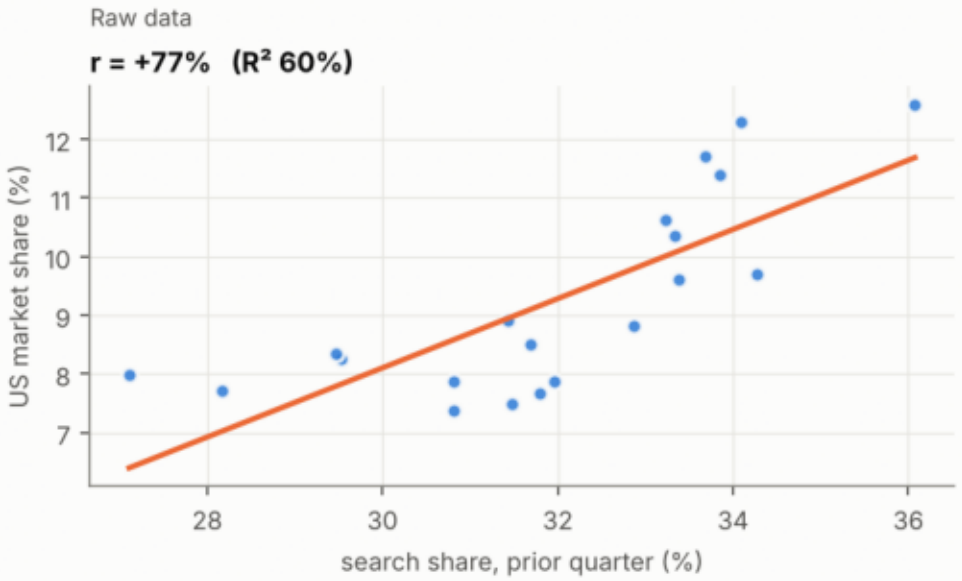
Test 40: Stellantis US share ~ Stellantis brands' search share vs Ford+Toyota

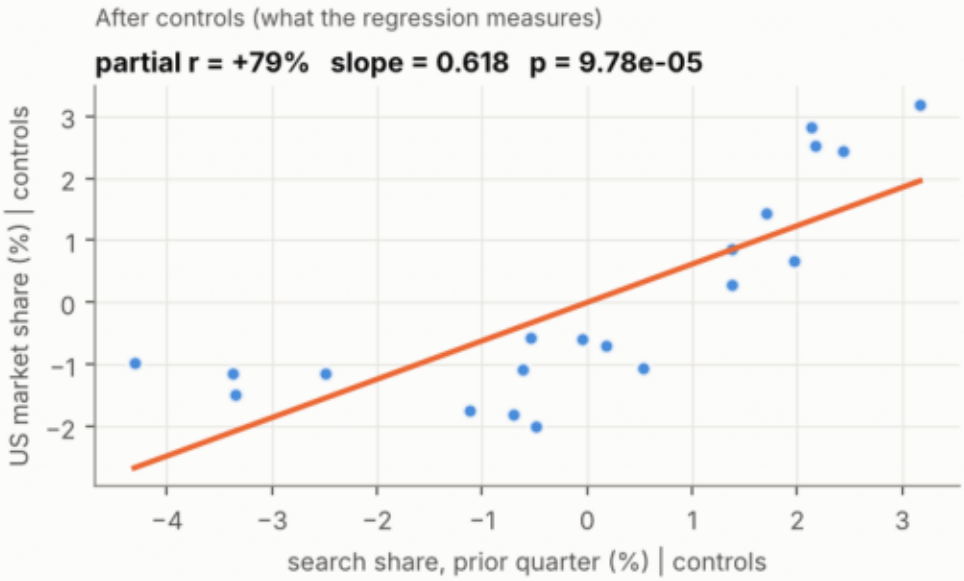




share% ~ search share% (Jeep+Dodge+Chrysler)/(those+Ford+Toyota) + quarter-of-year; HAC(2) n = 21 model R² = 66% [RUN]

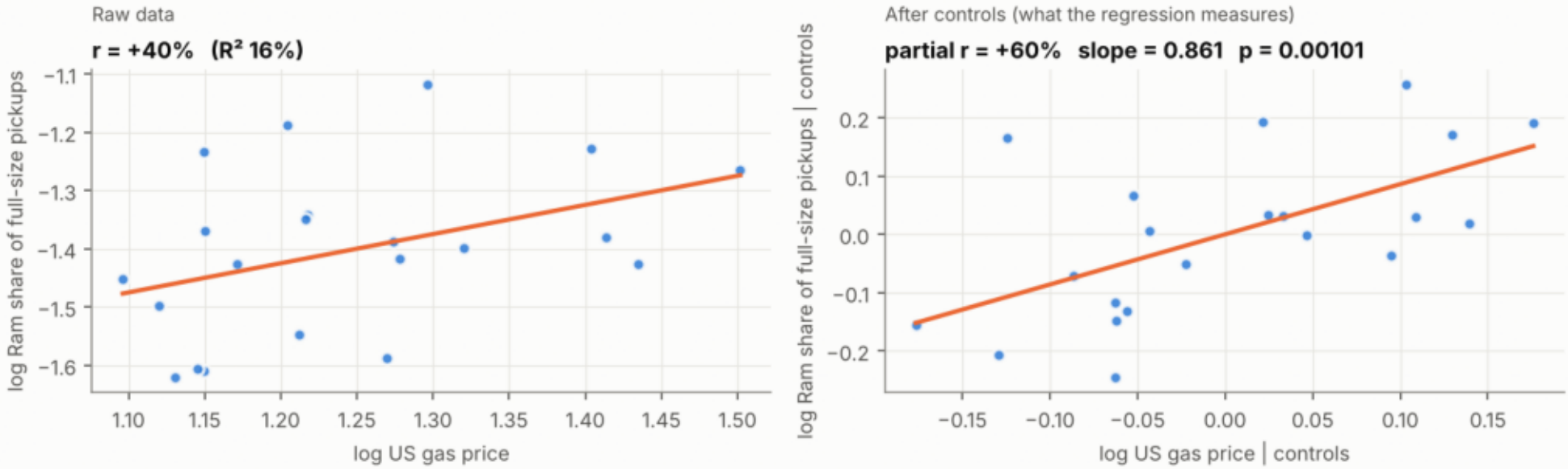
Test 40: Same, search share lagged one quarter





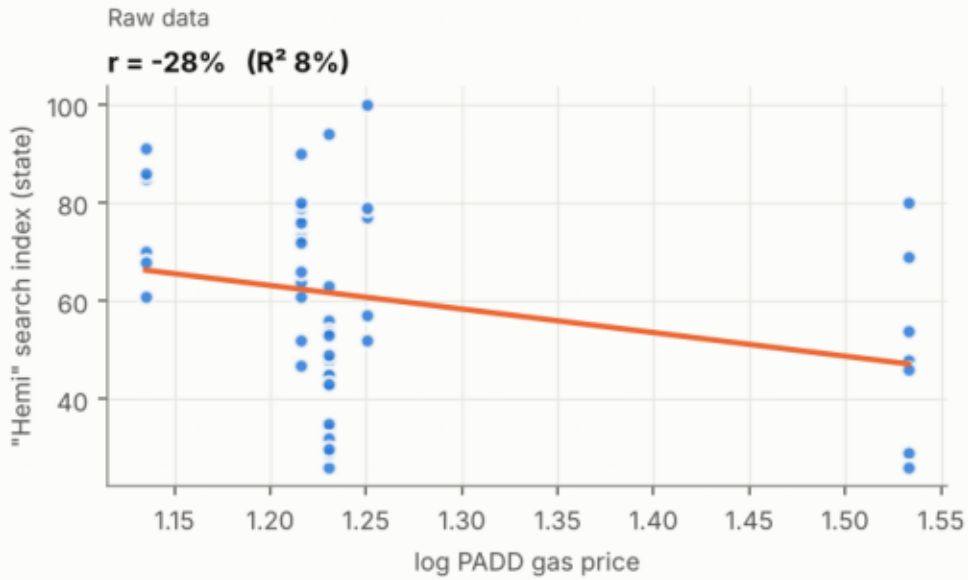
n = 20 model R² = 62% [RUN]

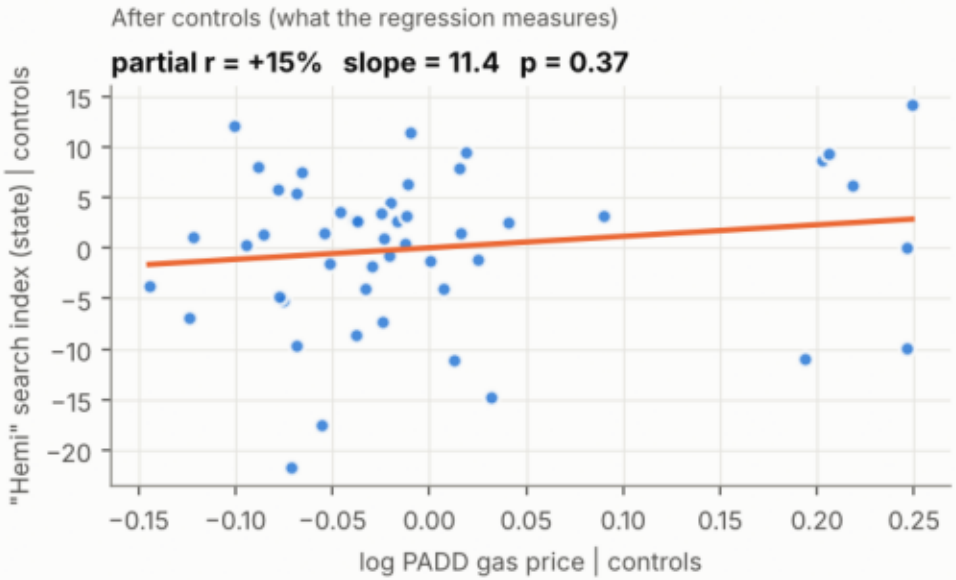
Test 41: PROXY: Ram share of full-size pickups ~ US gasoline price



log(Ram/(Ram+F-Series+Silverado)) ~ log(EIA retail regular) + quarter-of-year; HAC(2) n = 21 model R² = 40% [RUN (proxy)]

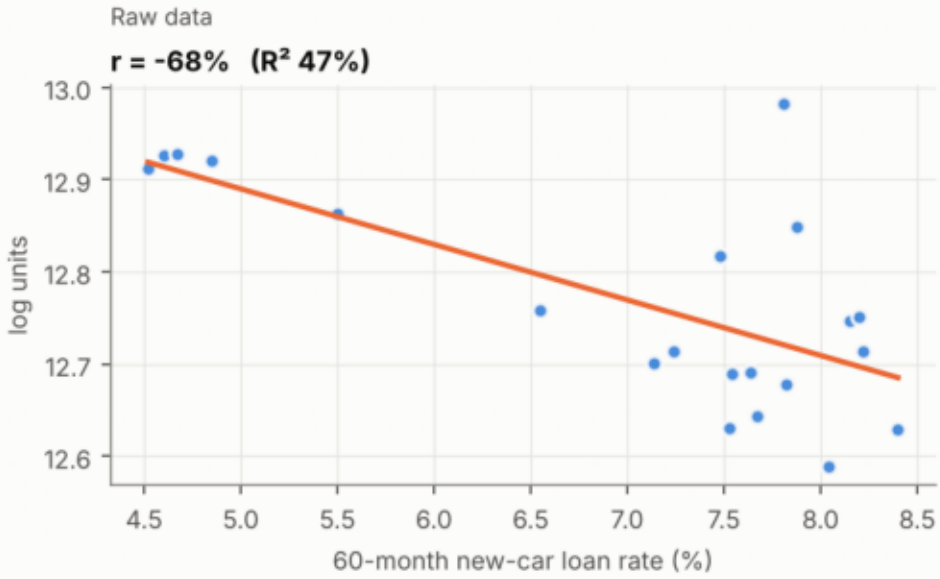
Test 41: Cross-state: "Hemi" search intensity ~ regional (PADD) gas price

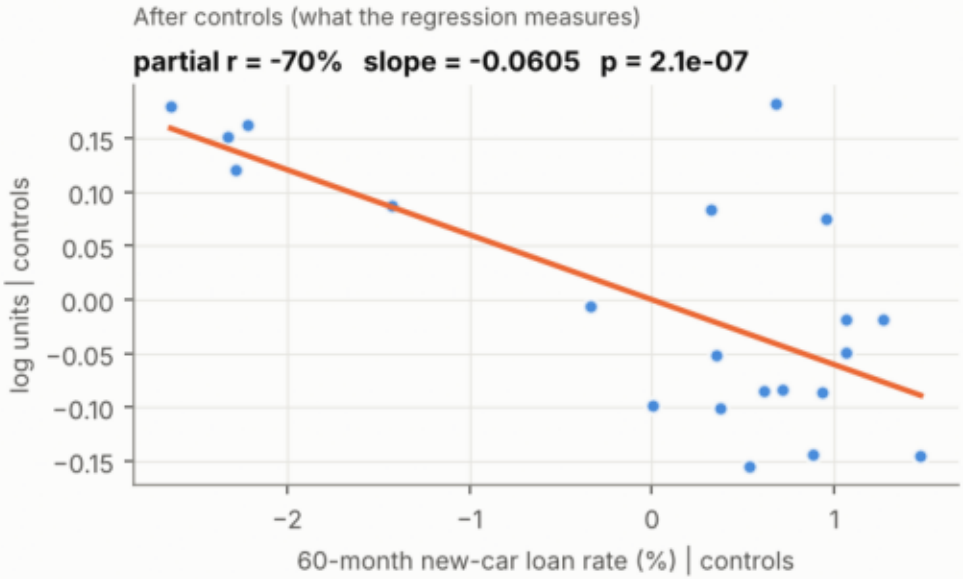




state Trends index ~ log PADD avg gas price (Jun 2025-Oct 2026) + state covariates; HC1 n = 50 model R² = 85% [RUN (proxy)]

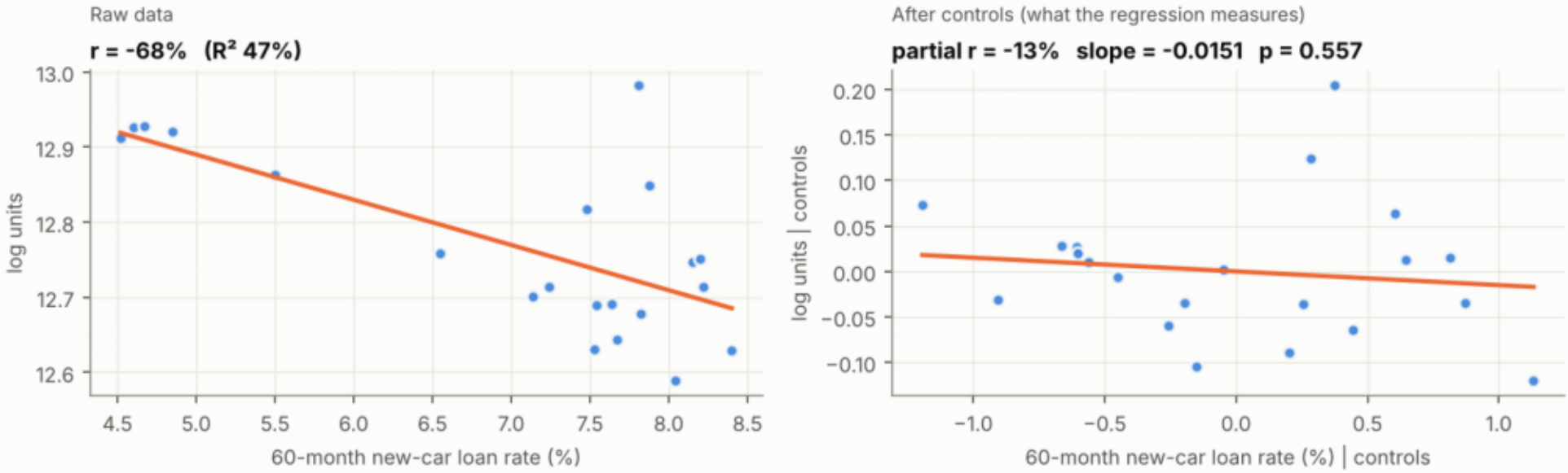
Test 43: Stellantis sales ~ 60-month new-car loan rate





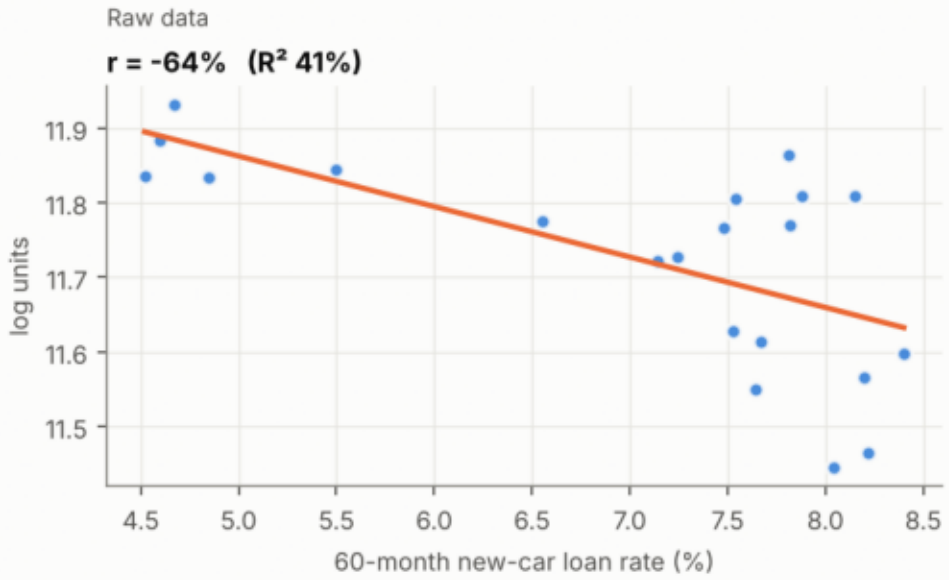
log(units) ~ rate (pp) + quarter-of-year; HAC(2) n = 21 model R² = 51% [RUN]

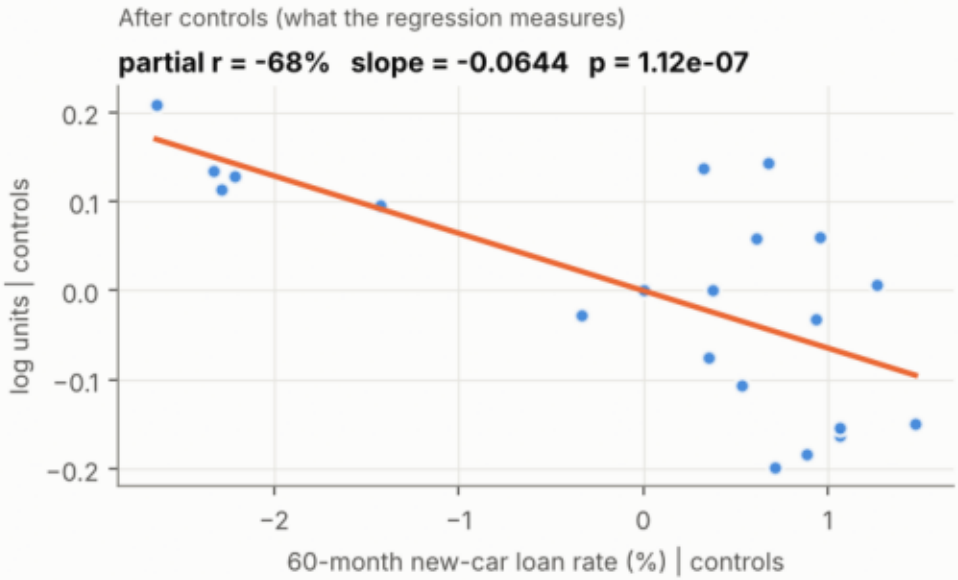
Test 43: Stellantis sales ~ loan rate, controlling for industry volume (excess sensitivity)



log(units) ~ rate + log(industry units) + quarter-of-year; HAC(2) n = 21 model R² = 59% [RUN]

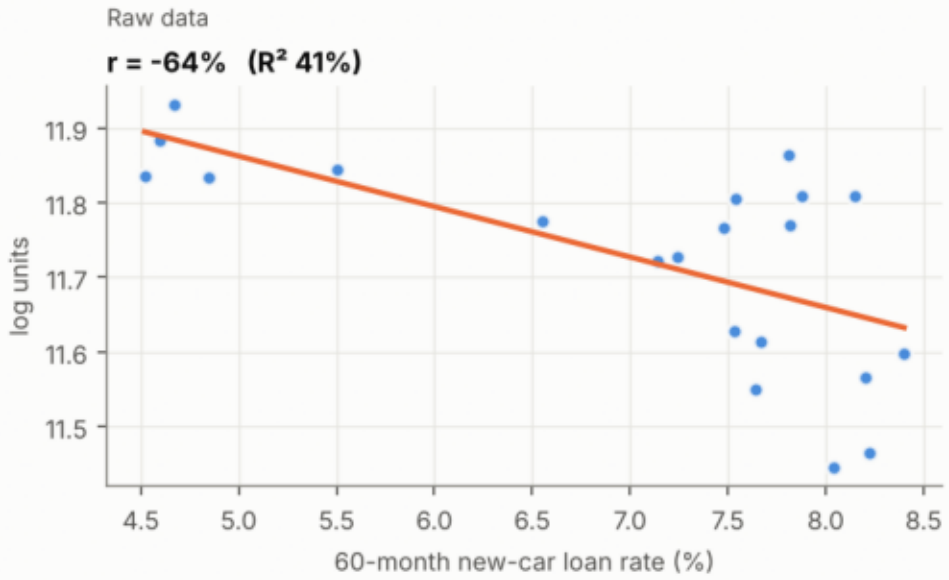
Test 43: Ram sales ~ 60-month new-car loan rate

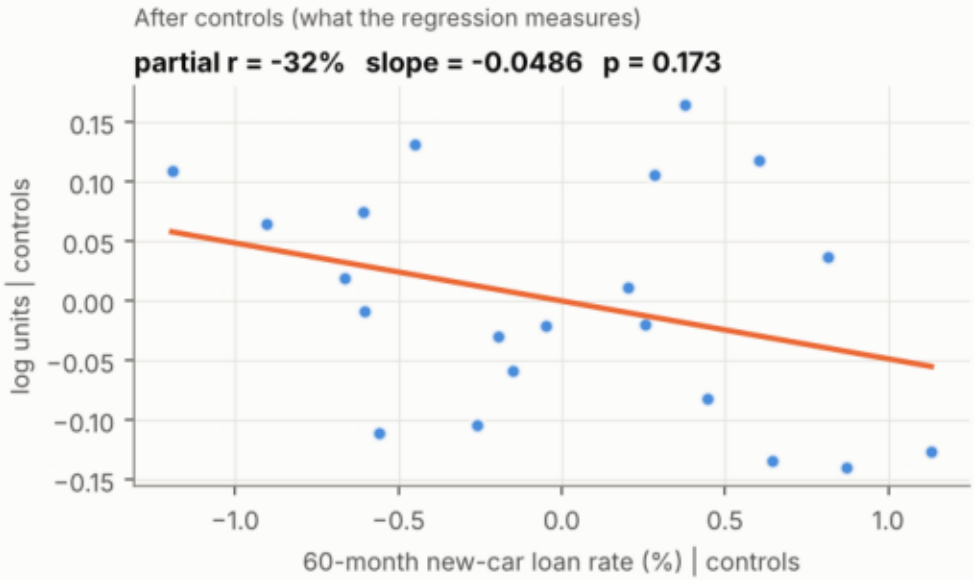




log(units) ~ rate (pp) + quarter-of-year; HAC(2) n = 21 model R² = 58% [RUN]

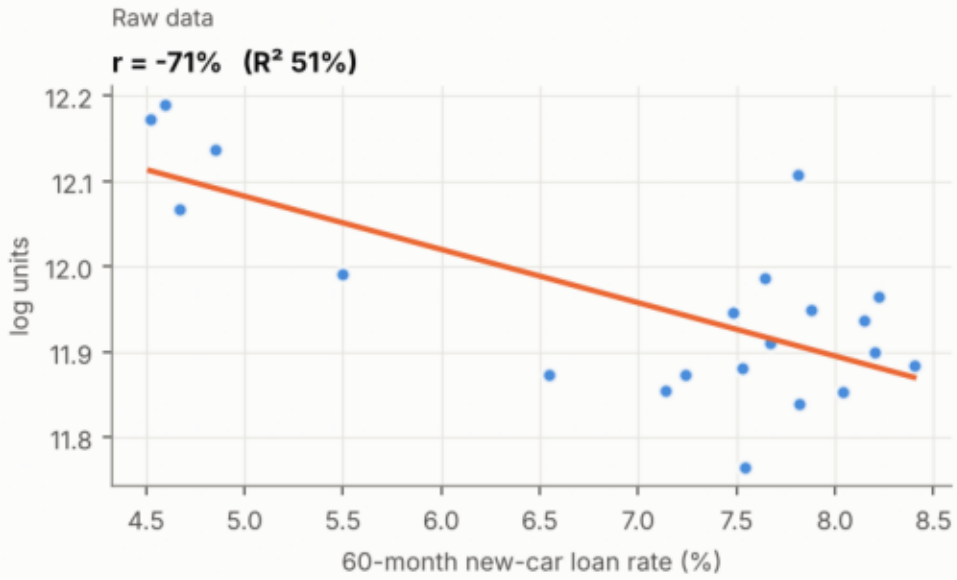
Test 43: Ram sales ~ loan rate, controlling for industry volume (excess sensitivity)

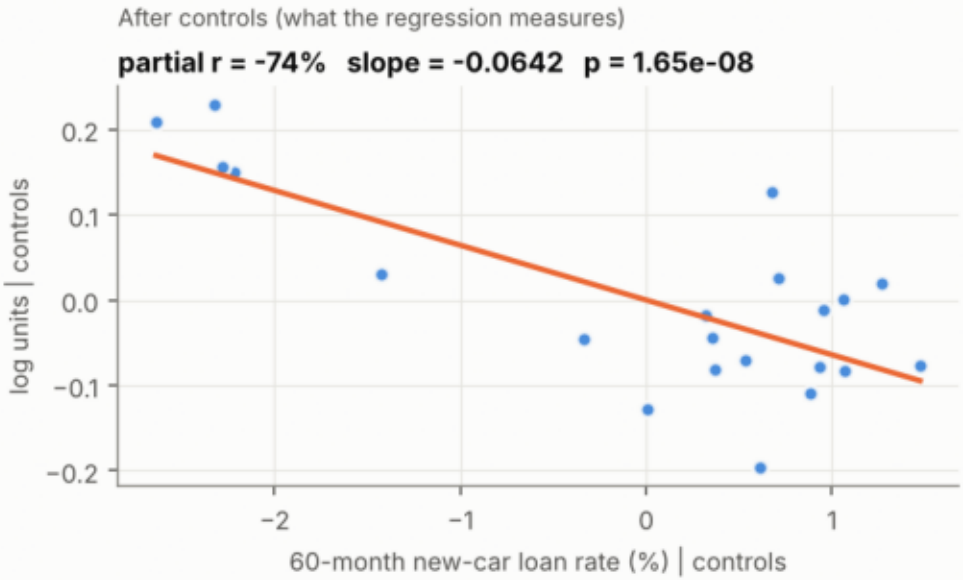




log(units) ~ rate + log(industry units) + quarter-of-year; HAC(2) n = 21 model R² = 59% [RUN]

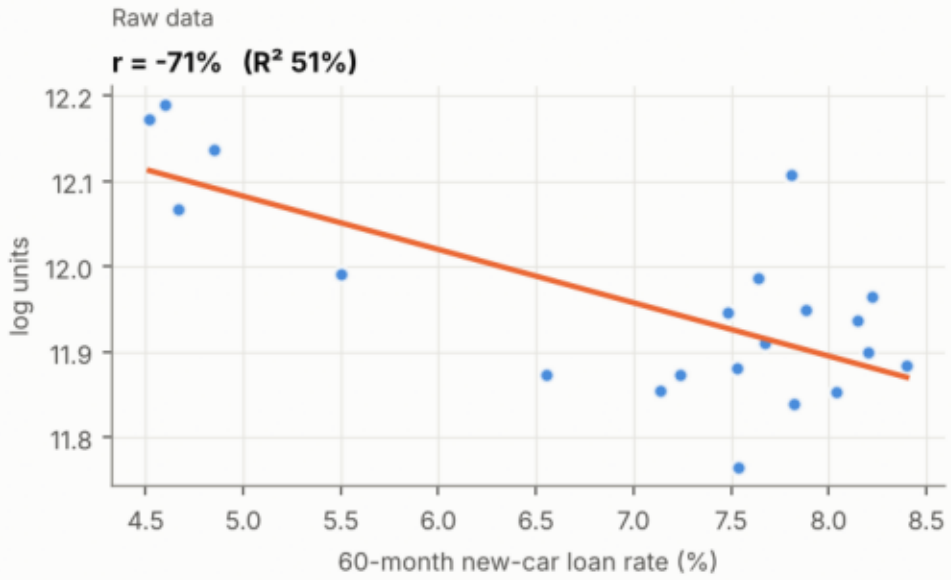
Test 43: Jeep sales ~ 60-month new-car loan rate

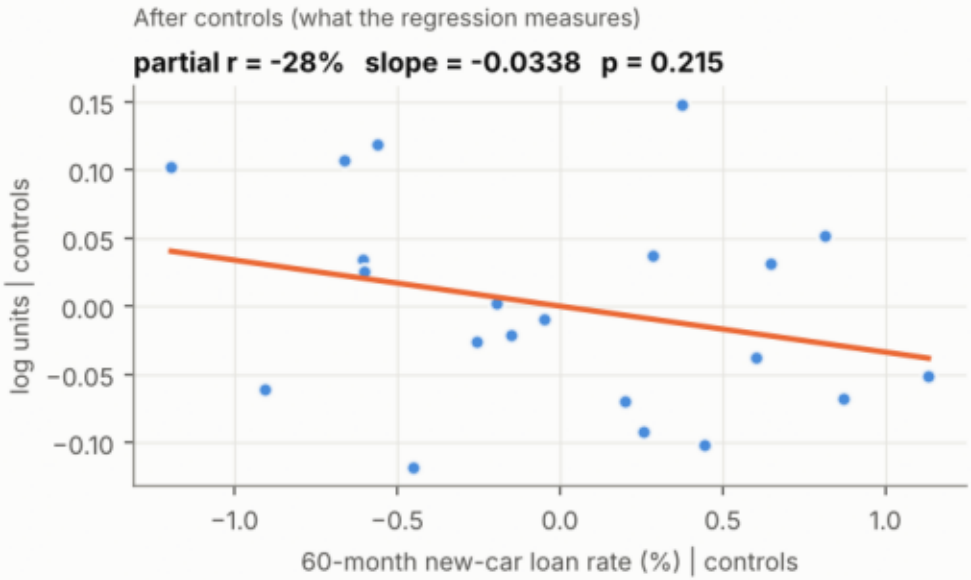




log(units) ~ rate (pp) + quarter-of-year; HAC(2) n = 21 model R² = 57% [RUN]

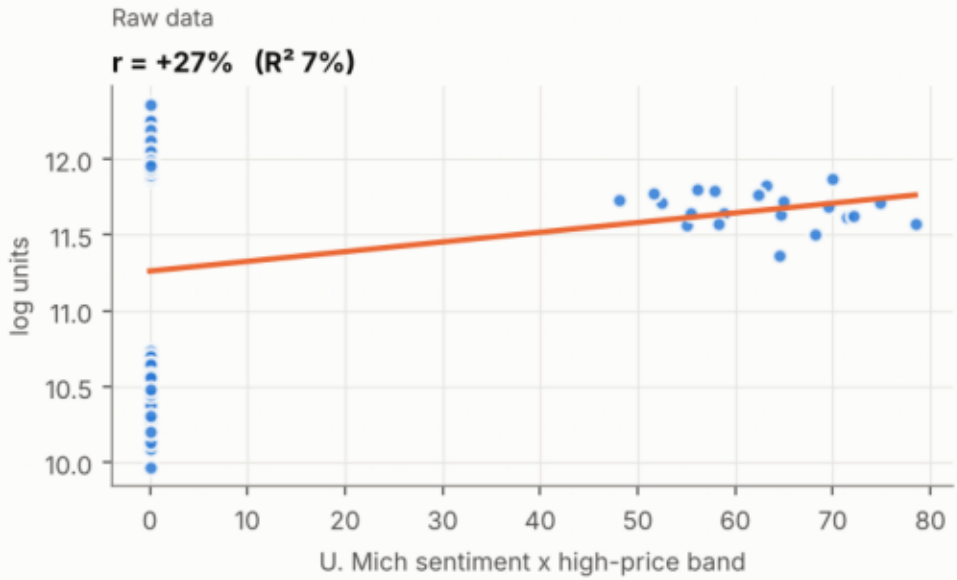
Test 43: Jeep sales ~ loan rate, controlling for industry volume (excess sensitivity)

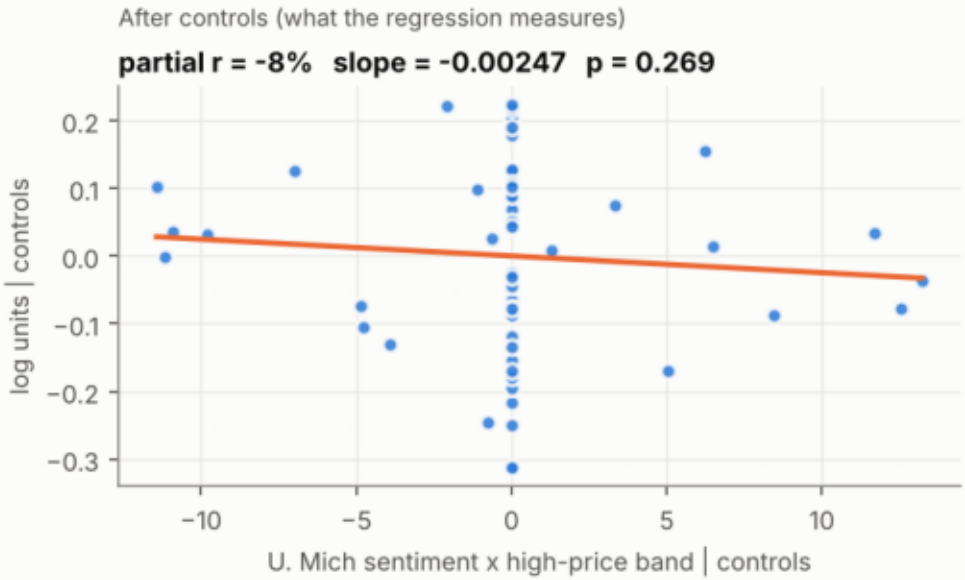




log(units) ~ rate + log(industry units) + quarter-of-year; HAC(2) n = 21 model R² = 61% [RUN]

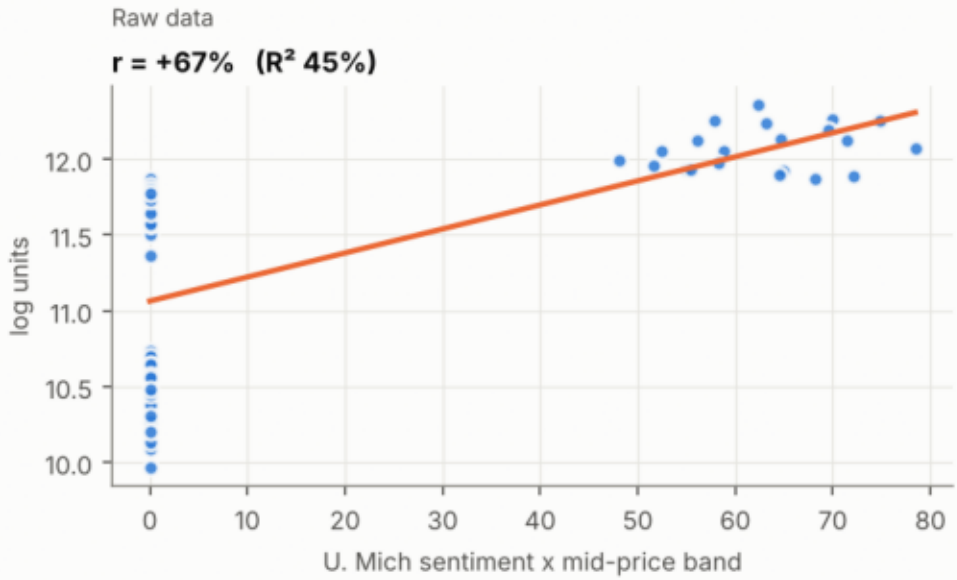
Test 44: Stellantis sales by price band ~ U. Michigan sentiment

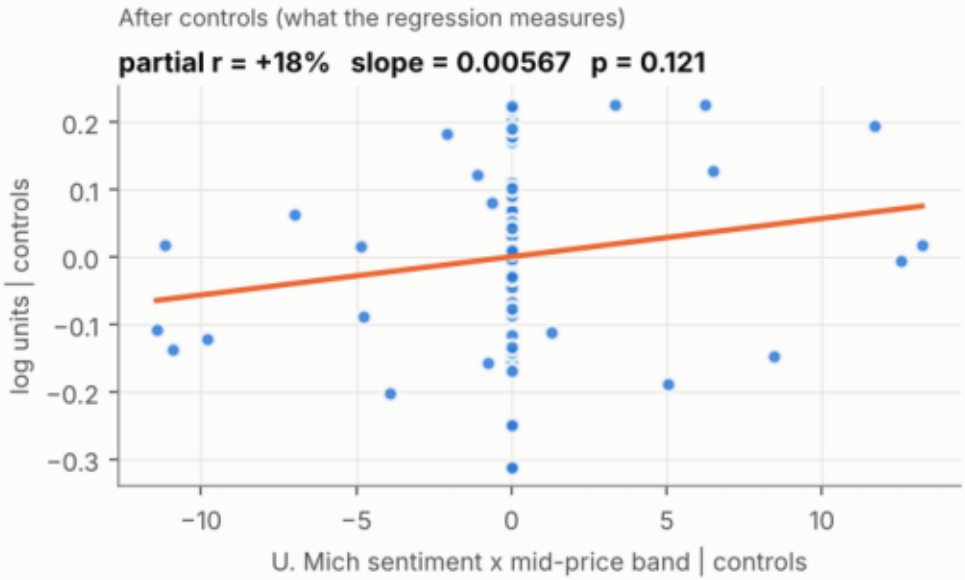




log(band units) ~ sentiment x band + band FE + band x quarter-of-year; HAC(2) n = 63 model R² = 96% [RUN]

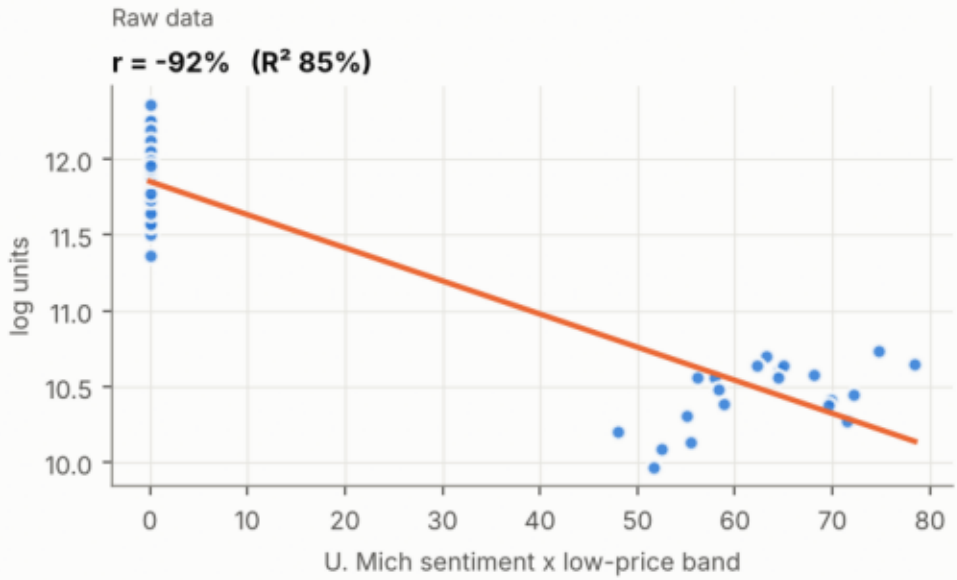
Test 44: Stellantis sales by price band ~ U. Michigan sentiment

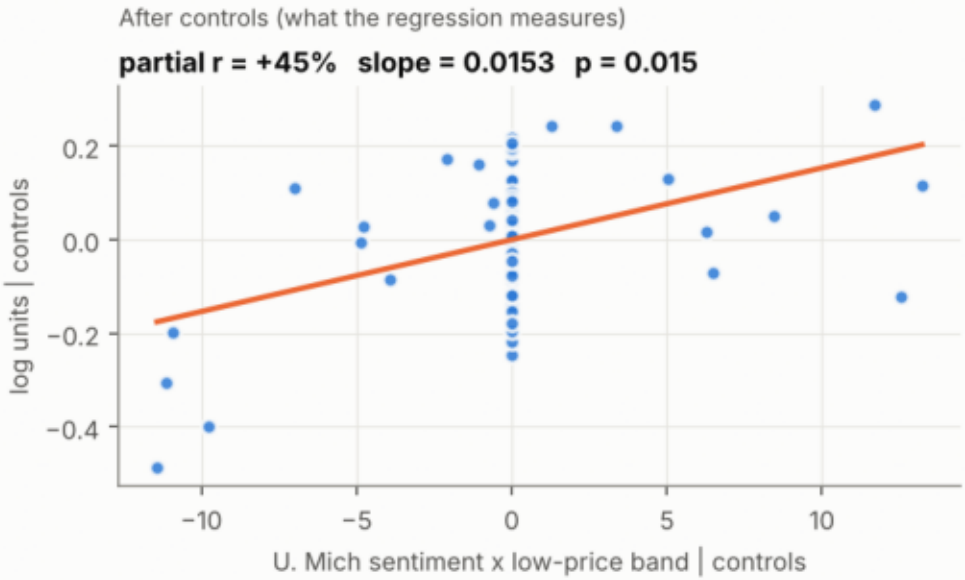




log(band units) ~ sentiment x band + band FE + band x quarter-of-year; HAC(2) n = 63 model R² = 96% [RUN]

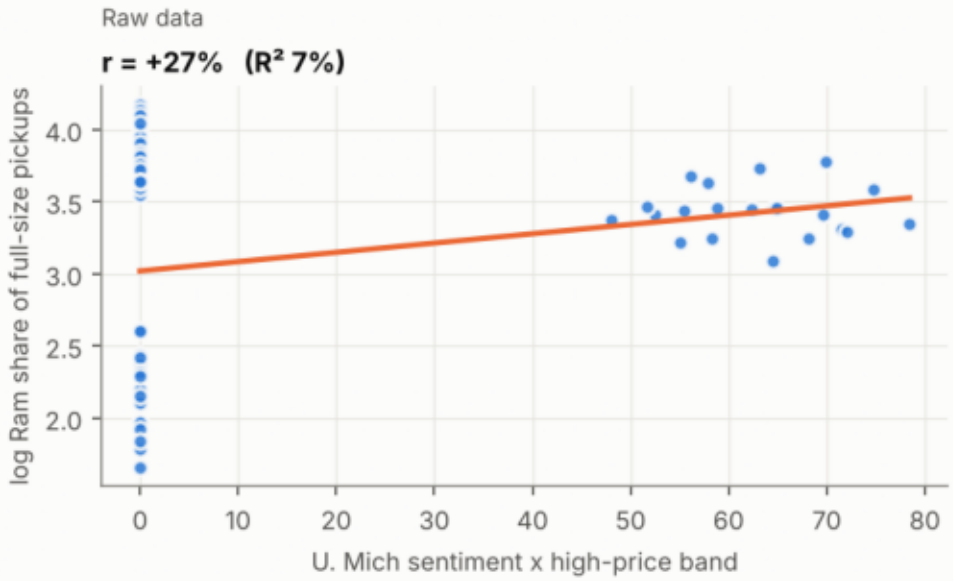
Test 44: Stellantis sales by price band ~ U. Michigan sentiment

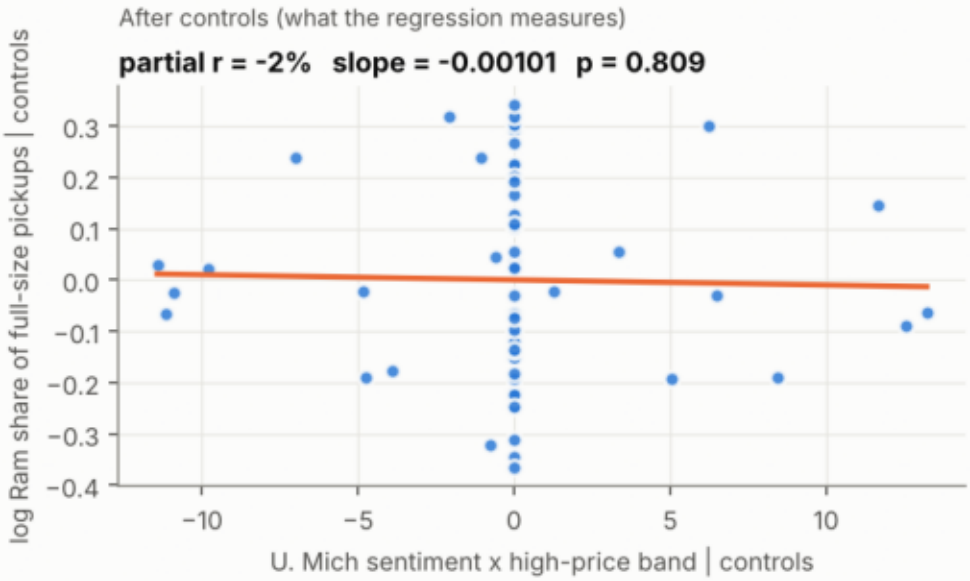




log(band units) ~ sentiment x band + band FE + band x quarter-of-year; HAC(2) n = 63 model R² = 96% [RUN]

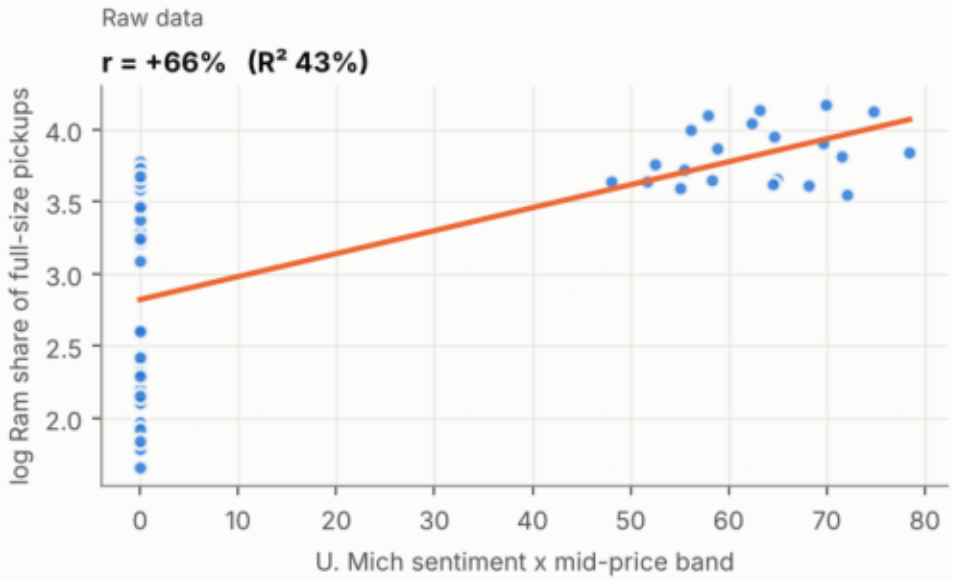
Test 44: Band sales relative to industry ~ sentiment (does Stellantis feel shocks more than the market?)

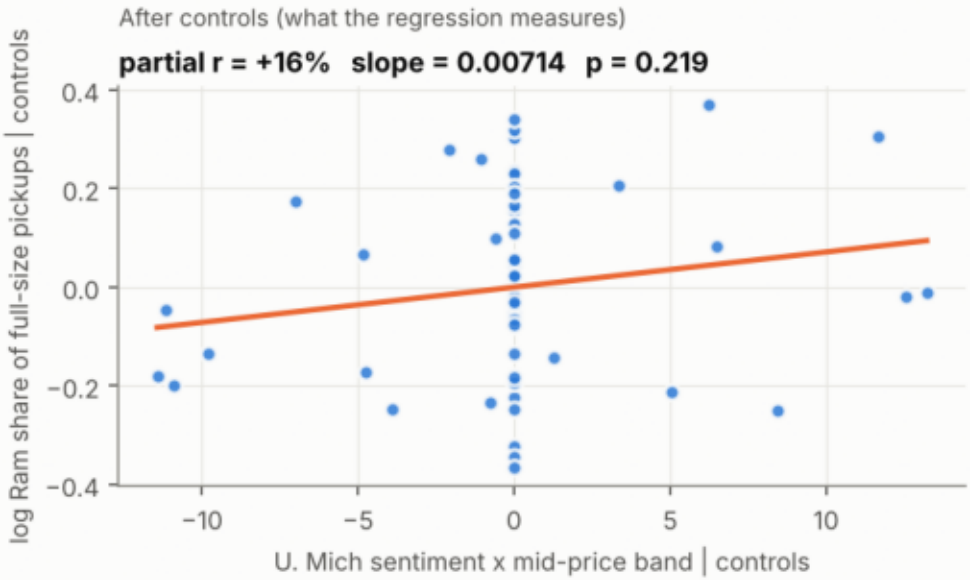




log(band units / industry units) ~ sentiment x band + FE n = 63 model R² = 93% [RUN]

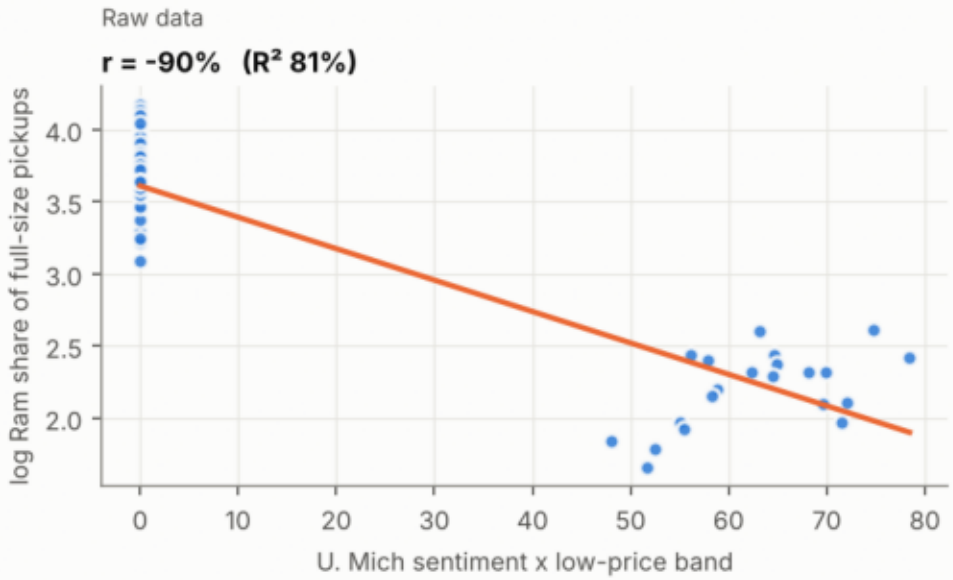
Test 44: Band sales relative to industry ~ sentiment (does Stellantis feel shocks more than the market?)

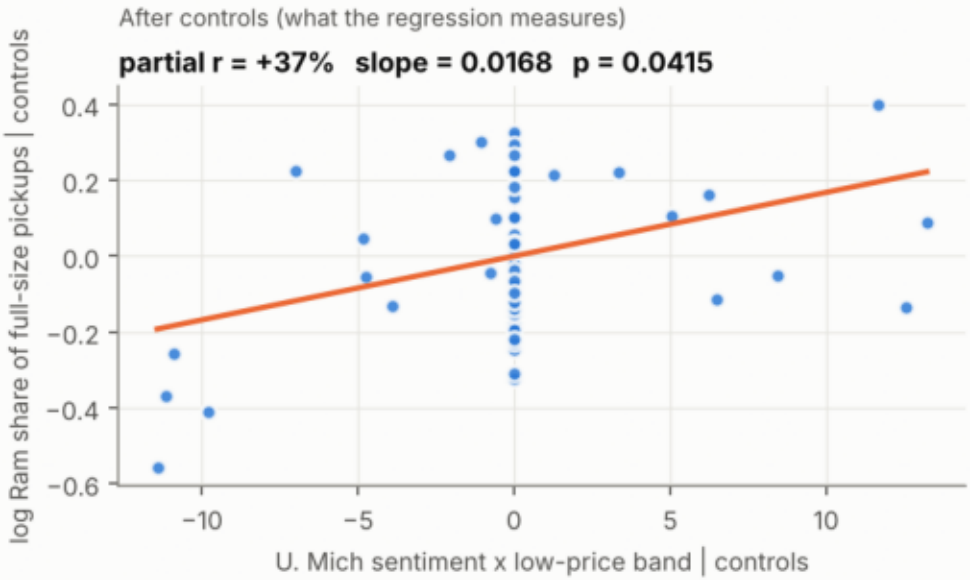




log(band units / industry units) ~ sentiment x band + FE n = 63 model R² = 93% [RUN]

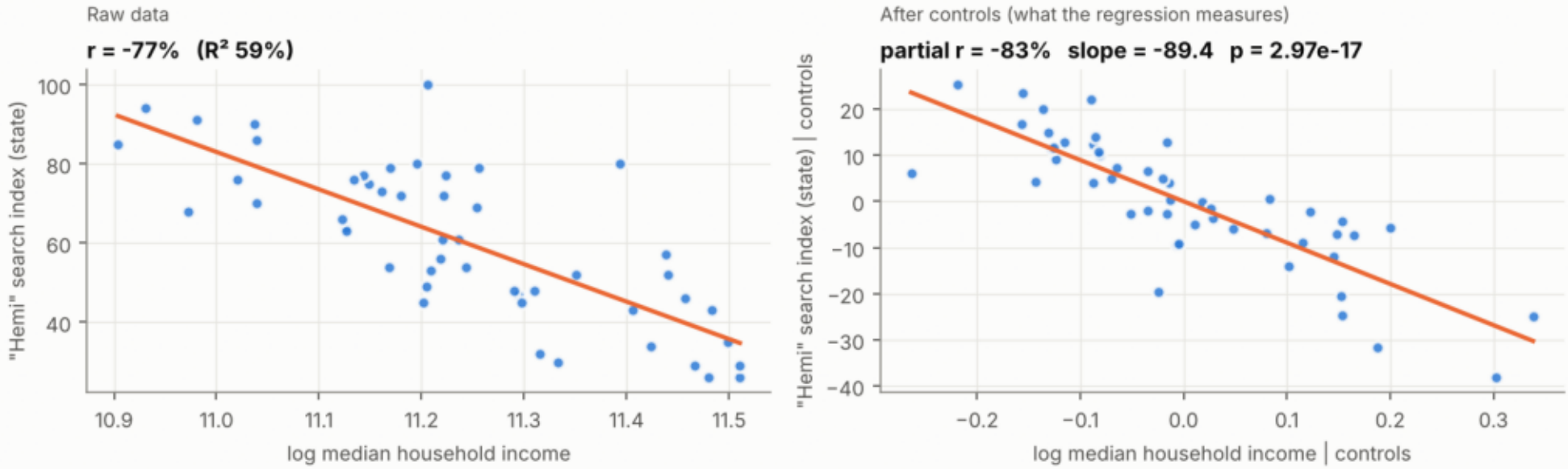
Test 44: Band sales relative to industry ~ sentiment (does Stellantis feel shocks more than the market?)





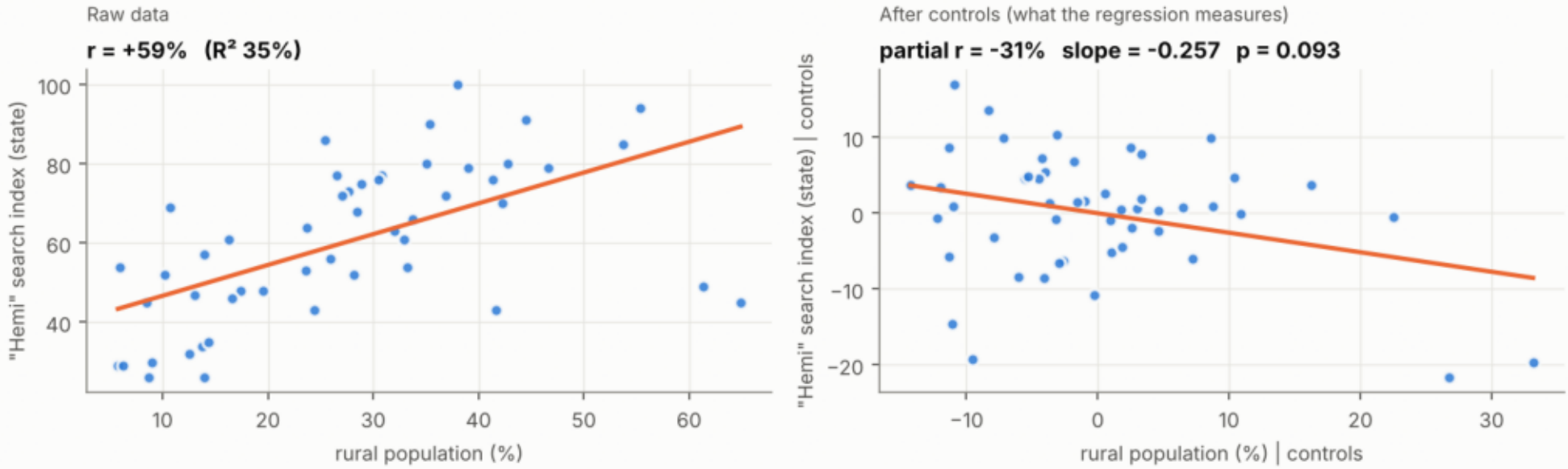
log(band units / industry units) ~ sentiment x band + FE n = 63 model R² = 93% [RUN]

Test 45: State "Hemi" search intensity ~ income, rurality, truck share



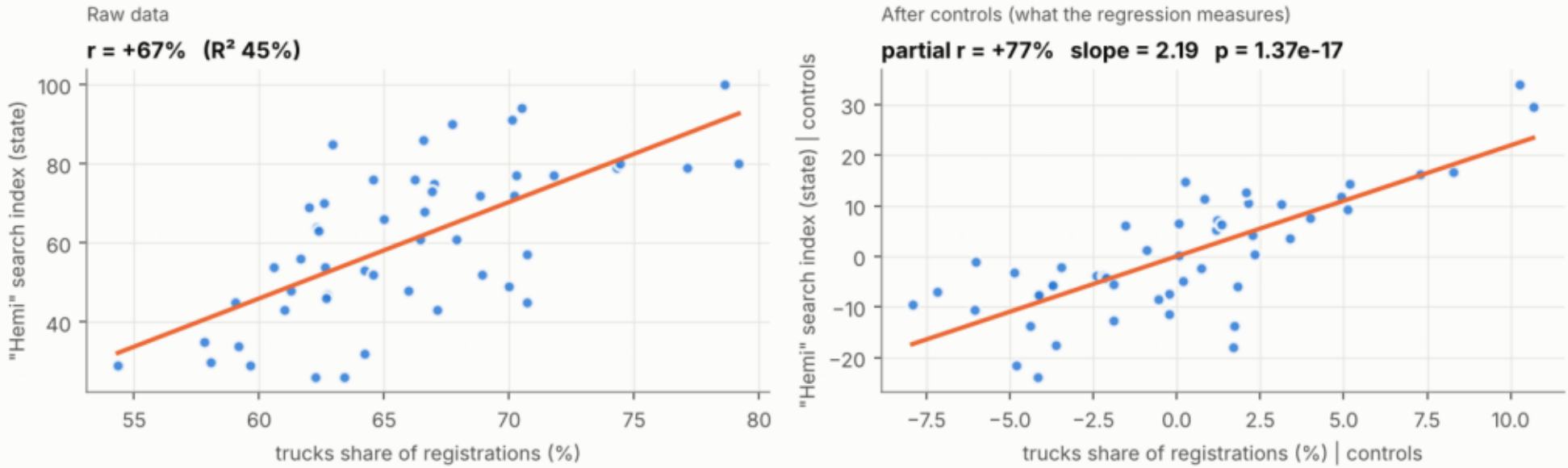
Trends state index (Jun 2025-Oct 2026) ~ log median HH income + rural % + truck registration %; HC1 n = 50 model R² = 85% [RUN (proxy)]

Test 45: State "Hemi" search intensity ~ income, rurality, truck share



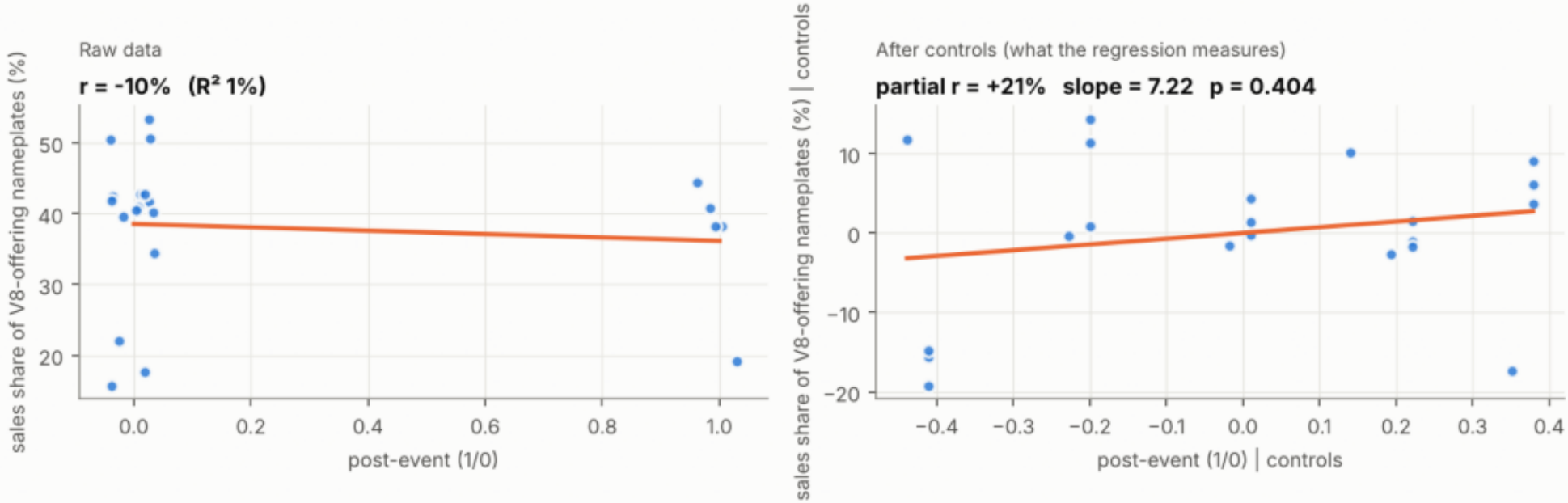
Trends state index (Jun 2025-Oct 2026) ~ log median HH income + rural % + truck registration %; HC1 n = 50 model R² = 85% [RUN (proxy)]

Test 45: State "Hemi" search intensity ~ income, rurality, truck share



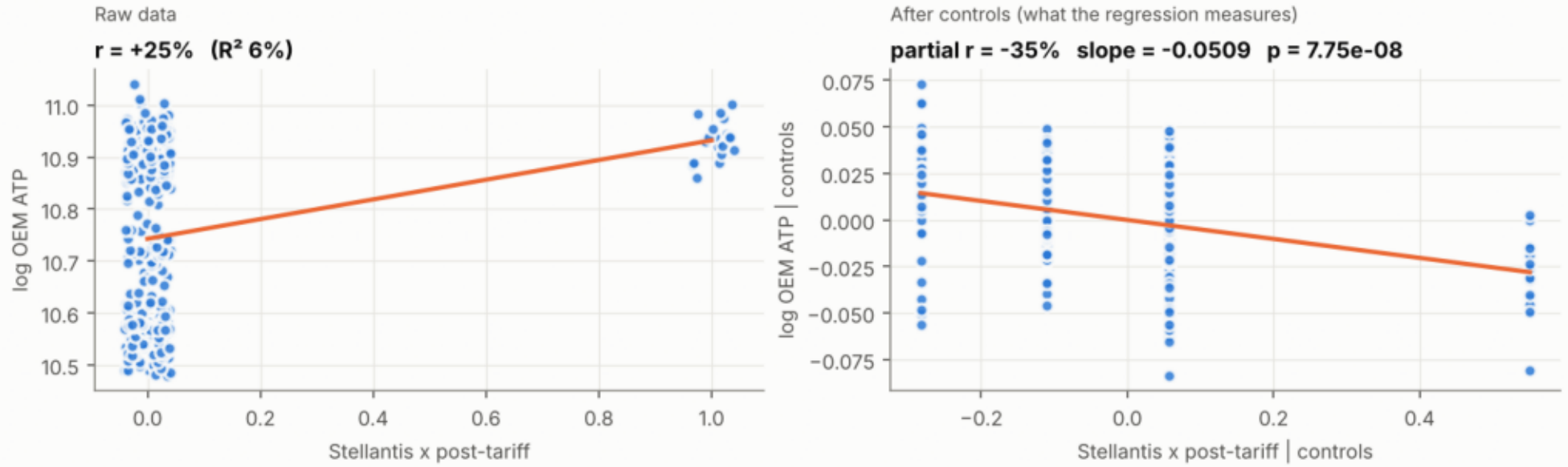
Trends state index (Jun 2025-Oct 2026) ~ log median HH income + rural % + truck registration %; HC1 n = 50 model R² = 85% [RUN (proxy)]

Test 47: Event study: share of Stellantis sales from V8-offering nameplates ~ post CAFE-penalty zeroing (Jul 2025)



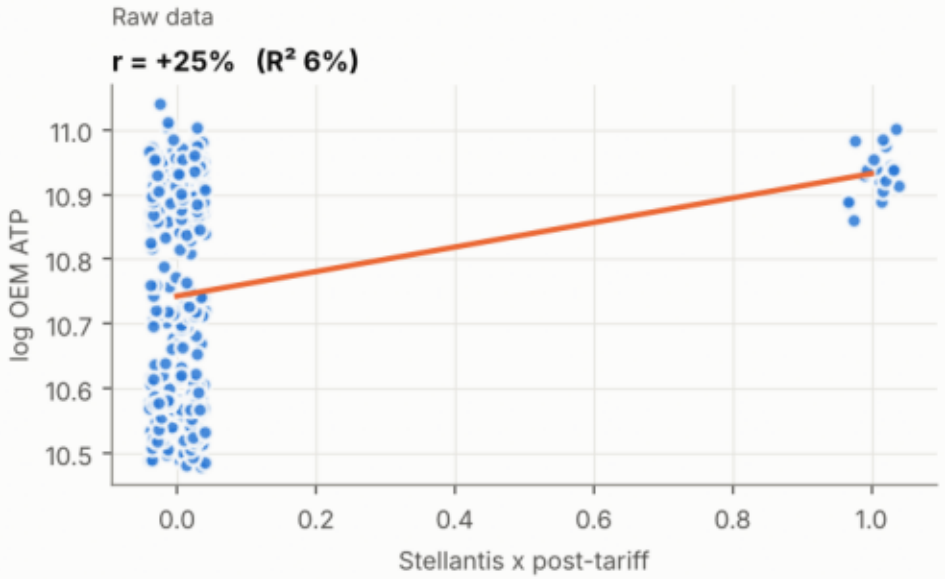
V8-available sales share % ~ post(>=2025Q3) + trend + quarter-of-year; HAC(2) n = 21 model R² = 20% [RUN (proxy)]

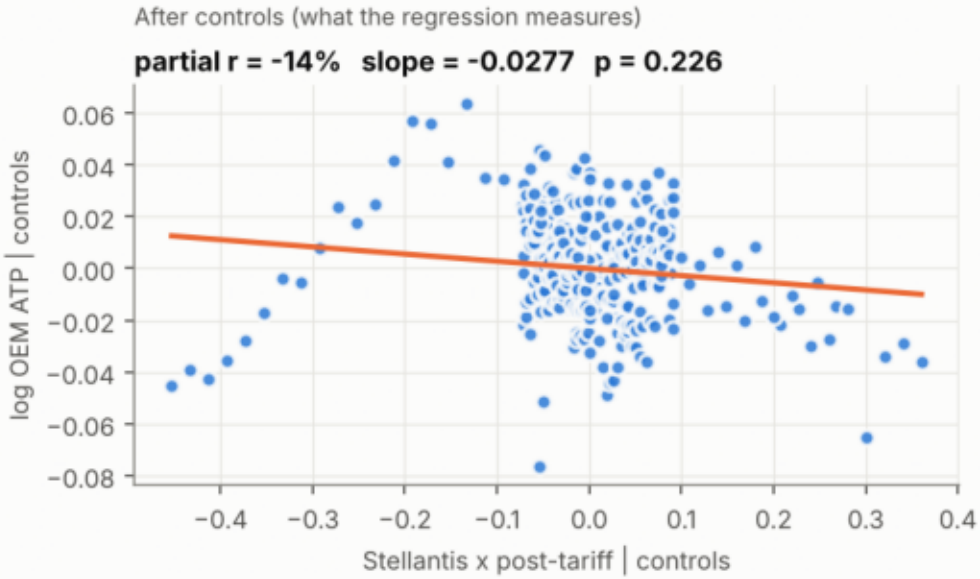
Test 48: Tariffs: Stellantis ATP vs peer OEMs after April 2025 auto tariffs



log(ATP) ~ Stellantis x Post(>=Apr 2025) + OEM FE + month FE; cluster month n = 336 model R² = 98% [RUN]

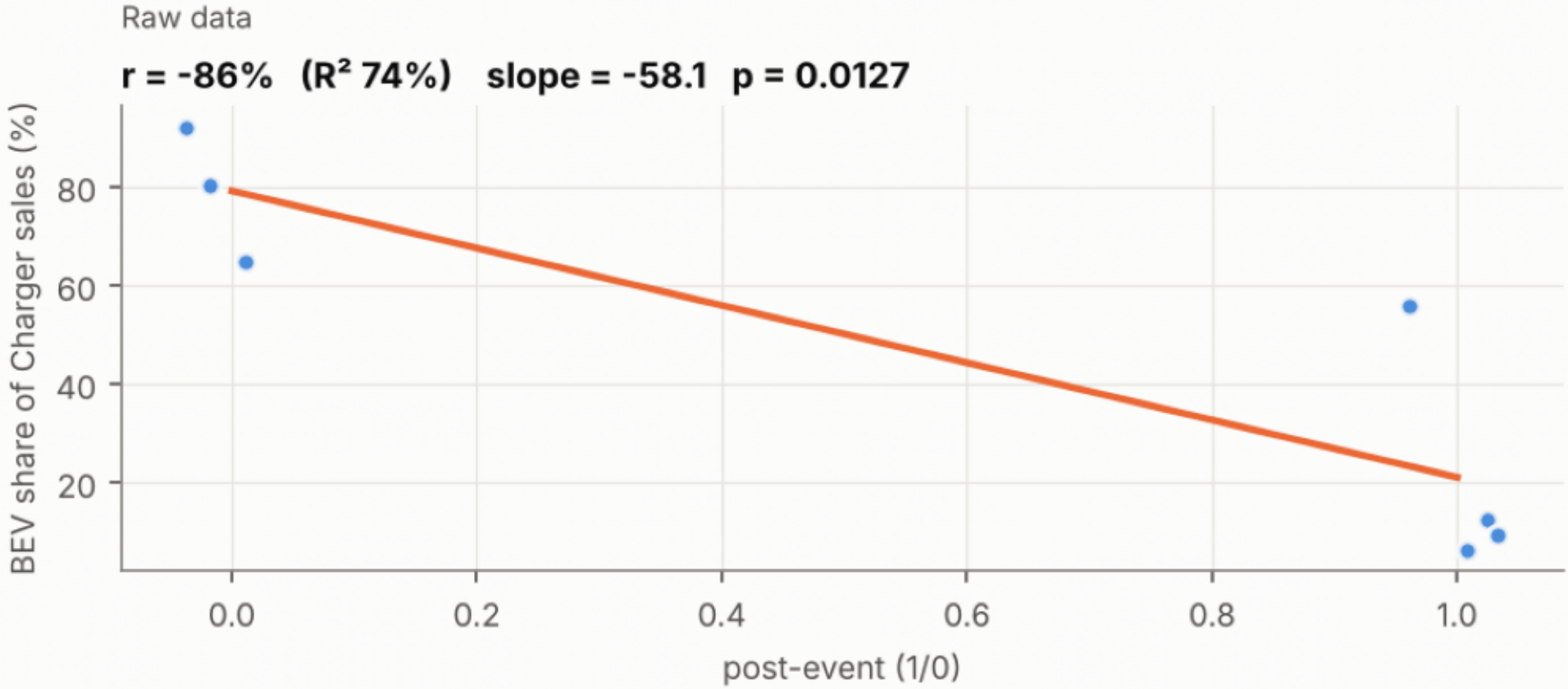
Test 48: Same, with OEM-specific linear trends





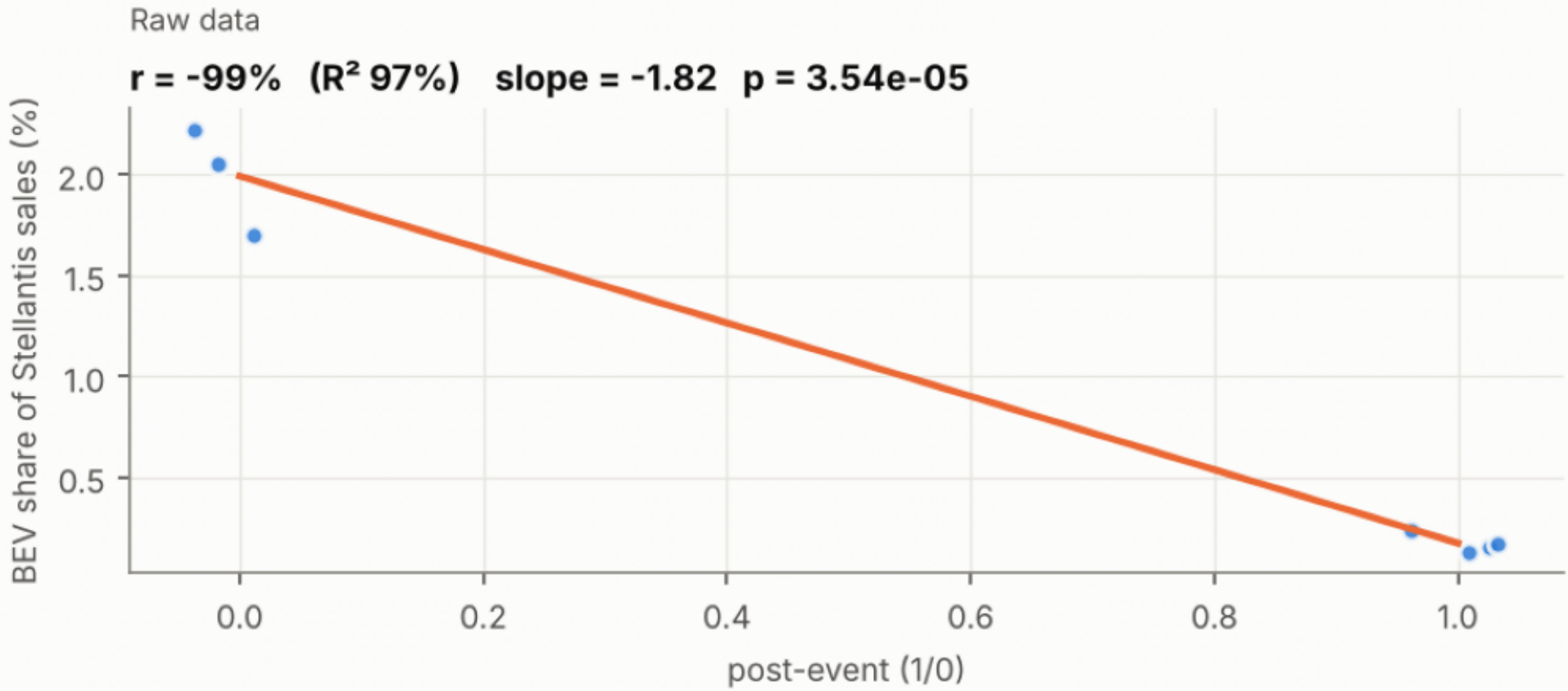
+ OEM x linear trend n = 336 model R² = 99% [RUN]

Test 49: Charger BEV share of Charger sales ~ post EV-credit expiry (Sep 30 2025)



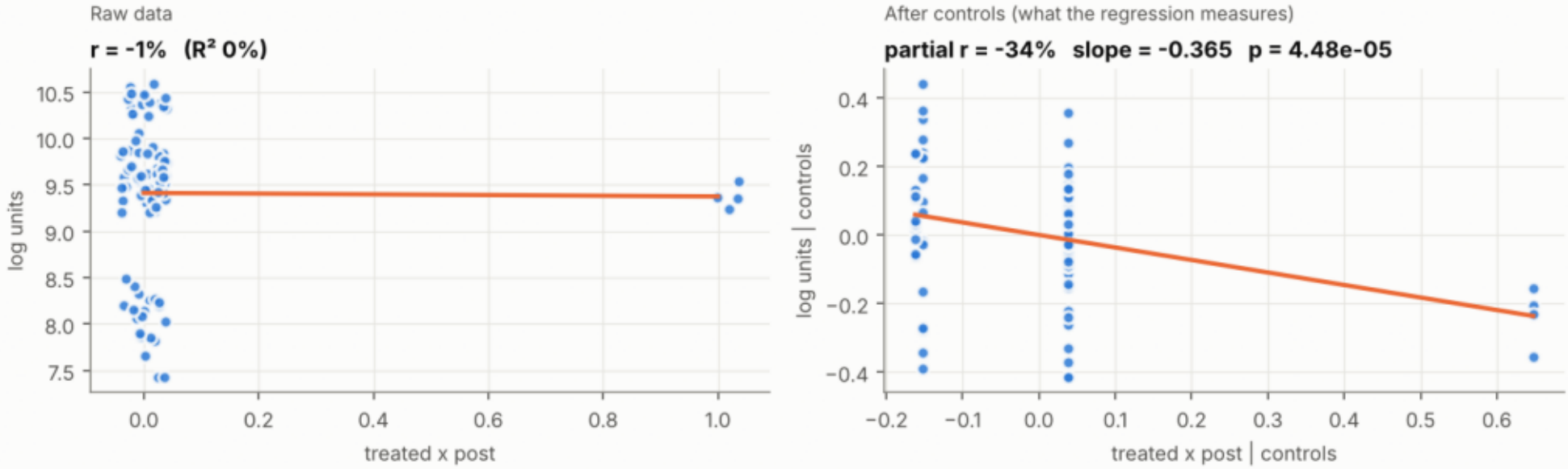
BEV share % ~ post(>=2025Q4); HAC(1); 2025Q1-2026Q3 n = 7 model R² = 74% [RUN]

Test 49: Stellantis BEV share (Charger BEV + Wagoneer S + 500e) ~ post credit expiry



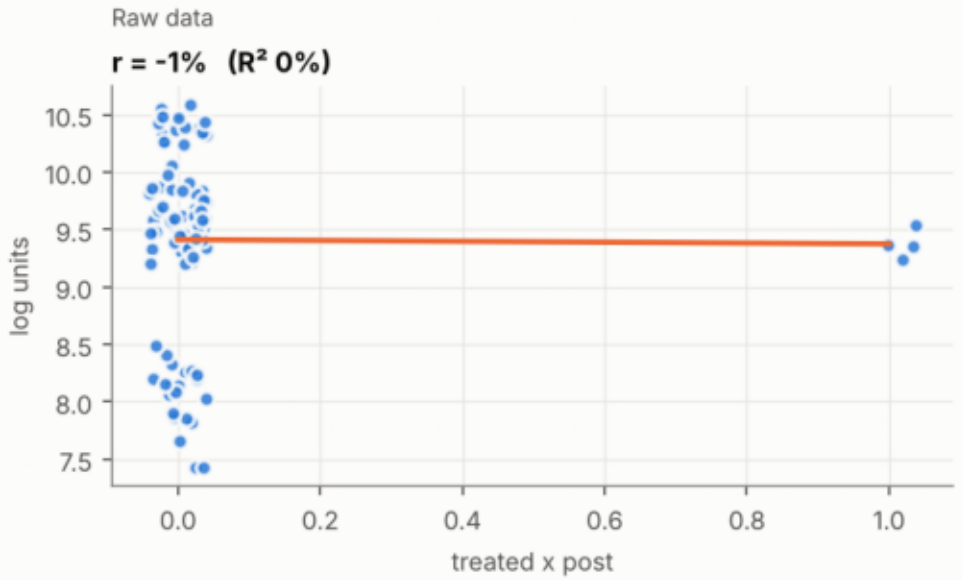
BEV % of Stellantis US ~ post n = 7 model R² = 97% [RUN]

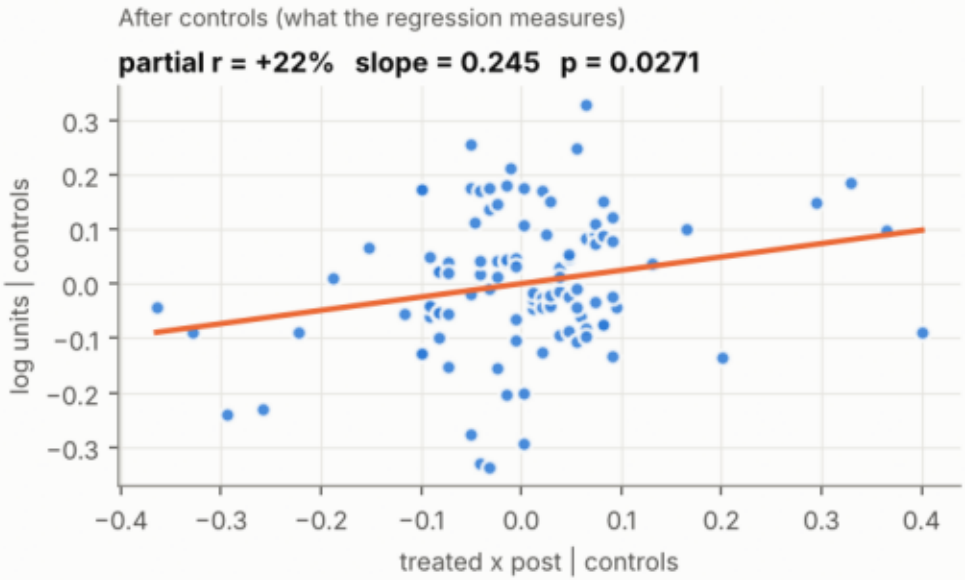
Test 50: Canada: Ram pickups vs Canadian full-size rivals after US Hemi return



log(units) ~ Ram x Post(>=2025Q4) + unit FE + quarter FE; HC1 n = 105 model R² = 96% [RUN]

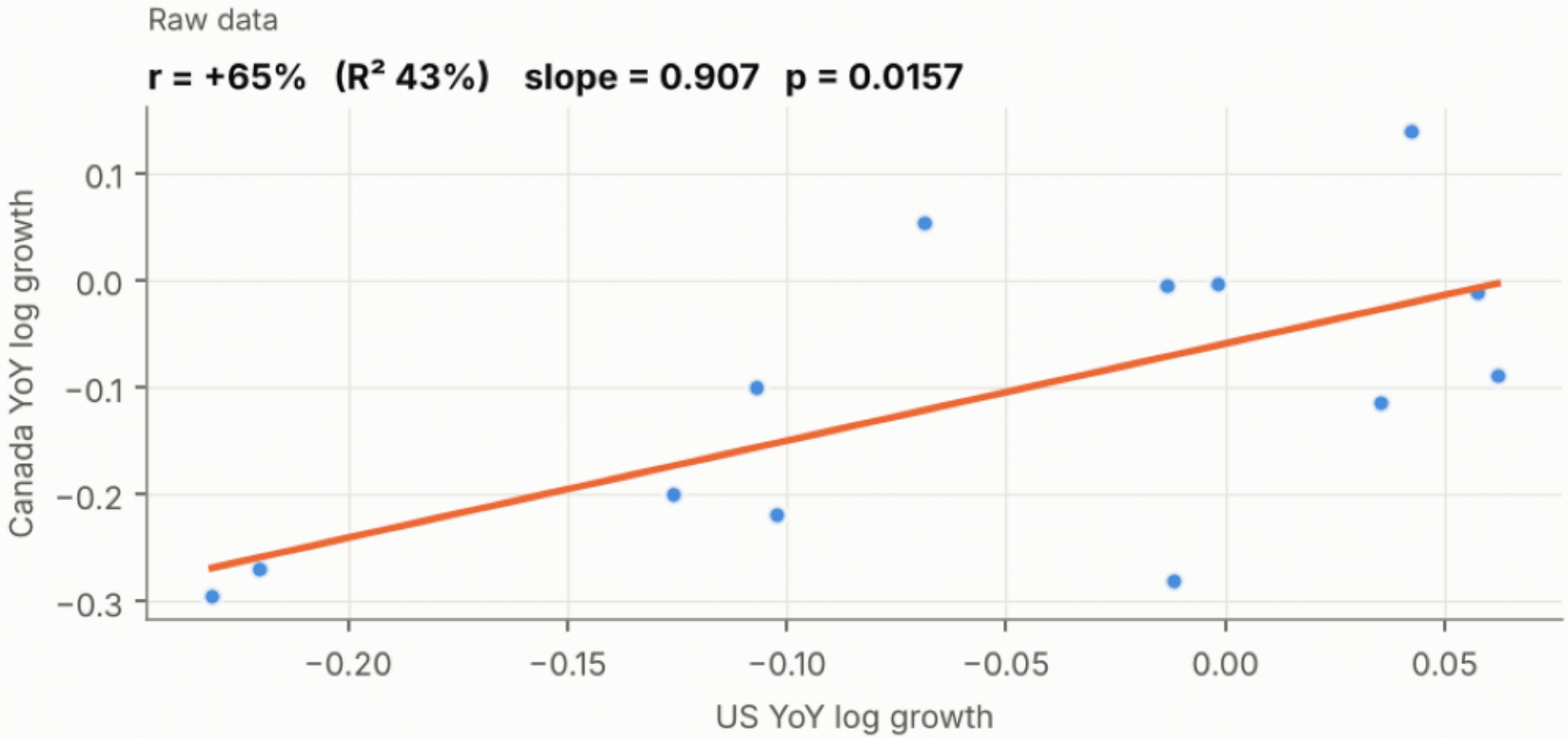
Test 50: Same, with Ram-specific trend





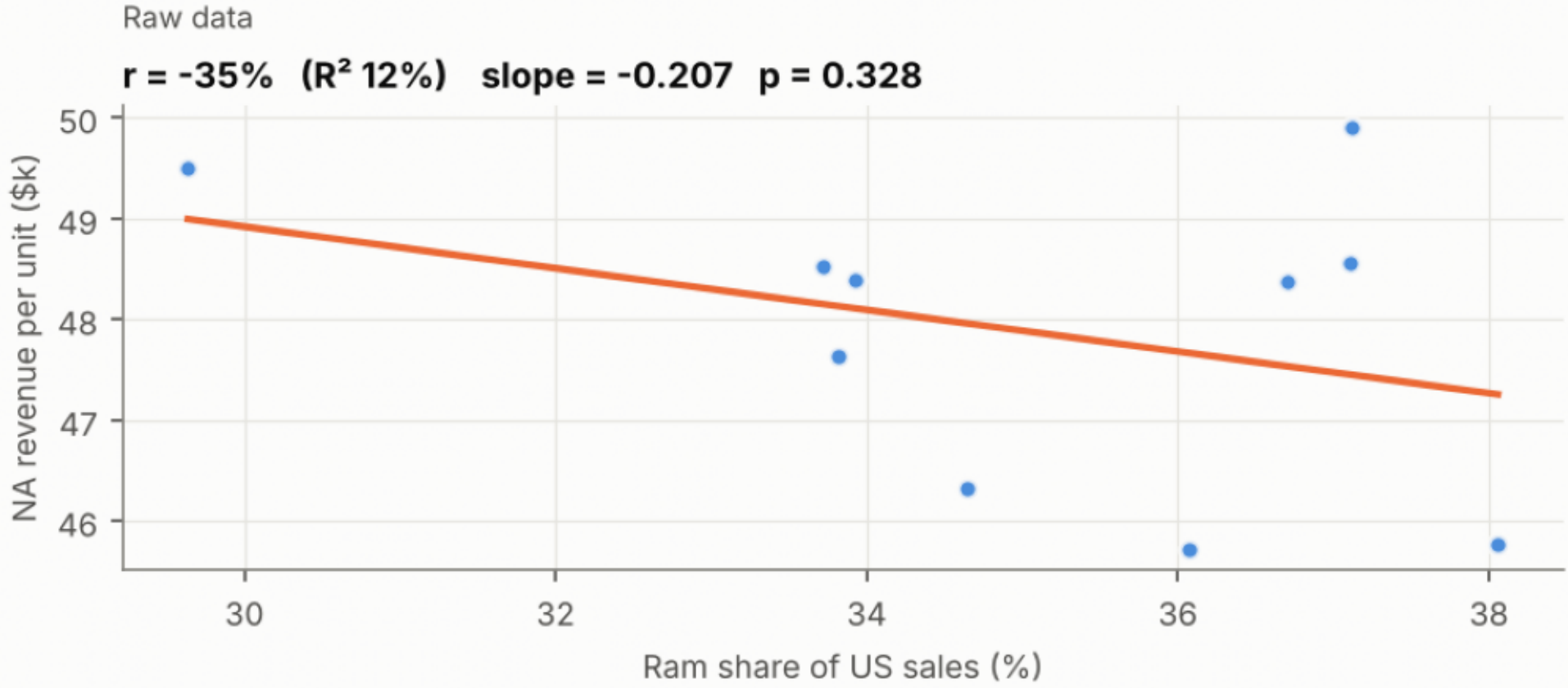
+ Ram x trend n = 105 model R² = 98% [RUN]

Test 50: Stellantis Canada YoY growth ~ US YoY growth



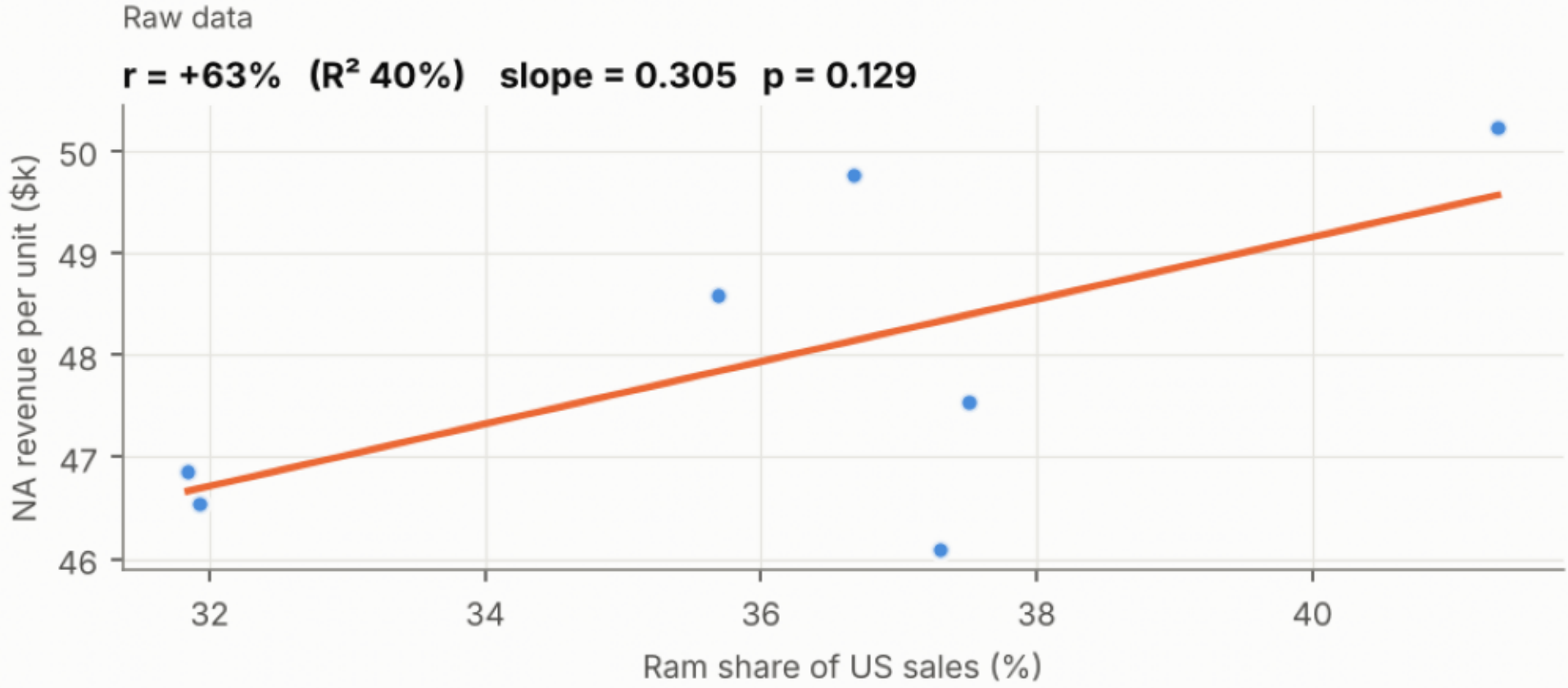
d4 log(Canada total) ~ d4 log(US total); HAC(2) n = 13 model R² = 43% [RUN]

Test 51: NA revenue per unit (USD k) ~ Ram share of US sales (half-years 2021-2026)



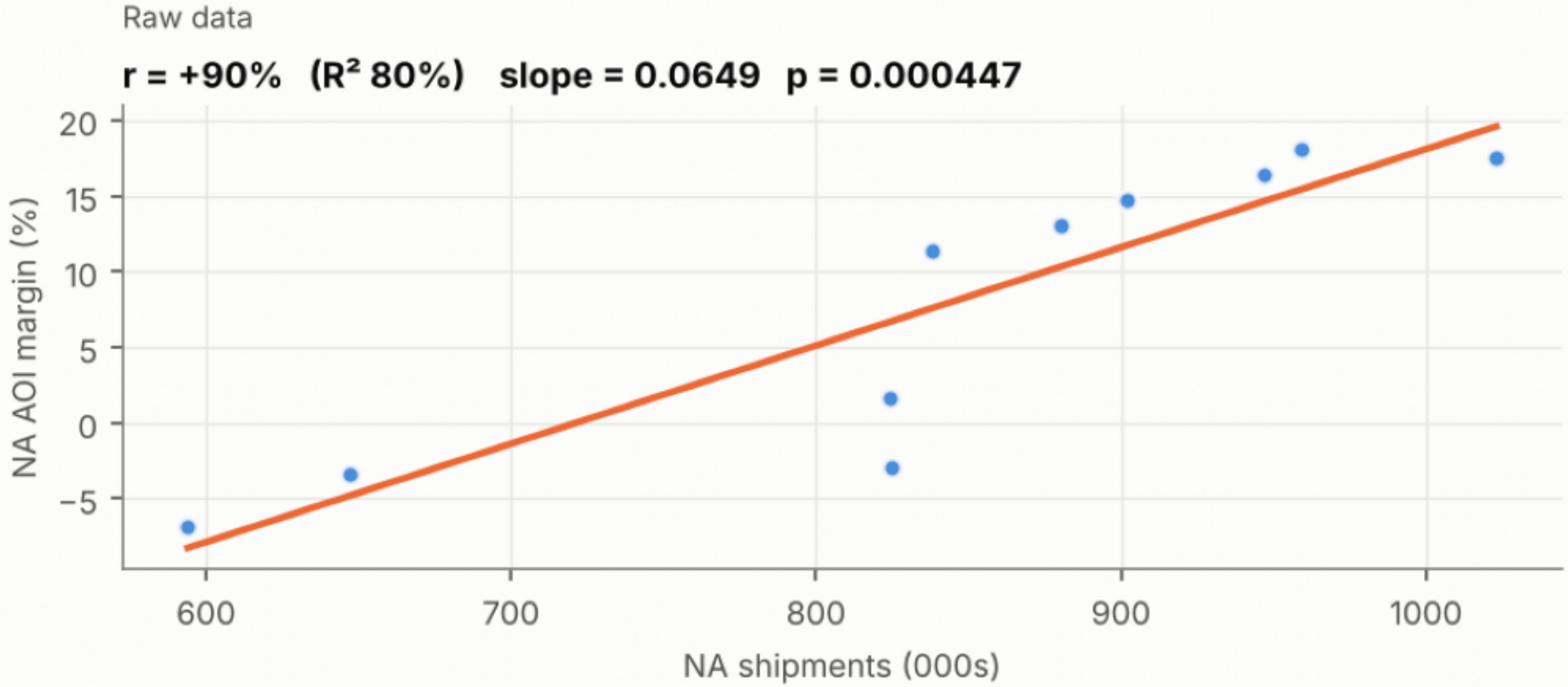
NA net revenue x EURUSD / NA shipments ~ Ram % of Stellantis US sales; HC1 n = 10 model R² = 12%
[RUN]

Test 51: NA revenue per unit (USD k) ~ Ram share of US sales (quarters 2025Q1-2026Q3)



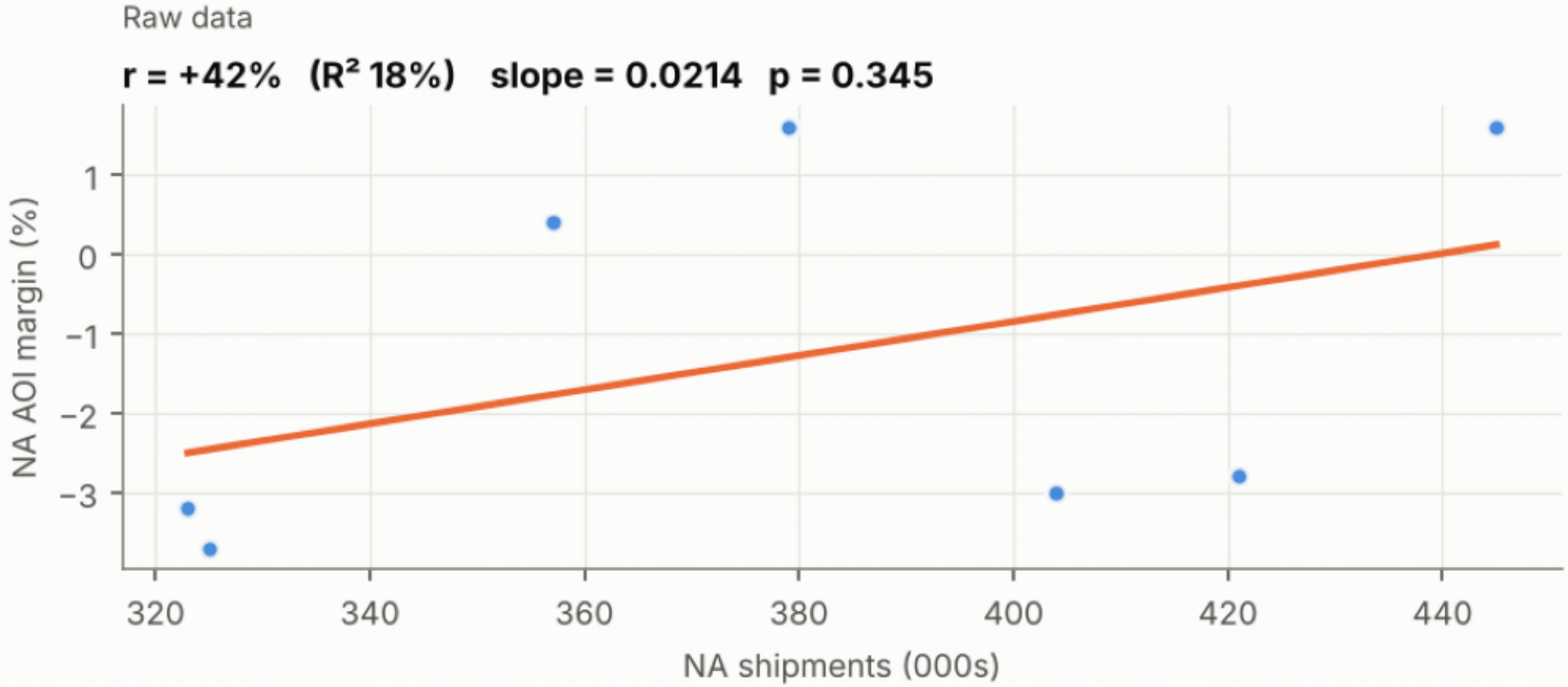
NA net revenue x EURUSD / NA shipments ~ Ram % of Stellantis US sales; HC1 n = 7 model R² = 40%
[RUN]

Test 52: NA AOI margin (%) ~ NA shipments (000s) (half-years 2021-2026)



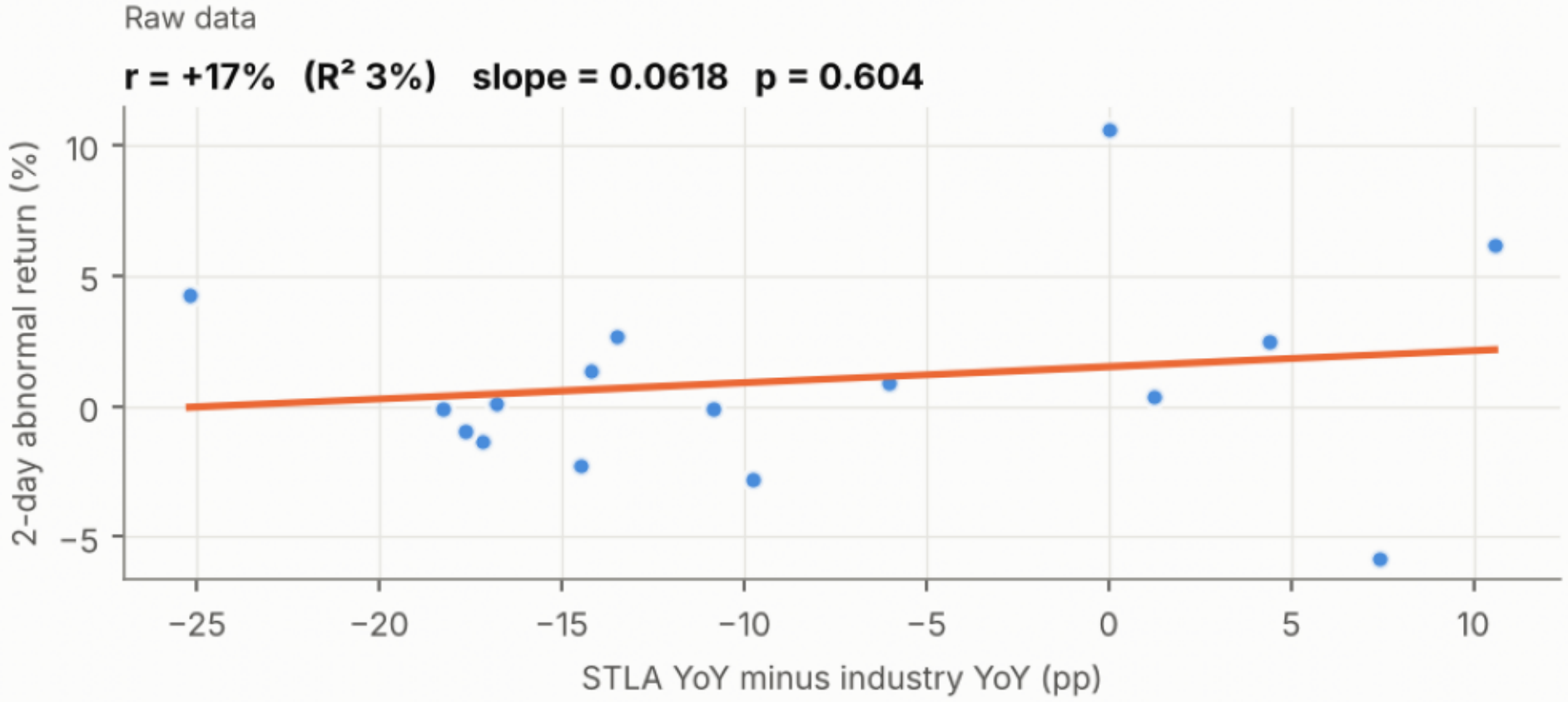
AOI margin ~ shipments; HC1 n = 10 model R² = 80% [RUN]

Test 52: NA AOI margin (%) ~ NA shipments (000s) (quarters 2025Q1-2026Q3)



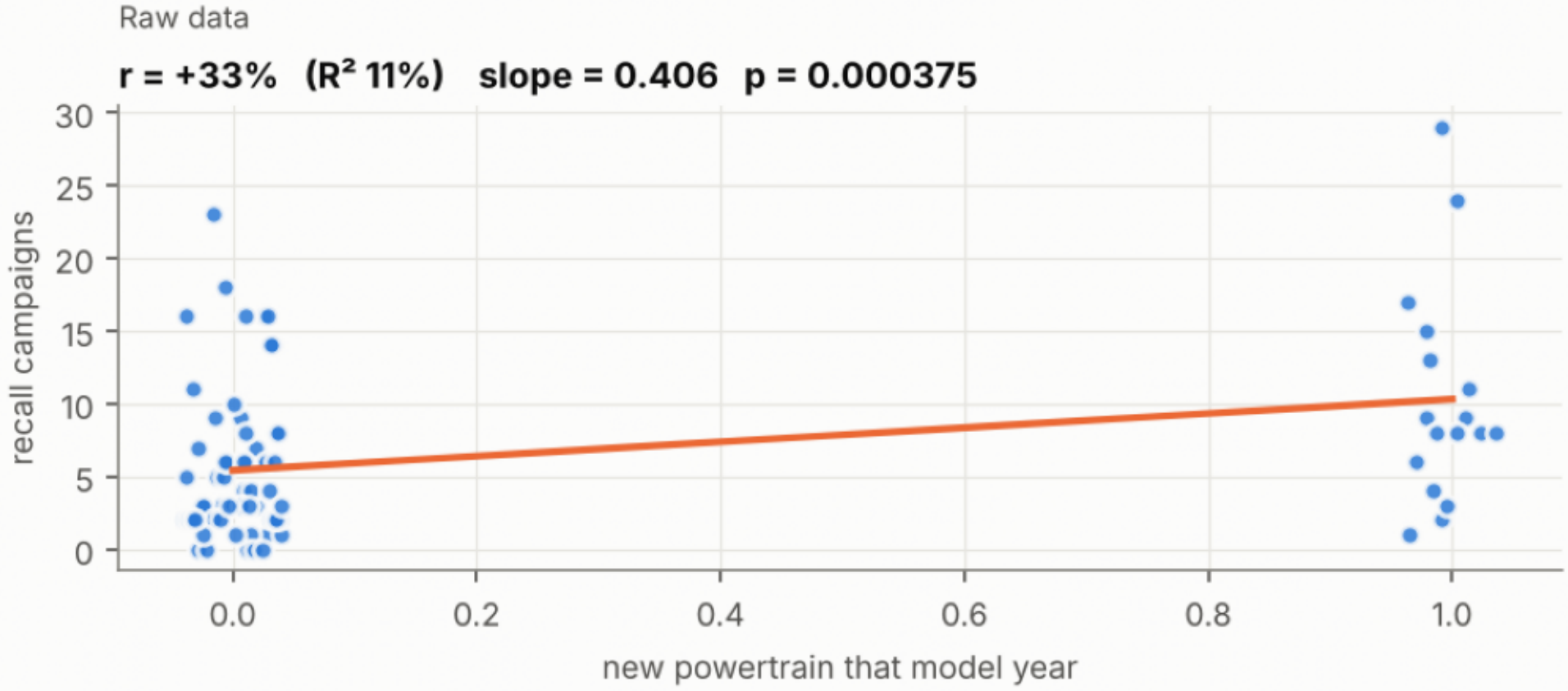
AOI margin ~ shipments; HC1 n = 7 model R² = 18% [RUN]

Test 55: STLA abnormal return on US sales-release days ~ sales growth vs industry



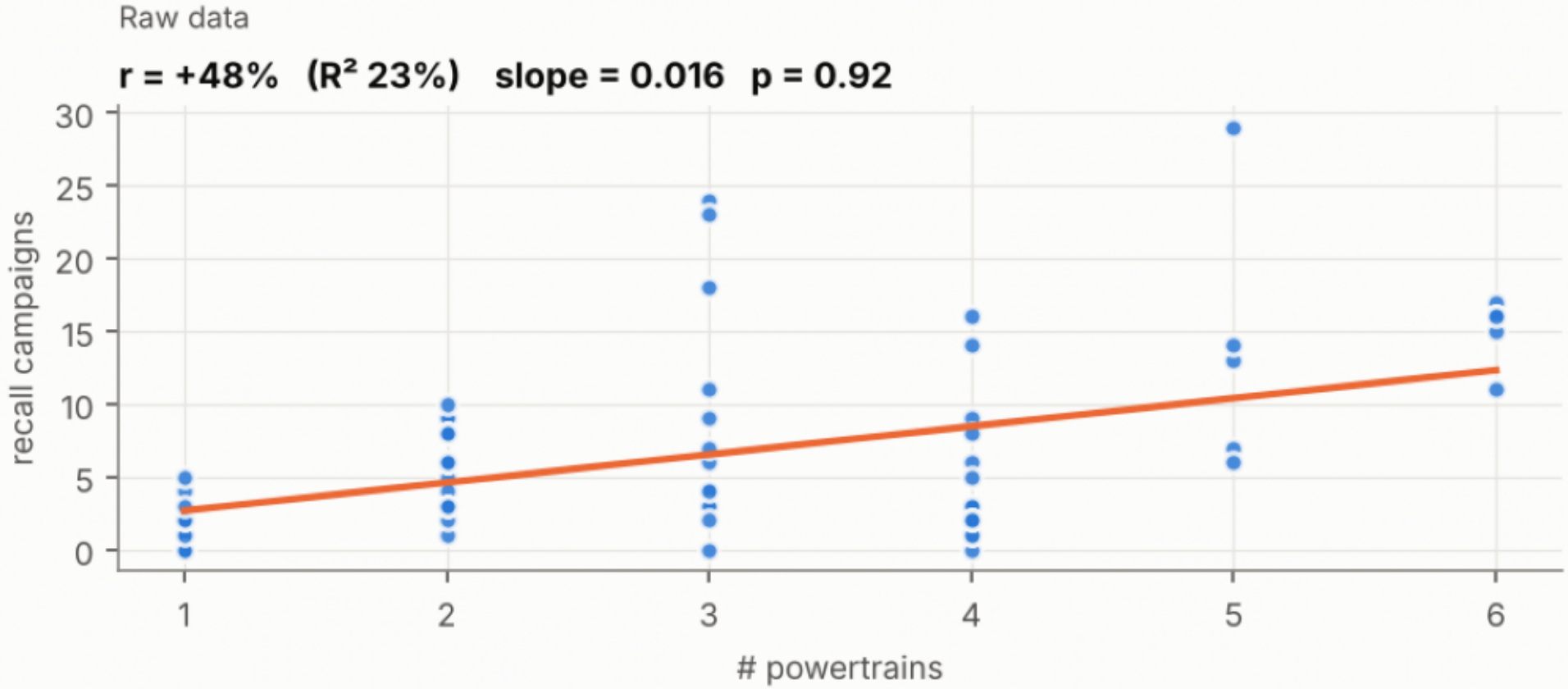
CAR[0,+1] (market model vs SPY, est. window t-130..t-10) ~ (STLA YoY - industry YoY) pp; HC1 n = 16
model R^2 = 3% [RUN]

Test 56: NHTSA recall campaigns per nameplate-model-year ~ new powertrain introduced that MY



Poisson: campaigns ~ new-powertrain dummy + # powertrains + nameplate FE + MY FE; cluster nameplate (coef = log rate ratio) n = 80 (Poisson model; raw counts shown) [RUN]

Test 56: NHTSA recall campaigns per nameplate-model-year ~ new powertrain introduced that MY



Poisson: campaigns ~ new-powertrain dummy + # powertrains + nameplate FE + MY FE; cluster nameplate (coef = log rate ratio) n = 80 (Poisson model; raw counts shown) [RUN]